



Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

**New Arts, Commerce and Science College (Autonomous),
Ahmednagar**

PG Department of Statistics

Project report on

Title: Analysis of WPI data of India

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**Academic Year
2022-2023**

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**New Arts, Commerce and Science College
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(M. Sc. II)

CERTIFICATE

This to certify that _____,
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Of WPI data of India** during the year 2022-2023.

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ACKNOWLEDGEMENT:

I am thankful to our respected principal Prof. Dr.B.H.Zaware and Head of Department Dr.A.A.Kulkarni for their support and co-operation towards the successful completion of this project. I avail this opportunity to express our deep sense of gratitude and wholehearted thanks to project guide Prof. Marziya Shaikh for giving her valuable guidance, inspiration, and affectionate encouragement to embark on this project. We are also extremely thankful to Dr. B.K. Thorve, Prof. K.B. Mane, Prof. M.D. Rohakale, Prof. B.P Kharat and Prof. Komal Kawade for their support and guidance for this project. Finally, I would like to thank all those who helped us directly or indirectly in endeavour and infused their help for the success of the project.

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Abstract

In this study we attempted to analyse Wholesale Price Index of India by considering the base year 2011-12 i.e., the percent rise or decline in price index at wholesale level is compared with financial year 2011-12. In India, the Wholesale Price Index (WPI) is the main measure of inflation. The WPI measures the price of representative basket of wholesale goods.

The data contains 697 commodities over the span of last 10 years.

In India Wholesale Price Index is divided into 3 groups:

- Primary Articles category (22.6% of total weight)
- Fuel & Power (13.2% of total weight)
- Manufactured Products (64.2% of total weight)

Since these 3 groups contains different commodities with different weights, therefore we have analysed all these categories separately in order to find insights. We have applied different cluster algorithms that give best results in identifying patterns and trends in different groups that might not be apparent when looking at the individual commodity.

We have applied Stochastic model to find overall system behaviour for past 10 years and to predict future probabilities of price index for these 3 groups and applied Time series analysis for predicting future price moments (point estimates) based on historical data which will help in forecasting inflation rate of India.

KEYWORDS:

K-Means, K-medoids, Hierarchical, Markov-Chain ,Holt-Winter, SARIMA.

Introduction

Index Numbers:

Economic phenomena are dynamic, we observe that the prices of commodities change from time to time, place to place, wages of workers, prices of shares, exhibit up and down moment. Index number measures such changes in fact they measure pulse of economy like inflationary or deflationary tendencies.

The apparatus barometer reflects the changes in atmospheric pressures likewise, index numbers reflect the changes in economic activities. Hence index numbers are rightly called as economic barometers.

For determination of index numbers, we need to consider several commodities we know that a series of observation can be reduced to a single observation or two series can be compared with the help of averages. Thus, it is essential to find the average change in prices or quantities to find index numbers but to calculate an average we require the observation measured in same unit. However, the prices and quantities differ in their units.

For example: Prices of sugar, rice is expressed in Rs per kg whereas prices of Milk, Petrol, etc are in Rs per litre. Thus, average of prices of such quantities is meaning less and it no longer remain useful. In order to overcome this difficulty, changes are measured in percentages which are unit less numbers and the average change is calculated. The average so obtained is an index number.

Definition: “Index number is a number designated to measure the average change in values of a group of related variables over two different situations.”

What is Wholesale Price Index(WPI)?

The Wholesale Price Index (WPI) is an economic indicator that measures the average change in prices of goods at the wholesale level over a specific period of time. It is used to track inflationary pressures and price trends in the economy.

The WPI calculates the price changes of a basket of goods and commodities commonly traded in wholesale markets, such as raw materials, intermediate goods, and finished products. It reflects the price fluctuations faced by producers and wholesalers before the goods reach the final consumers.

The index is typically calculated using a weighted average formula, where each item in the basket is assigned, a weight based on its relative importance or value in the wholesale market. The weights reflect the significance of different goods or commodities in the overall economy.

By monitoring changes in the WPI over time, economists, policymakers, and businesses can assess the level of inflation or deflation in the economy, analyse price trends, and make informed decisions regarding monetary and fiscal policies, pricing strategies, and market conditions.

It's important to note that the WPI is different from the Consumer Price Index (CPI), which measures the average change in prices of goods and services at the consumer level. The WPI focuses on wholesale prices, while the CPI represents the prices paid by consumers for a specified basket of goods and services.

Interpretation of WPI:

WPI: March 2023 is 120

In this example WPI value of 120 indicates that wholesale prices, on average, were 20% higher in March 2023 compared to the base year of 2011. The base year is typically assigned a value of 100 for ease of comparison.

Weighting structure

Weight for items under different groups have been assigned on the basis of the total wholesale transaction in the economy.

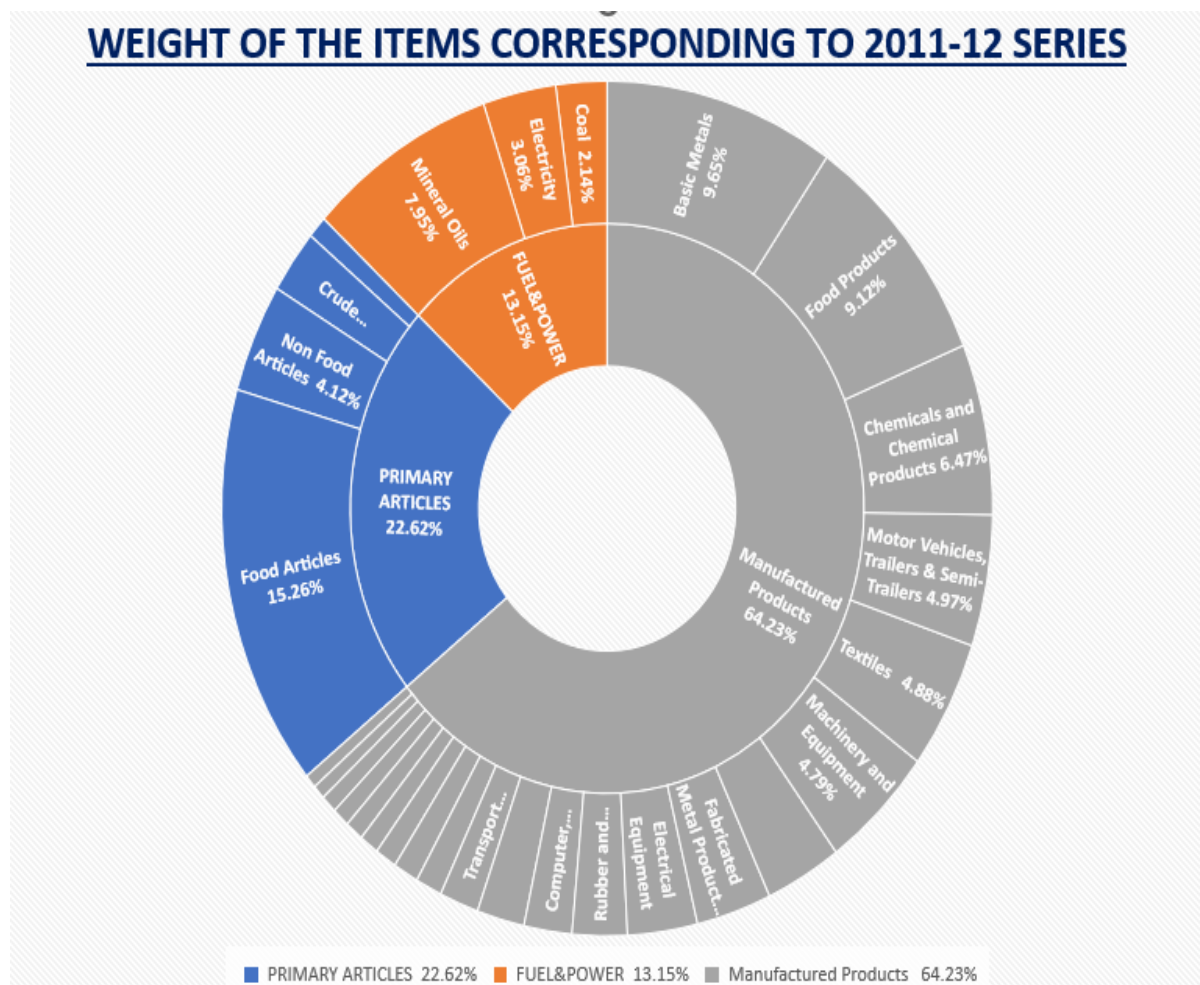


Figure no.1

The inner most ring of sunburst chart represent the main categories of WPI data as well as their weightage, followed by outermost ring which represent subcategories within each main categories , whereas size of each segment represents the weightage of that specific subcategories.

It can be observed that high weightage in WPI data is given to manufacturing product category of 64.23% followed by category of primary article of weight 22.62% and then fuel and power with weightage of 13.15%.

Visual Representation of No. of Commodities:

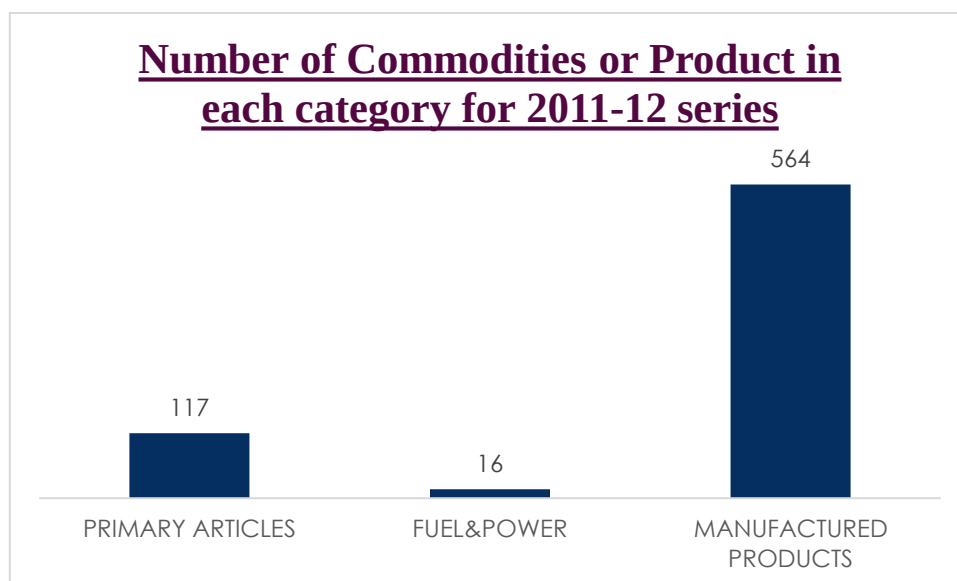


Figure no.2

There are 697 commodities in WPI data which is divided into 3 main Categories where 117 products belong to primary articles, 16 to Fuel and Power and remaining to Manufacturing Category.

Quotations:

‘Quotations’ refer to the prices at which goods or commodities are quoted or offered in wholesale markets.

To calculate the WPI, data collectors regularly gather price information from selected wholesale markets or establishments across various sectors of the economy. These prices are collected through surveys or direct reporting from wholesalers, manufacturers, or importers. The prices collected are typically referred to as "Quotations."

Visual Representation of No. of Quotations for each of 3 main categories in WPI:

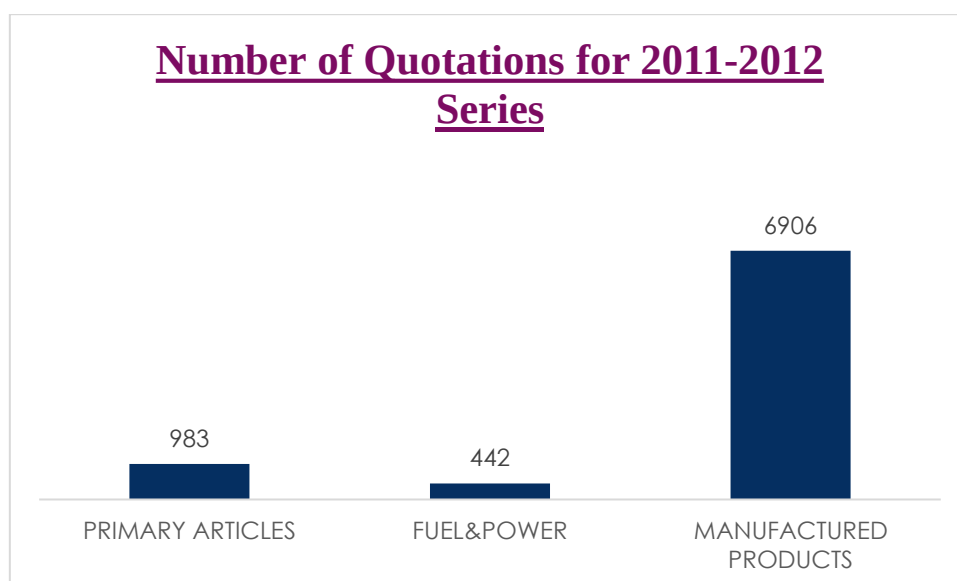


Figure no.3

Method of Calculation of WPI :

The Compilation of new WPI series broadly consists of two stages –

(i) First, the item level indices (i.e., elementary price index) are calculated using “Jevons Index formula”, which uses the Geometric Mean (GM) of Price Relatives (i.e., the price change). Price relatives are calculated as the percentage ratios, i.e., by dividing the current price by the base period price and multiplying the quotient by 100. These elementary indices are the lowest level of aggregation where prices are combined into price indices.

(ii) Compilation of WPI at Group Level:

In the second stage, these elementary price indices are aggregated using weighted arithmetic mean to obtain higher level (at sub-group/ group/ major group/ All Commodities level) indices using Laspeyre’s index formula, which has a fixed base-year weighting diagram operative through the entire life span of the series. The formula used is:

$$I = \sum (I_i \times W_i) / \sum W_i$$

Where,

$\sum$ represents the summation operation, I = Index Number of wholesale prices of a sub-group/group/major group/All commodities

W_i = weight assigned to the i th item/sub- group/group/major group

I_i = Index of the i th item/sub- group/group/major group.

In the new WPI series, elementary price index (i.e. at the item level) has been computed using the geometric mean of the price relatives (Jevons' Index) since

Jevons' Index gives equal weight to upward and downward movements within the elementary aggregate. . Since it is not desirable to allow the item level indices to be swayed by price movements of higher priced items especially when number of quotations is less, Jevons' index seems to be a reasonable choice for computing item level index.

Reason for missing values in WPI data:

There are 6 items, especially in fruits and vegetables group which are of seasonal nature. When a particular seasonal item disappears from the market and its prices are not quoted therefore values are missing for these 6 items(Mango, Orange, grapes, litchi, peas, Cauliflower), the index of such an item is not compiled and its weight is distributed over the remaining items in the concerned sub-group, on a pro rata basis.

What is inflation and its calculation:

Wholesale Price Inflation (WPI) refers to the rate of change in the average prices of goods and commodities at the wholesale level over a specific period. It measures the inflationary pressures and price trends in the wholesale market.

WPI inflation is calculated by comparing the WPI values for two different periods. The formula to calculate WPI inflation is as follows:

WPI Inflation Rate = $\frac{(\text{WPI in Current Period} - \text{WPI in Previous Period})}{\text{WPI in Previous Period}} \times 100$

For example, let us say the WPI in the previous period was 150 and the WPI in the current period is 160. Applying the formula, the WPI inflation rate would be:

WPI Inflation Rate = $\frac{(160 - 150)}{150} \times 100 = 6.67\%$

This means that the average wholesale prices have increased by 6.67% over the given period.

Interpreting wholesale price inflation is like interpreting general inflation. Positive WPI inflation indicates an increase in average wholesale prices,

which can be a sign of inflationary pressures in the economy. It implies that producers and wholesalers are facing higher costs, which could potentially lead to increased prices for end consumers.

On the other hand, negative WPI inflation, also known as deflation, suggests a decrease in average wholesale prices. This can have implications for businesses and economic growth, as it may indicate weak demand, excess supply, or reduced production costs.

Data Description

We have WPI monthly (Wholesale Price Index) data which is considered as univariate panel data [Univariate panel data refers to a type of dataset that combines elements of both time series data and cross-sectional data. It involves observations collected over multiple time periods for a single variable (univariate) across different entities or individuals (cross-sectional)] because it includes observation of a single variable namely the wholesale price index for different commodities over time. in our data we have 697 commodities over period of 10 years with base year as 2011-12.

Here is a snapshot of our data:

	COMM_NAME	COMM_CODE	COMM_WT	INDX042011	INDX052011	INDX062011	INDX072011	INDX082011	INDX092011	INDX102011	...	INDX032022
1	I PRIMARY ARTICLES	1100000000	22.61756	95.6	95.7	97.6	98.6	100.4	102.1	102.4	...	170.9
2	(A). FOOD ARTICLES	1101000000	15.25585	94.5	95.5	98.0	99.6	101.5	103.6	104.7	...	169.6
3	a. FOOD GRAINS (CEREALS+PULSES)	1101010000	3.46238	97.6	98.7	98.8	99.6	100.2	101.1	100.9	...	169.1
4	a1. CEREALS	1101010100	2.82378	98.5	99.7	99.7	100.5	100.8	100.7	99.8	...	167.7
5	Paddy	1101010101	1.43052	97.5	98.9	99.4	100.1	101.3	101.5	101.5	...	163.1
...	...	...	...	...	...	...	...	...	...	...	...	...
864	Plastic moulded-others toys	1322040003	0.00210	99.4	99.3	99.5	99.4	99.7	99.5	100.2	...	142.4
865	Non Mechanical Toys	1322040004	0.00014	99.9	99.9	99.9	99.9	99.9	99.9	99.9	...	202.9
866	e. Manufacture of medical and dental instrumen...	1322050000	0.04941	99.6	99.6	99.6	99.6	99.6	99.6	99.6	...	177.8
867	Intraocular Lens	1322050001	0.04941	99.6	99.6	99.6	99.6	99.6	99.6	99.6	...	177.8
868	IV FOOD INDEX	2000000000	24.37758	95.8	96.6	98.4	99.3	100.6	102.2	103.0	...	168.2

868 rows × 144 columns

Aim and Objectives

Wholesale Price Index(WPI) is an economic indicator that measures the changes in prices of goods at wholesale level.

Inflation Analysis:

- The WPI is often used as an indicator of inflation in an economy by tracking changes in index over time, we can identify inflationary trends in wholesale markets.

Sectorwise Analysis:

The WPI is compiled for various sectors, such as Primary Articles, Fuel & Power and Manufactured goods by analysing the WPI for each sector we can identify which sector are experiencing price changes and to what extent.

Forecasting Analysis:

The WPI can also be used to forecast future price trend in wholesale market. By analysing historical data and identifying patterns we can make prediction about future price movements.

EDA & Data Visualisation

Visualizing sub-main & their sub-categories of Primary Articles with Sunburst chart

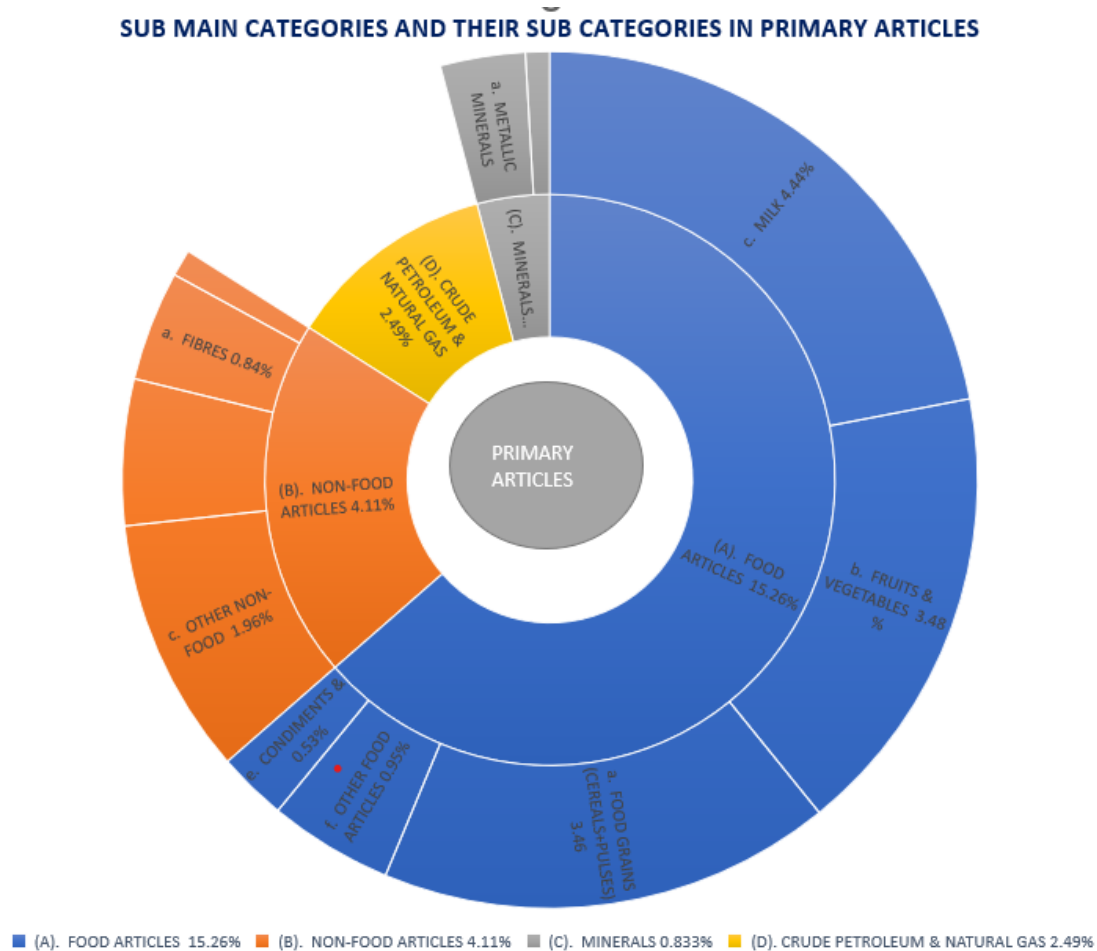


Figure no.4

Interpretation: The inner most ring of sunburst chart represents the sub-main categories of Primary Articles as well as their weightage, followed by outermost ring which represent subcategories within each sub-main categories , whereas size of each segment represents the weightage of that specific subcategory.

It can be observed that high weightage in Primary Articles Category is given to sub-main category 'Food Articles' with 15.26% share followed by sub-main category 'Non-food articles' with 4.11% share, sub-main category 'Crude Petroleum & Natural gas with 2.49% share and then sub-main category 'Minerals' with 0.83% share .

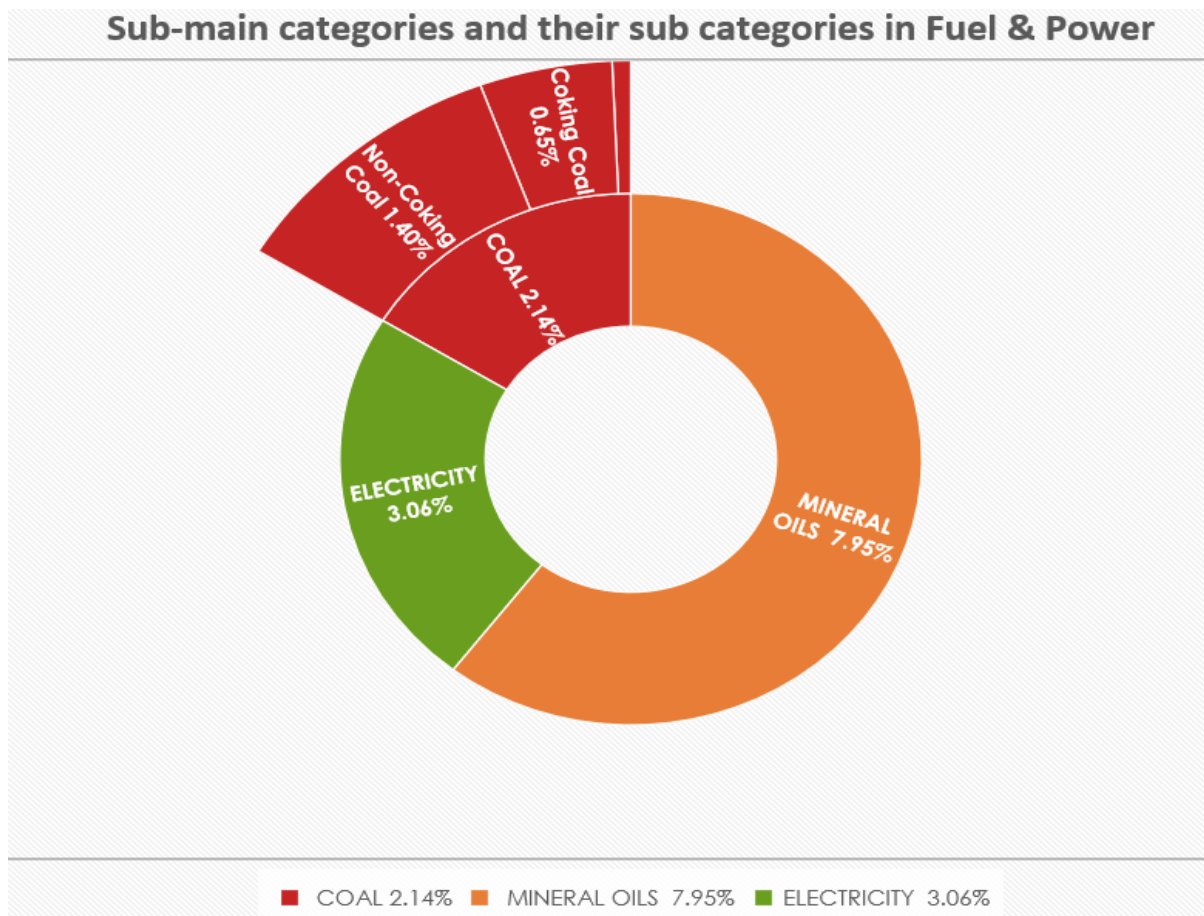


Figure no.5

Interpretation: The inner most ring of sunburst chart represents the sub-main categories of Fuel & Power as well as their weightage, followed by outermost ring which represent subcategories within each sub-main categories, whereas size of each segment represents the weightage of that specific subcategory.

It can be observed that high weightage in fuel & Power Category is given to sub-main category 'Mineral oils' with 7.95% share followed by sub-main category 'Electricity' with 3.06% share and then sub-main category 'Coal' with 2.14% share.

The most important components of manufactured products groups are basic metals (9.7 percent of total weight) food products (9.1 percent) chemicals and chemical products (6.5 percent) and textiles (4.9 percent).

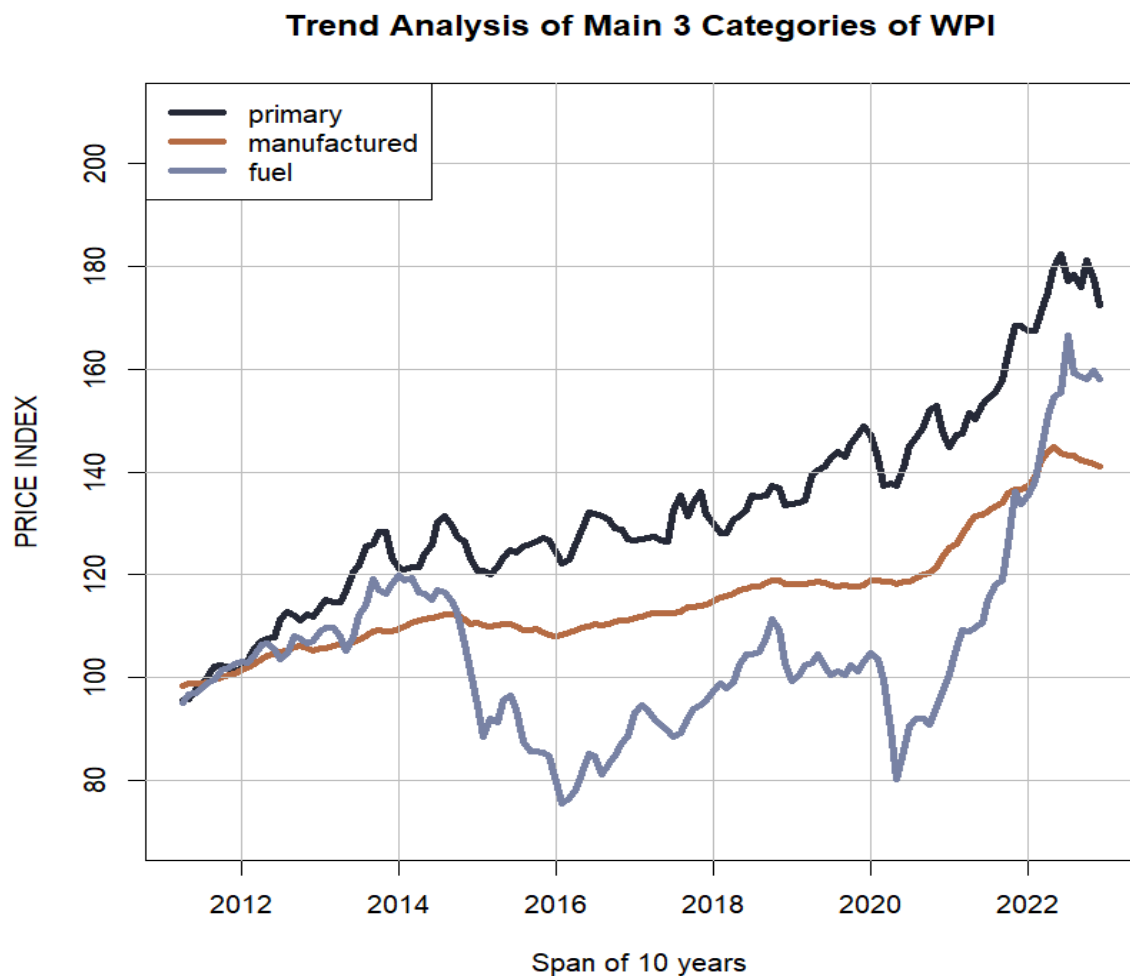


Figure no.6

Interpretation: It seems that primary article category shows increasing trend with random fluctuations over the time whereas there seems a unusual spike in middle of year 2021, the manufactured product category shows steady trend over the years but unusual spike is observed for same year for Fuel and power category the trend shows continuous fluctuating nature over the years additionally the spike is observed for the same year.

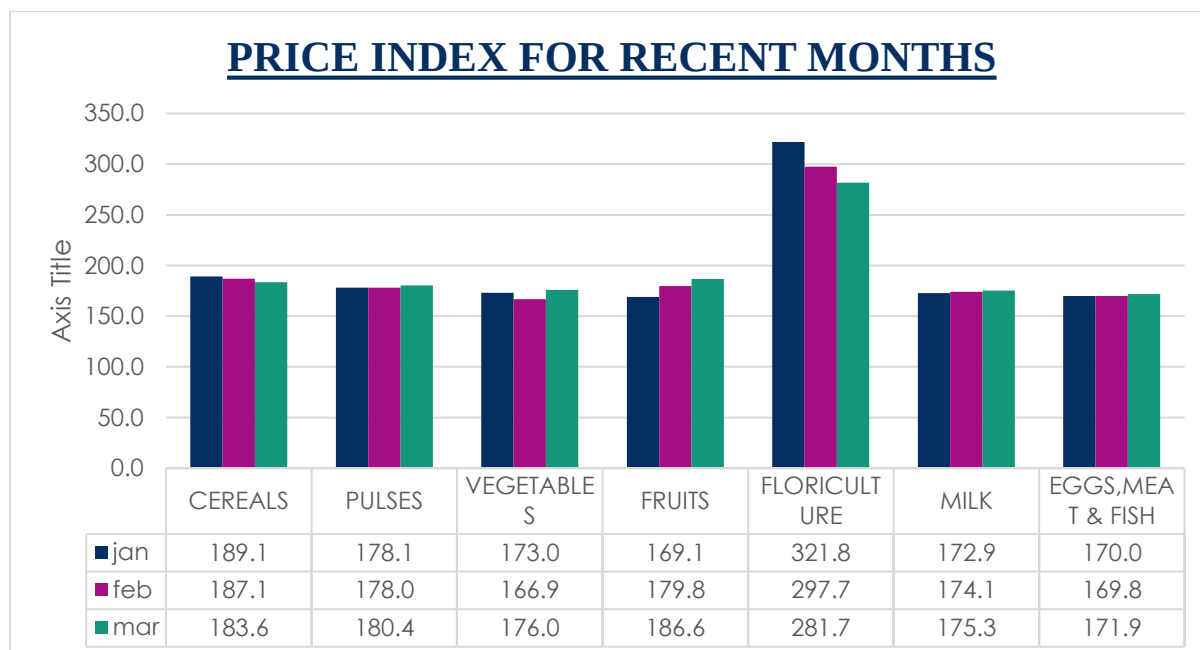


Figure no.7

Interpretation: Above chart displays the Price Index for recent 3 months that are January, February, and March 2023 for the sub categories of primary articles which constitutes the majority of variations.

- The WPI for cereals has shown a declining trend over the three months, indicating a decrease in the average selling prices of cereals at the wholesale level during these 3 months.
- The WPI for pulses has remained relatively stable, with a slight increase in March compared to January and February.
- The WPI for vegetables has shown some fluctuations, with a decrease in February followed by an increase in March compared to January.
- The WPI for fruits has displayed a consistent upward trend over the three months, indicating an increase in the average selling prices of fruits at the wholesale level.
- The WPI for floriculture has experienced a significant decline over the three months, suggesting a decrease in the average selling prices of floriculture products at the wholesale level.
- The WPI for milk has shown a gradual increase from January to March, indicating a slight rise in the average selling prices of milk at the wholesale level.

- And at last, the WPI for eggs, meat, and fish has displayed a relatively stable trend, with minor fluctuations but no significant overall change over the three months.

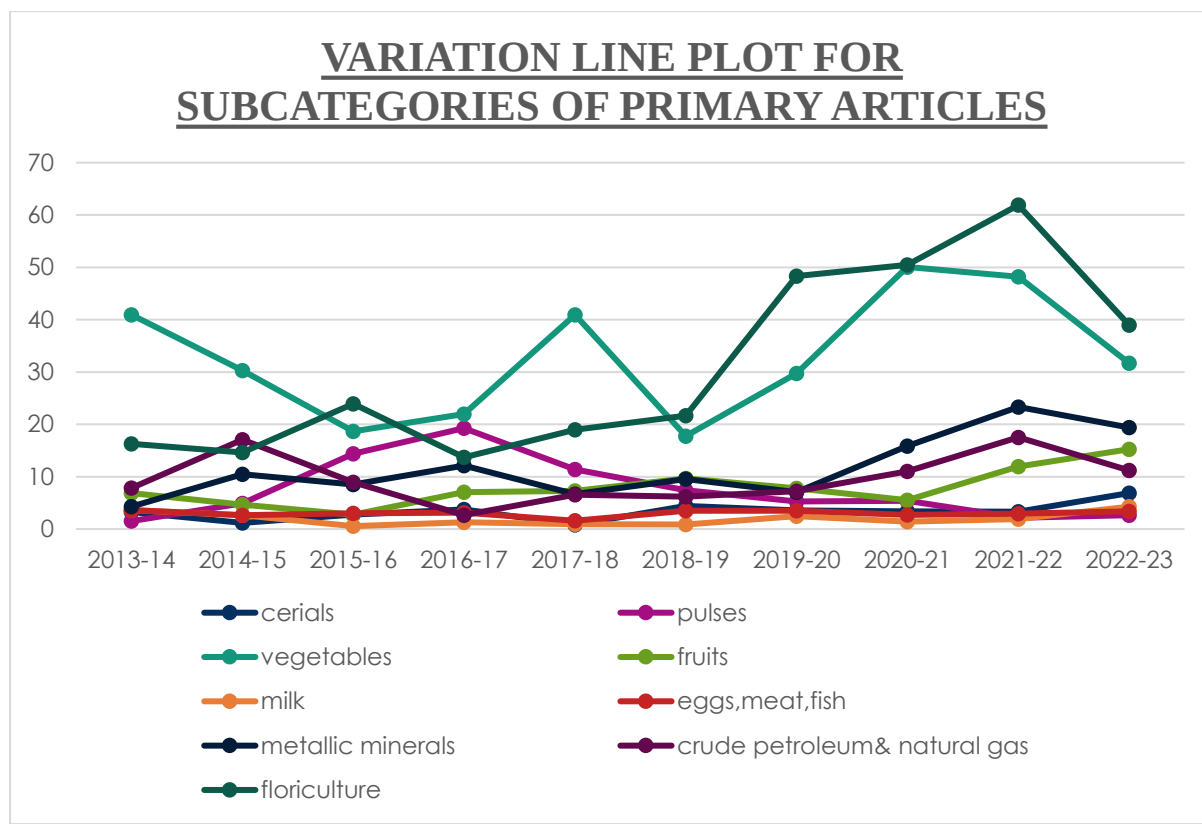


Figure no.8

Interpretation:

1. Cereals: The variation in the WPI for cereals has fluctuated over the years, with a general increasing trend. The highest variation was observed in 2022-23, indicating a significant increase in the average selling prices of cereals compared to the previous years.
2. Pulses: The variation in the WPI for pulses has also shown fluctuations, but the overall trend suggests a moderate increase in the average selling prices of pulses over the past 10 years.
3. Vegetables: The WPI for vegetables has displayed significant variations over the years. Notably, there was a sharp increase in the variation during 2020-21, indicating a substantial rise in the average selling prices of vegetables during that period.
4. Fruits: The variation in the WPI for fruits has generally shown an upward trend, with some fluctuations. The highest variation was observed in

2022-23, suggesting a notable increase in the average selling prices of fruits compared to the previous years.

5. Milk: The variation in the WPI for milk has remained relatively stable, with minor fluctuations over the years. There has been a gradual increase in the variation, indicating a slight rise in the average selling prices of milk.
6. Eggs, Meat & Fish: The variation in the WPI for eggs, meat, and fish has displayed some fluctuations, but the overall trend suggests a relatively stable market with moderate changes in average selling prices.
7. Metallic Minerals: The variation in the WPI for metallic minerals has shown fluctuations, but the overall trend indicates a moderate increase in average selling prices over the 10-year period.
8. Crude Petroleum & Natural Gas: The variation in the WPI for crude petroleum and natural gas has fluctuated, but there is no clear overall trend. Prices seem to have remained relatively stable or experienced minor changes over the years.
9. Floriculture: The WPI for floriculture has exhibited significant variations, with a notable increase in the variation in recent years. This suggests a significant rise in the average selling prices of floriculture products, particularly in 2020-21 and 2021-22.

The subcategory with the highest variation in average selling prices over the 10-year period was "Floriculture" followed by the sub-category "vegetables".

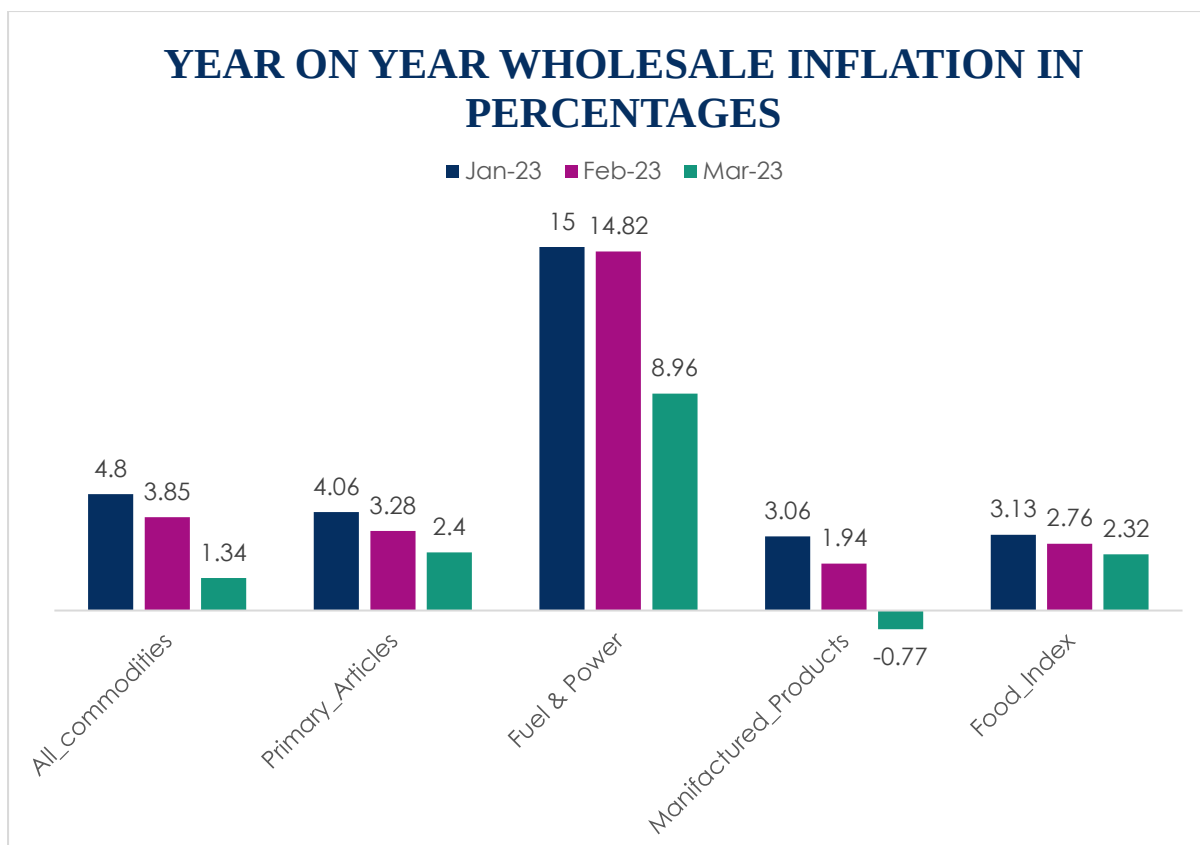


Figure no.9

- **Interpretation:** The overall wholesale inflation rate for all commodities decreased from January to March, indicating moderate decrease in prices of commodities from January to march 2023.
- The inflation rate for primary articles also decreased over the three months, suggesting moderate decrease in prices of primary goods at wholesale level.
- The inflation rate for fuel and power remained relatively high, but it showed a decline from January to March 2023, indicating decrease in prices for these energy-related commodities at the wholesale level.
- The inflation rate for manufactured products decreased over the three months, and it turned into deflation in March 2023. This indicates a decline in the average selling prices of manufactured goods at the wholesale level during this period.
- The inflation rate for the food index also decreased from January to March 2023, indicating a moderation in the price increase for food items at the wholesale level.

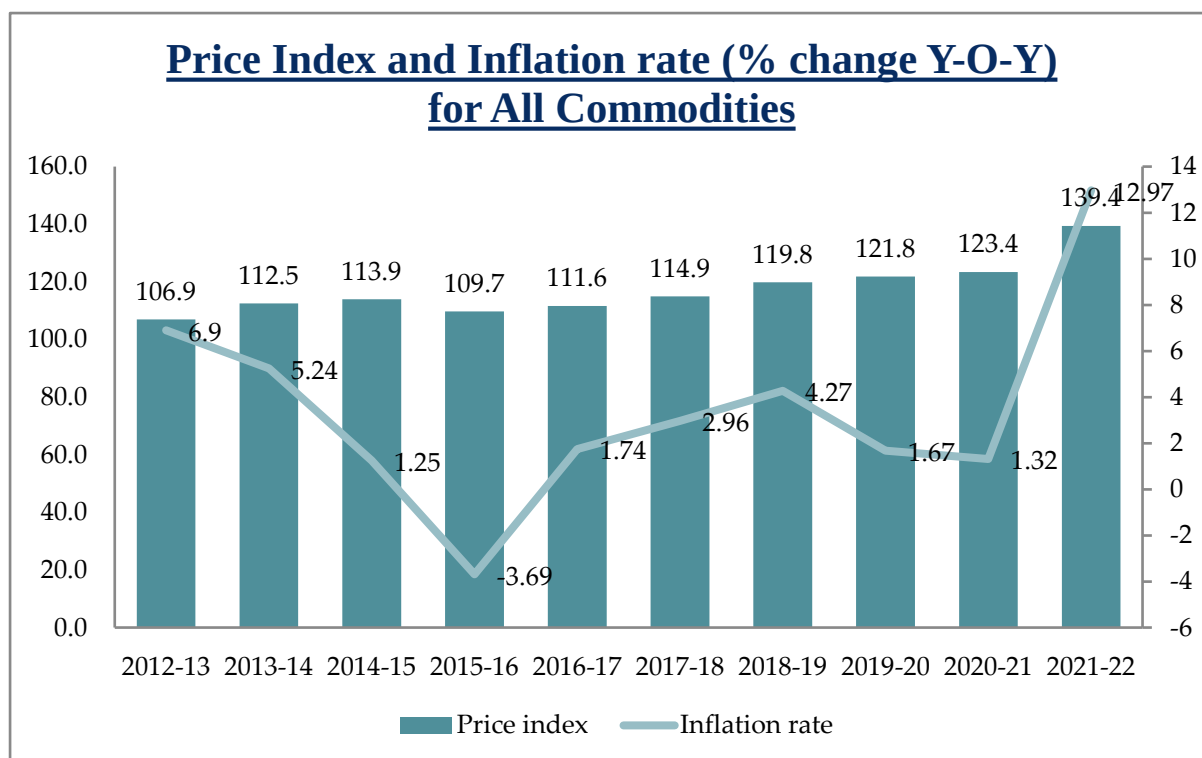


Figure no.10

Interpretation: From above plot we can say that over the span of 9 years, the prices have risen moderately for all commodities. But after the pandemic year i.e., 2020-21, the prices for all the commodities have risen significantly.

As for the inflation rate, there are fluctuations, eventually deflation for the year 2015-16 and again the inflation rate has increased moderately till 2020-21 and after 2020-21, there is a sudden spike in the inflation rate for all commodities.

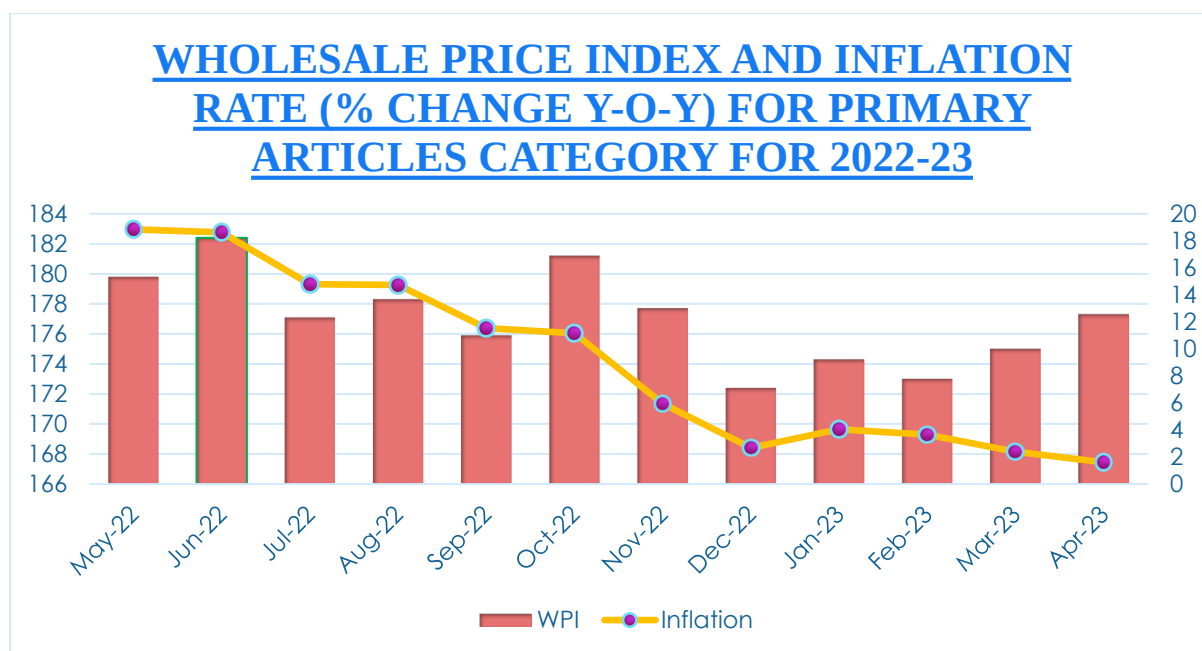


Figure no.11

• **WPI Trend:** Throughout the year 2022–2023, there are some changes in the WPI for primary articles. Starting at 179.8 in May 2022, it rises to its peak of 182.4 in June 2022, after which it progressively drops until it reaches a low of 172.4 in December 2022. Following that, it exhibits a modest increase in the months that follow, reaching 177.3 in April 2023.

• **Inflation Trend:** Over the specified years 2022–2023, the inflation rate for primary articles likewise varies. In May 2022, it is relatively high at 18.85%, and in the following months, it gradually declines. The inflation rate, however, has decreased to 11.55% as of September 2022 and will continue to drop steadily until it reaches its lowest level of 1.61% in April 2023.

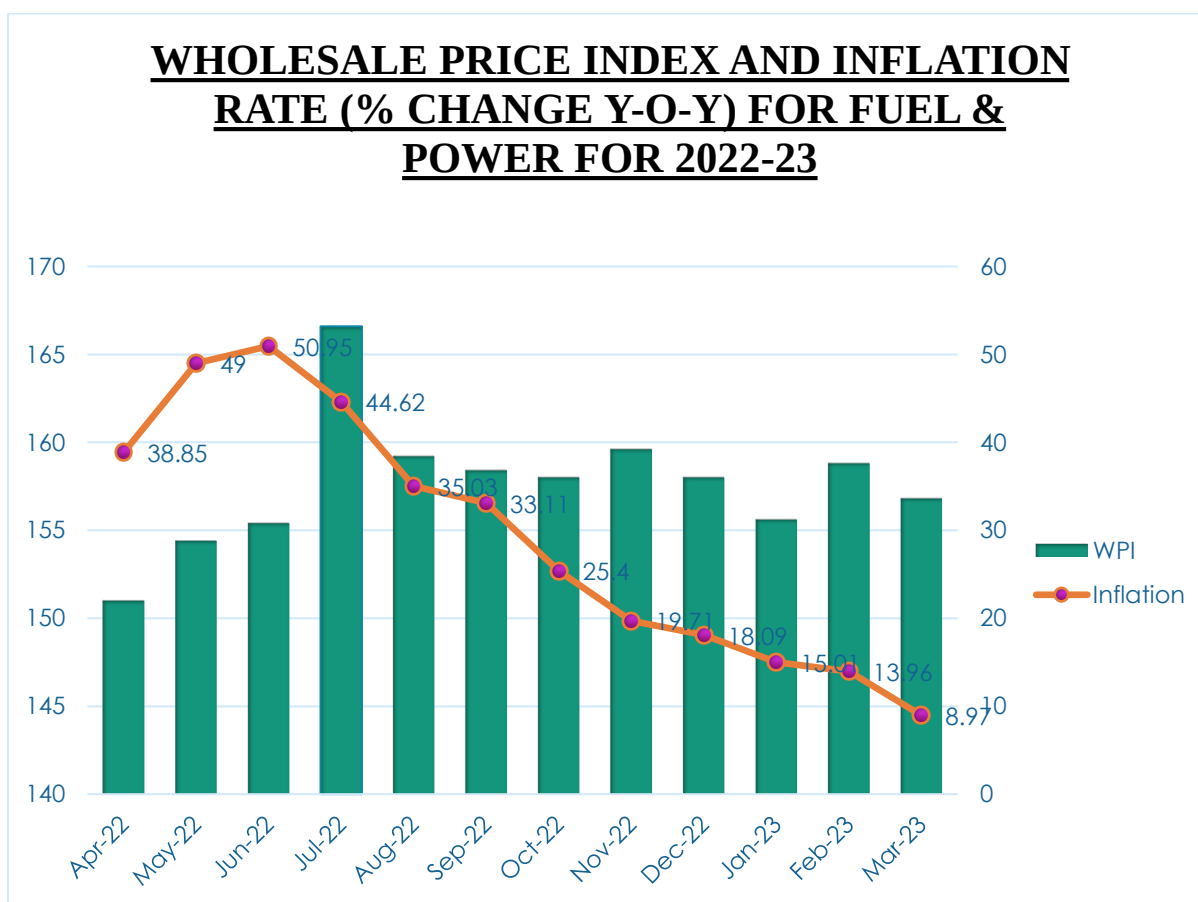


Figure no.12

• **WPI Trend:** Throughout the entire year, there are certain swings in the WPI for the Fuel & Power category. Starting at 151 in April 2022, it progressively rises to a maximum of 166.6 in July 2022, after which it begins to fall in August 2022. The WPI is comparatively constant from September 2022 to December 2022, hovering between 158 and 159.6. After that, it continues to fall, eventually hitting 156.8 in March 2023.

• **Inflation Trend:** The Fuel & Power category's inflation rate fluctuates between 2022 and 2023. It begins at a reasonably high level of 38.85% in April 2022 and then significantly rises to 49% and 50.95%, respectively, in May and June 2022. Additionally, the inflation rate fluctuates a little bit but gradually declines until it hits 8.97% in March.

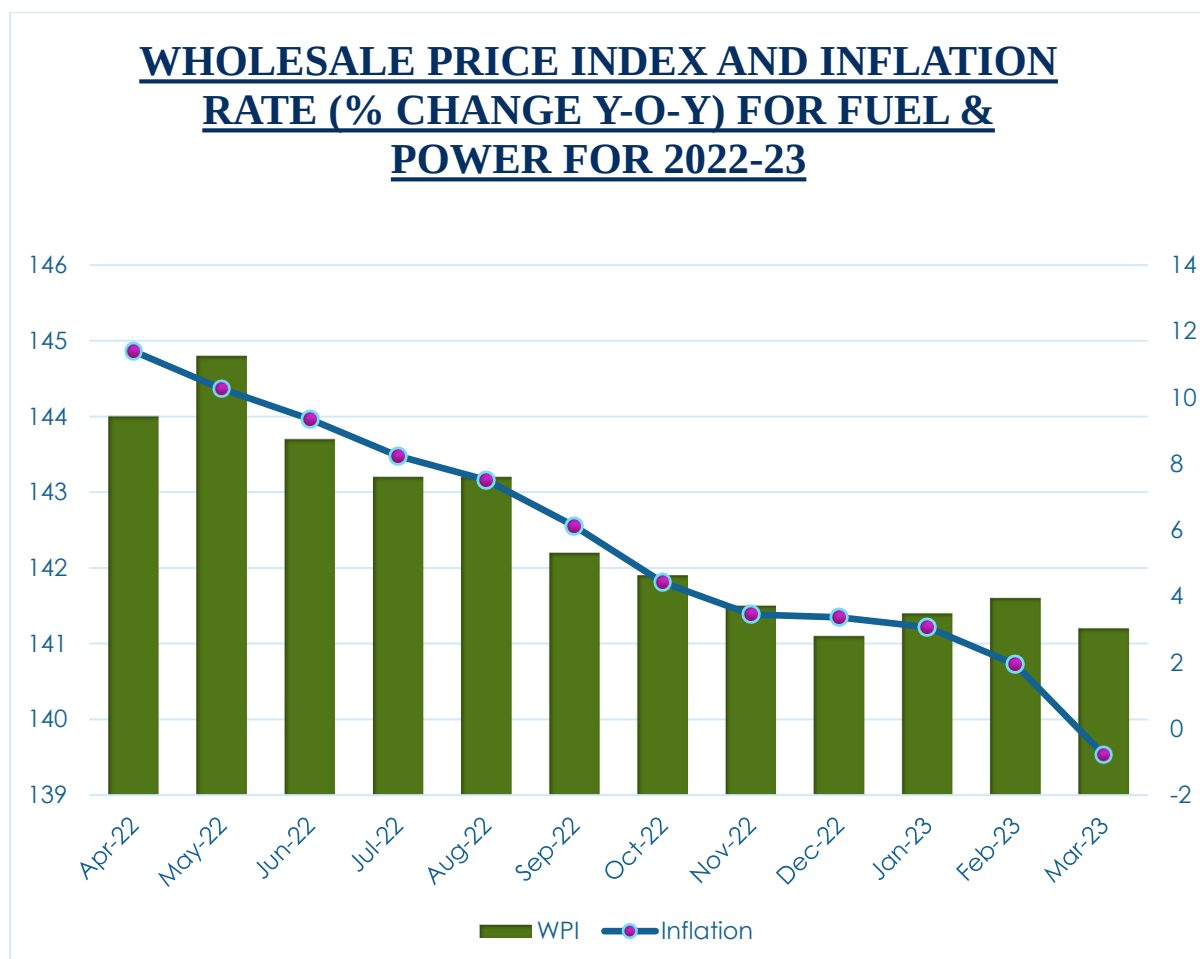


Figure no.13

Interpretation: From above plot we can say that the WPI value for Manufactured products category decreased gradually from 144 in Apr-22 to 141.2 in Mar-23. This indicates a reduction in the overall prices of manufactured products during the year 2022-23.

Similarly, the inflation rate for manufactured products also decreased gradually from 11.4 in Apr-22 to -0.78 in Mar-23. This indicates a decrease in the rate of increase in prices of manufactured products, and even a decrease in the overall price levels for manufactured products.

The decrease in inflation rates for manufactured products over the year 2022-23 could be due to a variety of reasons, such as the availability of cheaper raw

materials, increased productivity, changes in government policies , and shifts in consumer demand, among others.

Overall, there was a gradual reduction in the prices of manufactured products during the year 2022-23.

Methodology For Cluster Analysis

Clustering is a popular unsupervised machine learning technique used to group similar data points together based on their characteristics or properties. **The goal of clustering is to identify inherent patterns or structures in the data without any prior knowledge or labels.**

Before applying any clustering method on your data, it's important to evaluate whether the data sets contain meaningful clusters (i.e.: non-random structures) or not. If yes, then how many clusters are there. This process is defined as the assessing of clustering tendency or the feasibility of the clustering analysis.

The **Hopkins statistic** is used to assess the clustering tendency of a data set by measuring the probability that a given data set is generated by a uniform data distribution. In other words, it tests the spatial randomness of the data.

Determining the optimal number of clusters in a data set is a fundamental issue in partitioning clustering, such as k-means clustering, which requires the user to specify the number of clusters k to be generated.

Different method of determining optimal number of clusters for k means, k medoids (PAM) and hierarchical Clustering:

Direct methods: consists of optimizing a criterion, such as the within cluster sums of squares or the average silhouette. The corresponding methods are named **elbow and silhouette methods**, respectively.

Different clustering algorithm used:

K-means Clustering: It is one of the most widely used clustering algorithms. K-means aims to partition the data into a predetermined number of clusters, represented by their centroids. The algorithm iteratively assigns data points to the nearest centroid and updates the centroids based on the mean of the assigned points. It continues until convergence.

Hierarchical Clustering: This approach creates a hierarchy of clusters by iteratively merging or splitting them based on certain criteria. It can be agglomerative (bottom-up) or divisive (top-down). Agglomerative clustering starts with each data point as a separate cluster and merges the closest pairs of clusters until a stopping criterion is met.

Ward's linkage is a popular agglomerative hierarchical clustering method that is used to build a hierarchical cluster tree or dendrogram. It aims to minimize the within-cluster variance when merging clusters at each step. Ward's method is particularly effective for analysing data with continuous variables.

The PAM algorithm, also known as the K-medoids algorithm, differs from the popular K-means algorithm in that it uses medoids instead of centroids. A medoid can be thought of as the most representative or central object within a cluster. It is an actual data point from the dataset, unlike a centroid, which is a virtual point calculated as the mean of the data points.

The PAM algorithm has the advantage of being more robust to outliers compared to K-means because it uses actual data points as medoids.

The term cluster validation is used to design the procedure of evaluating the goodness of clustering algorithm results:

The silhouette method is a technique used to evaluate the quality or coherence of clusters obtained from a clustering algorithm. It provides a quantitative measure to assess how well data points within a cluster are separated from data points in other clusters.

The silhouette method calculates the silhouette coefficient for each data point, which is a value between -1 and 1. The silhouette coefficient considers two factors:

Cohesion: This measures how close a data point is to other points in its own cluster. It is calculated as the average distance between a data point and all other points within the same cluster.

Separation: This measures how far a data point is from points in other clusters. It is calculated as the average distance between a data point and all points in the nearest neighbouring cluster.

The silhouette method can be used to compare different clustering algorithms or different parameter settings of the same algorithm. By comparing silhouette coefficients, you can select the clustering solution that yields the highest average silhouette value, indicating better-defined and well-separated clusters.

Cluster Analysis

Objective of cluster analysis:

Clustering is a common technique used in data Analysis to identify Groups or Clusters of Objects that share similar Characteristics or properties.

In the Context of Wholesale Price Index data, Clustering can be used to identify patterns and trends in the WPI data that might not be apparent when looking at individual Commodities.

By Grouping Commodities based on their Price Movement over time, it is possible to identify Clusters that shows similar behaviour.

WPI data is complex and Difficult to analyse due to large number of Commodities included and the variability in price index of different Commodities over the time. Clustering can help to Simplify the analysis by reducing the number of Commodities Being Considered and allowing for Comparisons to be made between Group of Commodities rather than individual one.

Following analysis is for the Primary Article Category.

head(data)

	INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
Paddy	113.2	129.1	136.6	134.7	143.6
Wheat	115.3	124.9	123.4	128.4	142.1
Jowar	96.8	100.4	115.8	113.9	124.5
Bajra	120.0	133.0	129.6	133.8	155.1
Maize	116.0	123.1	118.5	125.5	137.0
Barley	115.6	119.2	126.8	129.1	155.2

	INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
Paddy	149.4	155.3	160.5	163.5	162.3
Wheat	138.7	149.4	159.2	152.9	160.1
Jowar	123.4	130.3	161.1	154.3	143.3
Bajra	139.1	150.4	193.7	161.2	167.9
Maize	124.7	133.2	175.4	143.3	152.1
Barley	140.6	149.8	172.9	154.6	169.0

Descriptive statistics:

INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
Min. : 38.3	Min. : 70.2	Min. : 62.5	Min. : 54.2	Min. : 51.0
1st Qu.:100.6	1st Qu.:107.8	1st Qu.:113.3	1st Qu.:111.9	1st Qu.:117.9
Median :110.2	Median :120.8	Median :126.6	Median :130.4	Median :137.2
Mean :110.6	Mean :124.3	Mean :131.1	Mean :132.3	Mean :137.9
3rd Qu.:118.4	3rd Qu.:135.9	3rd Qu.:149.0	3rd Qu.:151.1	3rd Qu.:155.1
Max. :186.8	Max. :280.2	Max. :266.2	Max. :255.5	Max. :248.0

INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
Min. : 62.3	Min. : 47.5	Min. : 64.8	Min. : 62.0	Min. : 73.3
1st Qu.:115.7	1st Qu.:118.3	1st Qu.:131.7	1st Qu.:127.6	1st Qu.:142.4
Median :135.0	Median :135.2	Median :151.2	Median :154.6	Median :166.3
Mean :138.4	Mean :140.5	Mean :158.1	Mean :158.0	Mean :174.8
3rd Qu.:157.1	3rd Qu.:152.7	3rd Qu.:180.9	3rd Qu.:182.3	3rd Qu.:207.3
Max. :236.5	Max. :311.2	Max. :350.8	Max. :371.6	Max. :372.2

Variation among price index of different commodities in Primary Article Category for every financial year from 2012-13 to 2021-22:

INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017	INDEX20172018
364.3921	899.2820	921.8447	1211.0647	1132.3786	1478.7174
INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022		
1687.7325	2650.2832	2481.6813	2927.3711		

Box Plot

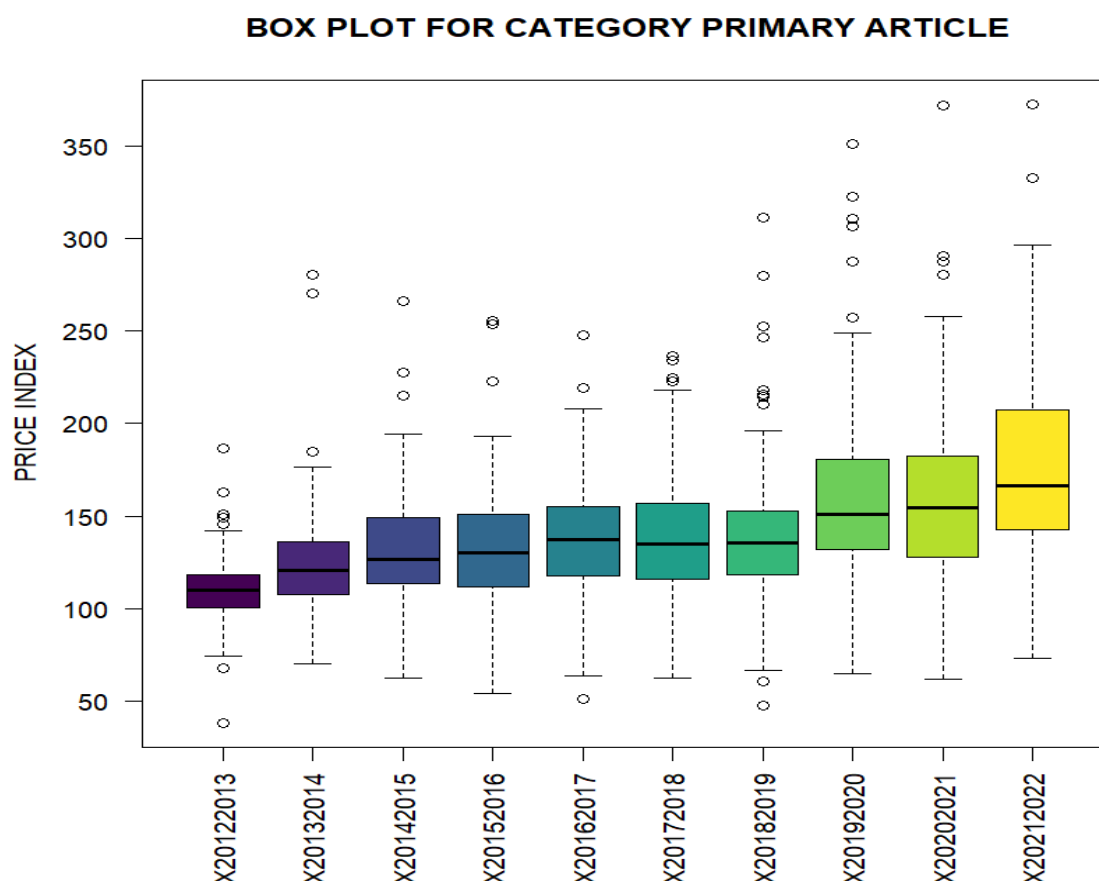


Figure no.14

INTERPRETATION:

Minimum value of price index in financial year 2021-2022 is 73.3 implies that there is at least one commodity whose price index for financial year 2021-22 is as low as 73.3, whereas maximum is 372.2 implies there is at least one commodity whose price index is as high as 372.2

The overall average of commodities in primary article category for financial year 2021- 2022 is 174.8.

Additionally, Q1 at 142.2 indicates 25% of 117 commodities had price index which is below 142.2 whereas 50% of 117 had price index below 166.3 whereas 75% of commodities had price index which is below 207.3.

For the financial year 2021-2022 the commodities having price index value

below

$Q1 - 1.5(IQR) = 44.55$ and above

$Q3 + 1.5(IQR) = 304.95$ is Considered to be outlier.

From box plot it seems there are two outlier or anomalies for 2021-2022

Commodities	Jasmine	Garnet
Price Index	372.2	332.8

Outliers are Values that are Significantly Higher or Lower than the rest of the Data , and they may be indicative of unusual events or trends.

In Our case Outliers are due to Spike or Decline in Price Index of Some Commodities for each Year it may be due to Shortage , High demand , Low Supply, or other Market Disruption.

Anomalies or Outlier Detection:

Year 2012-2013

Commodities	Rajma	potato	papaya	Garlic	Soyabean	Gaur Seed	Phosphorite
Price Index	151	162.9	67.8	38.3	49.4	186.8	145.6

Year 2013-2014

Commodities	Onion	Tapioca	Ginger Fresh
Price Index	270.5	184.9	280.2

Year 2014-2015

Commodities	Potato	Ginger Fresh	Guava
Price Index	227.3	266.2	215.3

Year 2015-2016

Commodities	Onion	Guava	Niger Seed
Price Index	253.6	255.5	222.8

Year 2016-2017

Commodities	Gram	Guava	Iron Ore
Price Index	219.2	248	51

Year 2017-2018

Commodities	Ragi	Tomato	Guava	Bauxite	Limestone
Price Index	234.1	222.5	234.3	236.5	224.4

Year 2018-2019

Ragi	Ginger fresh	Coconut fresh	Guava	Garlic	Copra coconut	Raw Rubber	Bauxite	Limestone	Garnet
214.7	215.9	217.9	311.2	47.5	210.1	60.5	252.4	246.3	280

Year 2019-2020

commodities	Onion	Ginger fresh	Guava	Jasmine	Bauxite	Garnet
Price Index	306.5	310.6	322.3	350.8	257.3	287.7

Year 2020-2021

commodities	Potato	Guava	Jasmine	Garnet
Price Index	290.7	280.3	371.6	287.7

ASSESSING CLUSTERING TENDENCY

Testing of hypothesis

H0: DATA SET CONTAINS NO MEANINGFUL CLUSTERS

VS

H1: DATA SET CONTAIN MEANINGFUL CLUSTERS

Calculated Hopkins Statistics

[1] 0.9289714

Calculated p value for Hopkins statistics

[1] 3.197344e-08

Conclusion: P-value for Hopkins statistics implies for rejection of null hypothesis and therefore we conclude that the commodities in Primary Article Category is significantly clusterable.

DETERMINING THE OPTIMAL NUMBER OF CLUSTERS:

Elbow Method

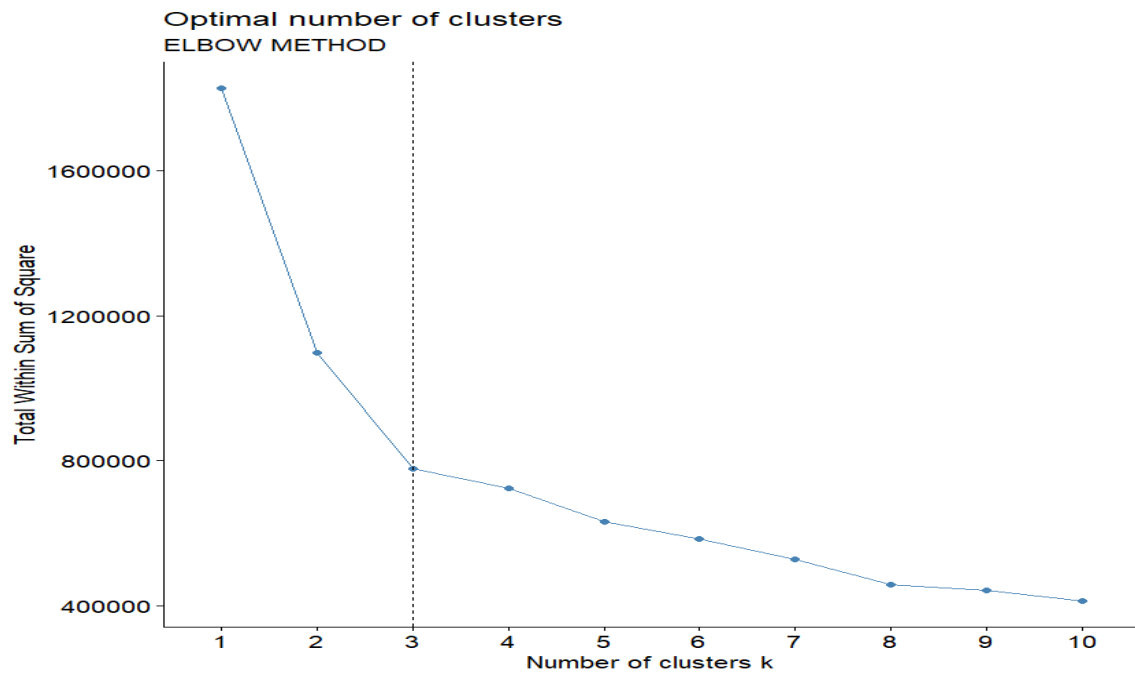


Figure no.15

Elbow method : 3 clusters solution suggested

Silhouette method

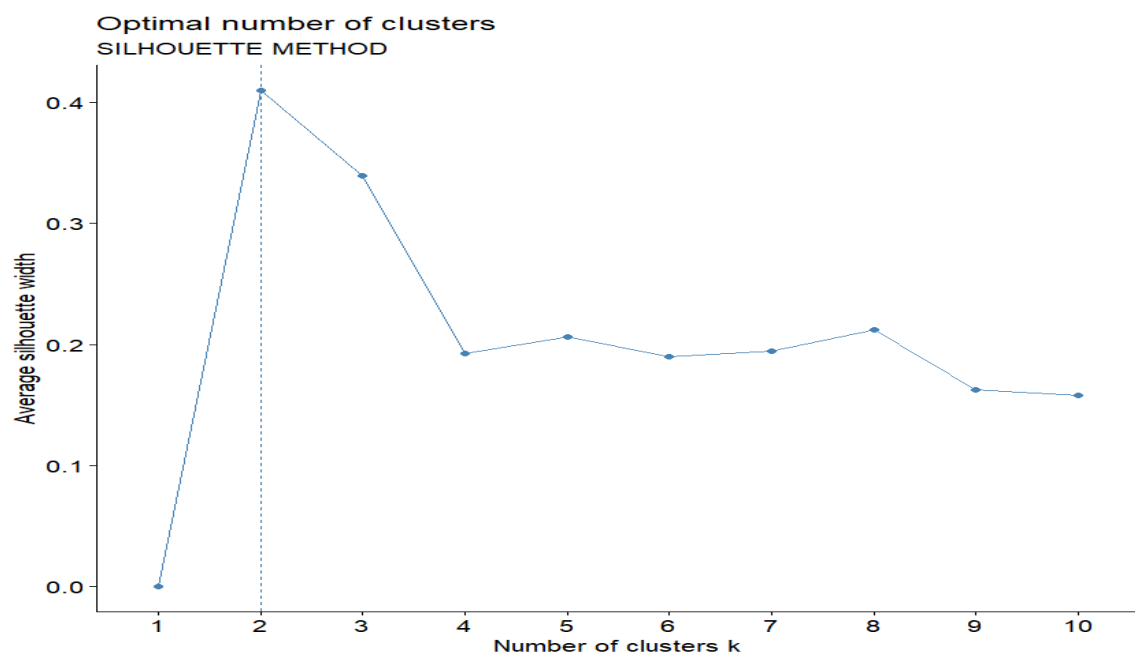


Figure no.16

Silhouette method : 2 clusters solution suggested.

OBSERVATION:

Because of these Conflicting results from above 2 methods, we will evaluate the Clustering results in detail by considering optimal number of clusters as 2 and 3 and will choose one that contribute to our problem-solving process.

Computing PAM clustering

Accessing the result:

Medoids: object that represent clusters.

Clustering: a vector containing the cluster number of each Object.

Cluster medoids : poultry chicken, Walnut, coffee

Medoids:

	ID	INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016
Poultry Chicken	61	118.1	131.0	133.7	129.1
walnut	54	116.0	144.9	164.6	168.7
Coffee	74	97.0	88.3	112.0	106.6

	INDEX20162017	INDEX20172018	INDEX20182019	INDEX20192020
Poultry Chicken	139.0	138.2	134.6	148.8
walnut	171.2	182.2	184.2	192.5
Coffee	99.5	98.2	94.0	96.7

	INDEX20202021	INDEX20212022
Poultry Chicken	149.8	162.6
walnut	221.4	225.1
Coffee	102.5	119.0

Size of each cluster

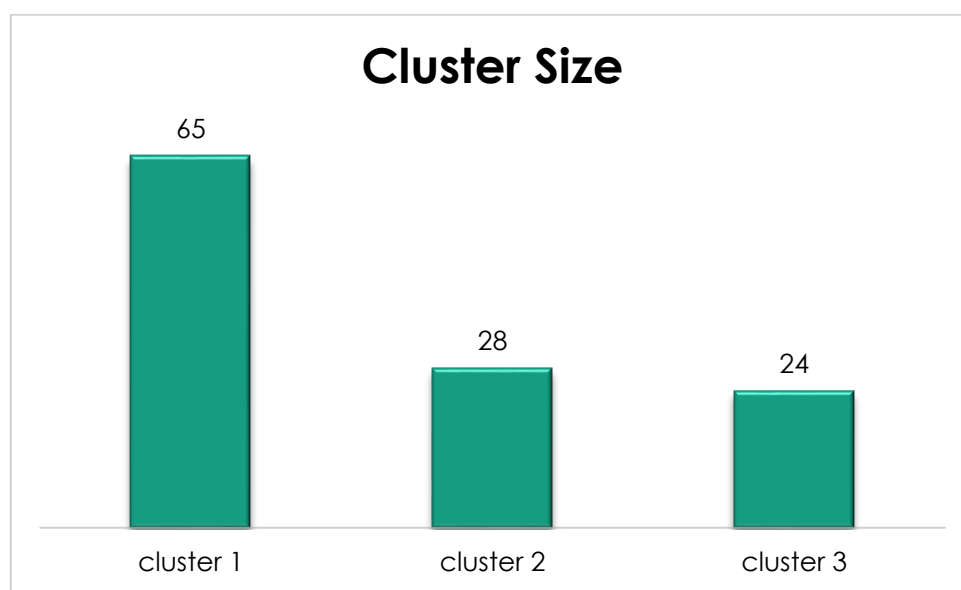


Figure no.17

Visualizing PAM Clusters:

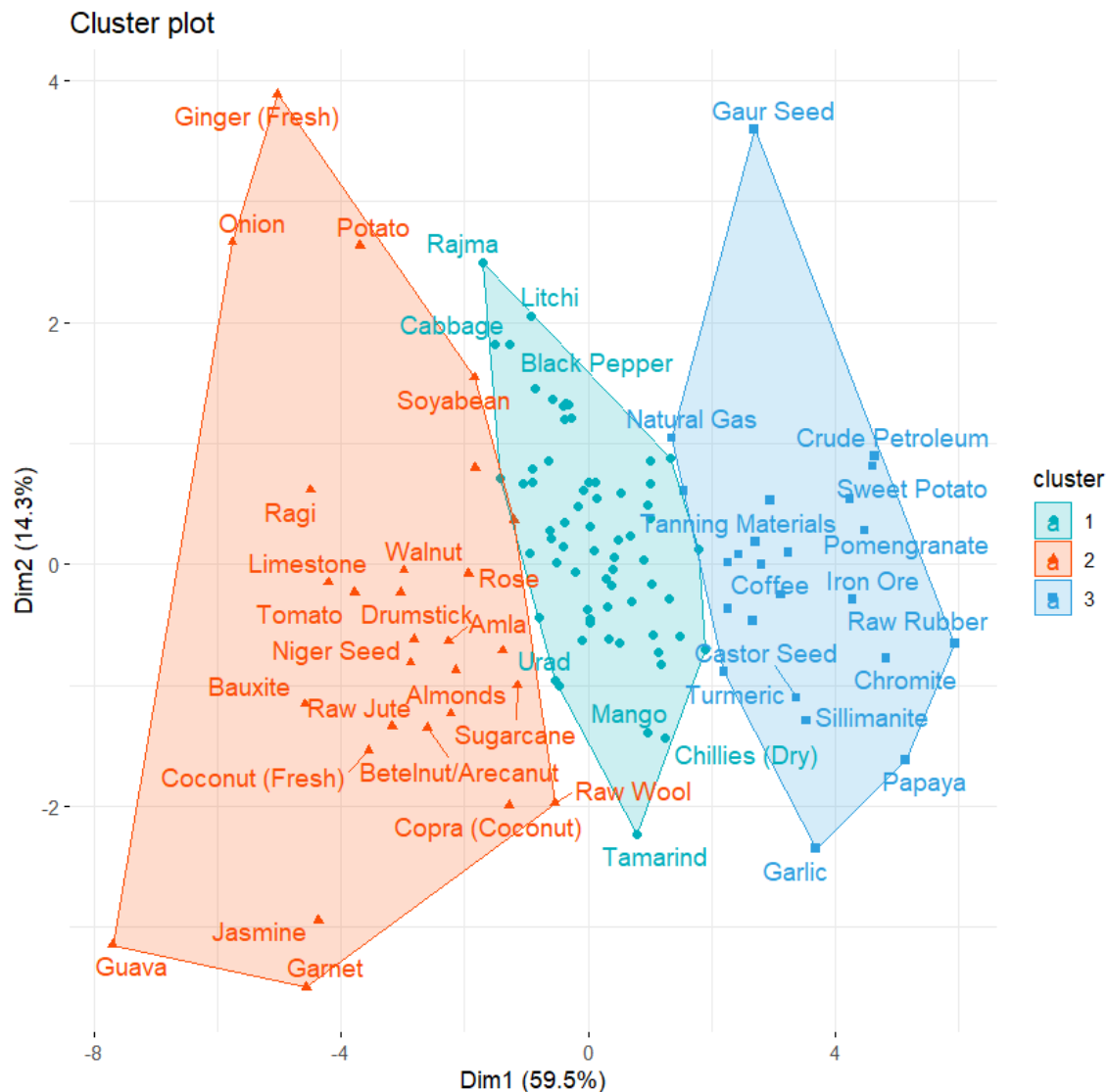


Figure no.18

These two components explain 73.87% of the point variability.

ASSESSING THE QUALITY OF CLUSTERING RESULTS:

Average silhouette width of each cluster:

cluster	size	ave.sil.width
1	65	0.41
2	28	0.11
3	24	0.40

The average Silhouette Width for Cluster 1 with 65 Commodities is 0.41 , for cluster 2 with 28 commodities is 0.11 whereas for Cluster 3 with 24 Commodities is 0.40. The Total average that is mean of all individual Silhouette Width is 0.34.

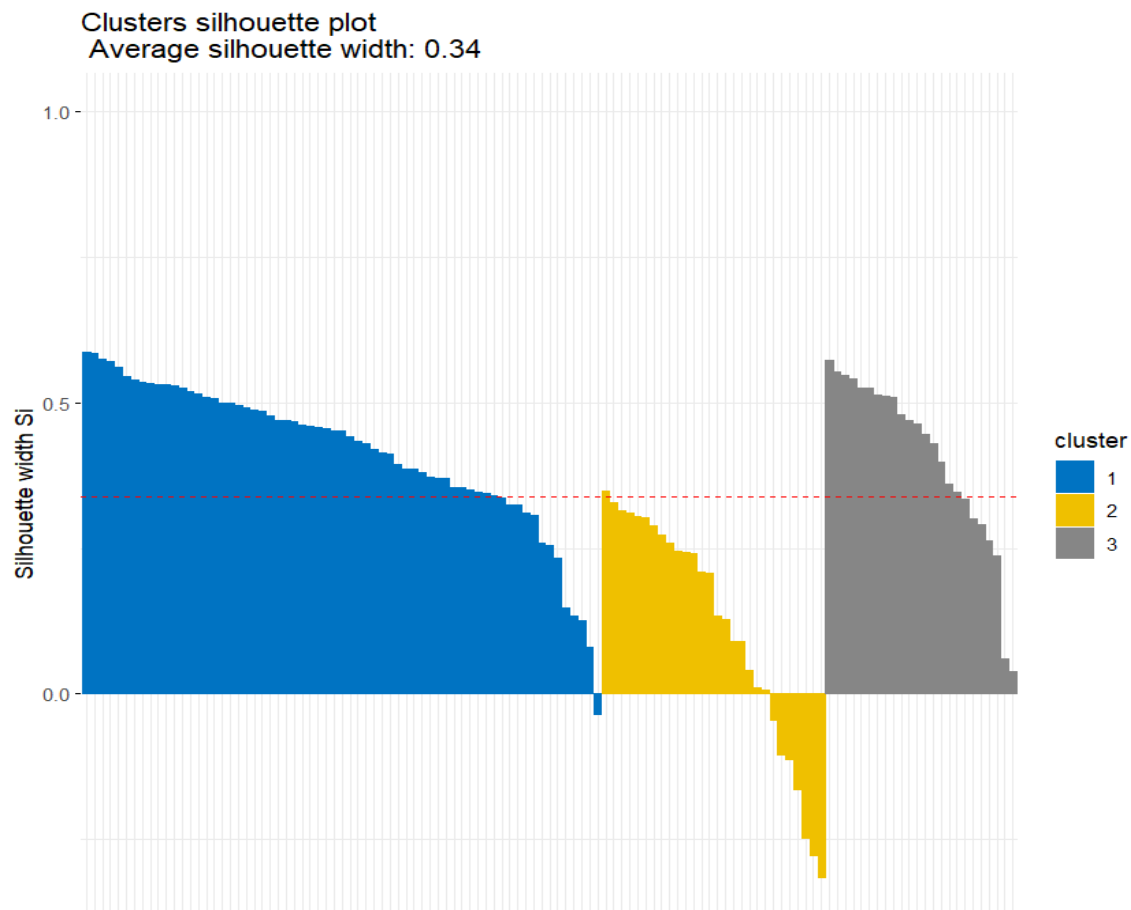


Figure no.19

It can be seen that several observations in cluster 2 have negative silhouette coefficient. This means that they are not in right cluster. We can find names of these observation(commodities) and determine the clusters they are closer (neighbour cluster).

	cluster	neighbor	sil_width
Lemon	1	3	-0.03492212
Copra (Coconut)	2	1	-0.04512199
Ginger (Dry)	2	1	-0.10500482
Soyabean	2	1	-0.11197871
Raw wool	2	1	-0.16327460
Peas (Green)	2	1	-0.24792956
Peas/Chawali	2	1	-0.27694155
Sugarcane	2	1	-0.31483198

Visualizing Cluster Medoids

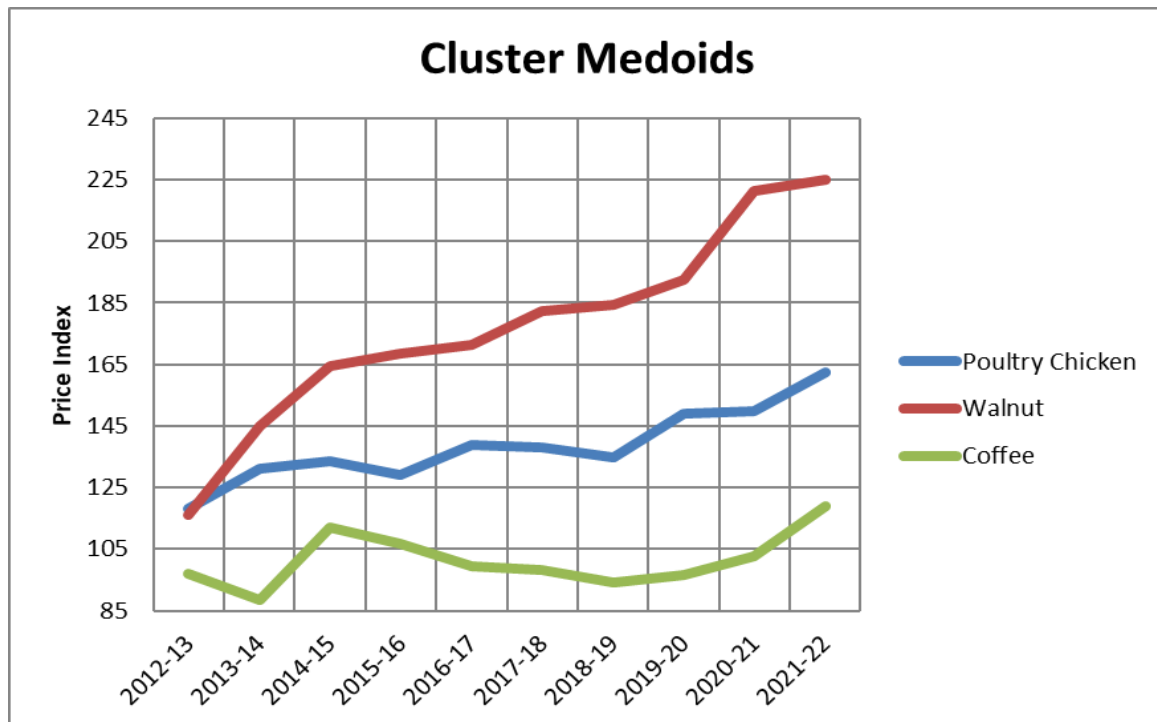


Figure no.20

- **Instead of using cluster centres we have used weighted average to represent clusters in our cluster analysis** Since by using Weighted average, we are considering the relative importance or significance of each commodity within the cluster. The weights represent the relative importance of each commodity in the overall cluster.
- We have calculated the weighted average using the assigned weights. multiplying the price index of each commodity by its corresponding weights, sum up the weight values and divide by the total weight of that particular cluster.
- The resulting weighted average represents a representative value for a cluster, considering the contribution of different commodities based on their weights.
- This average is used to analyse and compare clusters or to make inference about the overall behaviour or characteristic of the commodities within the clusters.

The inflation rate for each group of commodities can be interpreted as the percentage change in the average price of commodities in that group over a specified time period.

This will provide insight into which group of commodities experience the most significant price changes.

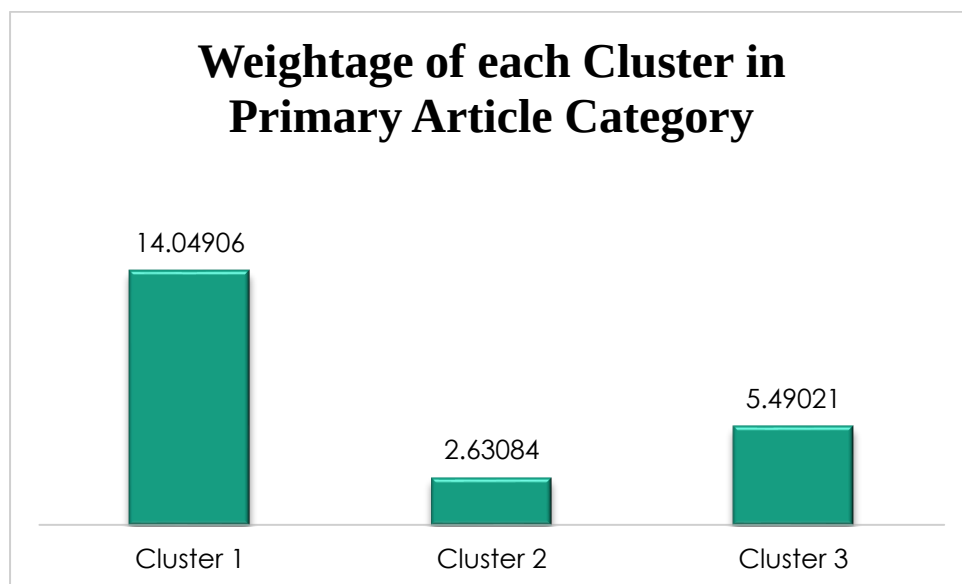


Figure no.21

CLUSTER1:

Financial year	Price index	Inflation rate (% change)
2012-13	111.79313	11.79313
2013-14	121.565	8.741027
2014-15	127.68653	5.035601
2015-16	132.53418	3.796529
2016-17	139.95848	5.601799
2017-18	140.14935	0.136375
2018-19	140.74082	0.422027
2019-20	150.48708	6.924975
2020-21	154.9822	2.987043
2021-22	164.45387	6.111458

CLUSTER1:

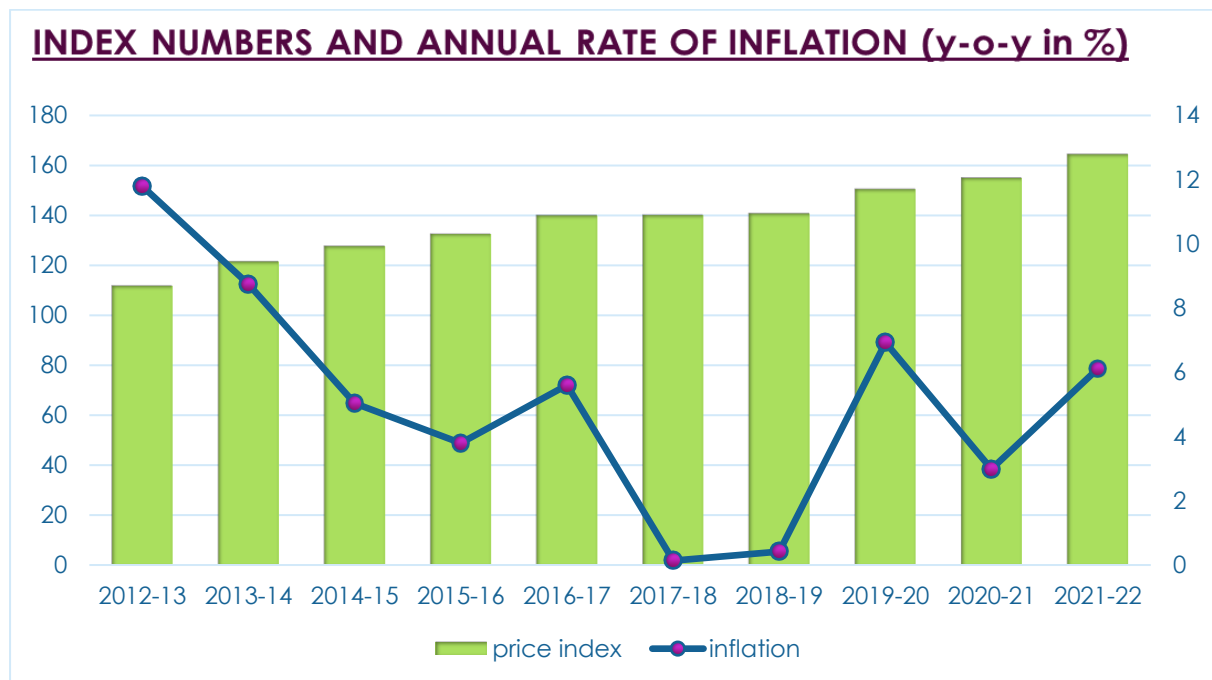


Figure no.22

Cluster 1:

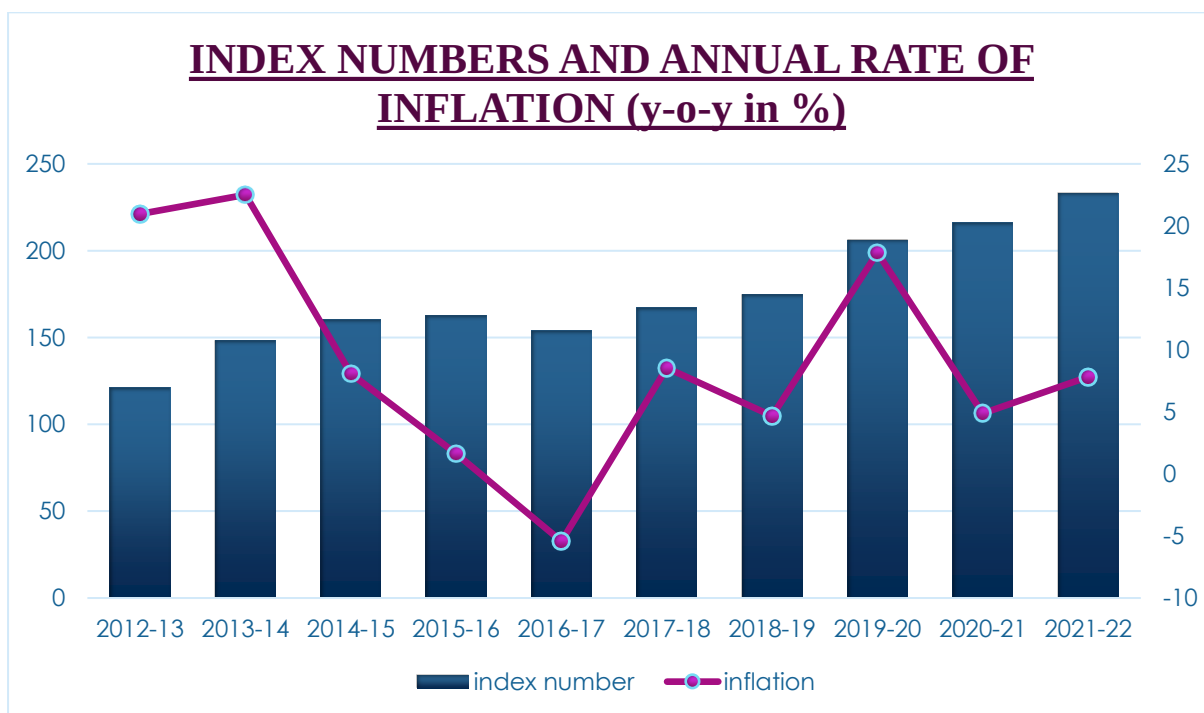
Commodities in cluster 1 appears to have a price index that has increased over time with some fluctuations as well as volatility but not as extreme as those seen in cluster 2 nor as low as those seen in cluster3 this suggest that commodities in cluster 1 experiencing a steady upward trend in price index with some ups and downs.

Inflation rate for Cluster 1:

It seems that Cluster 1 did not experience any deflation period since last 10 years also, it can be observed that the price index of cluster 1 consistently increased with some fluctuations. For initial year 2012-13 inflation rate was high around 11.79% .Later on for subsequent years it slow down for financial years 2017-2018 and 2018-19 very low inflation rate of 0.14 % and 0.42% was observed. For recent previous year 2021-22 inflation rate of 6.11% was observed for cluster 1.

Cluster 2:

Financial year	Price Index	Inflation rate (% change)
2012-13	120.9724	20.97235
2013-14	148.2002	22.50749
2014-15	160.1649	8.073352
2015-16	162.7941	1.641521
2016-17	153.9542	-5.43008
2017-18	167.1171	8.549898
2018-19	174.9275	4.673562
2019-20	206.1201	17.83176
2020-21	216.1835	4.882315
2021-22	233.1149	7.831933

Cluster 2:**Figure no.23****Interpretation for Cluster 2:**

commodities in cluster 2 are characterized by continuous fluctuations in price index with significant ups and downs every year these are the commodities which appears to be outliers in terms of distribution of every financial year .

These findings suggest that commodities in cluster2 may exhibit high level of price volatility and unpredictability with prices that are subject to large swings.

Inflation rate for Cluster 2:

From the above plot as well as table it can be observed that cluster 2 experienced high-level inflation of 20.97% and 22.50% in financial years 2012-2013 and 2013-2014 whereas for subsequent years 2014-2015 and 2015-16 the inflation rate falls to single digit of 8% and 1% respectively.

For financial year 2016-2017 the cluster 2 experienced deflation of 5.43% but again for year 2017-18 inflation rate rises to 8.54% afterward cluster 2 experience low inflation rate exception for year 2019-20 where it picked up reaching 17.83%. Overall we can say cluster 2 experienced increasing trend with some fluctuations and only deflation for year 2016-17 and high-level inflation rate in year 2013-14 and 2019-20.

Cluster 3:

Financial Year	Price index	Inflation Rate (% change)
2012-13	104.2083	4.20835
2013-14	109.9245	5.485277
2014-15	99.58079	-9.4098
2015-16	85.69988	-13.9393
2016-17	87.73813	2.378357
2017-18	86.92579	-0.92586
2018-19	96.9499	11.5318
2019-20	92.18509	-4.91471
2020-21	85.00118	-7.79293
2021-22	112.6529	32.53102

Cluster 3

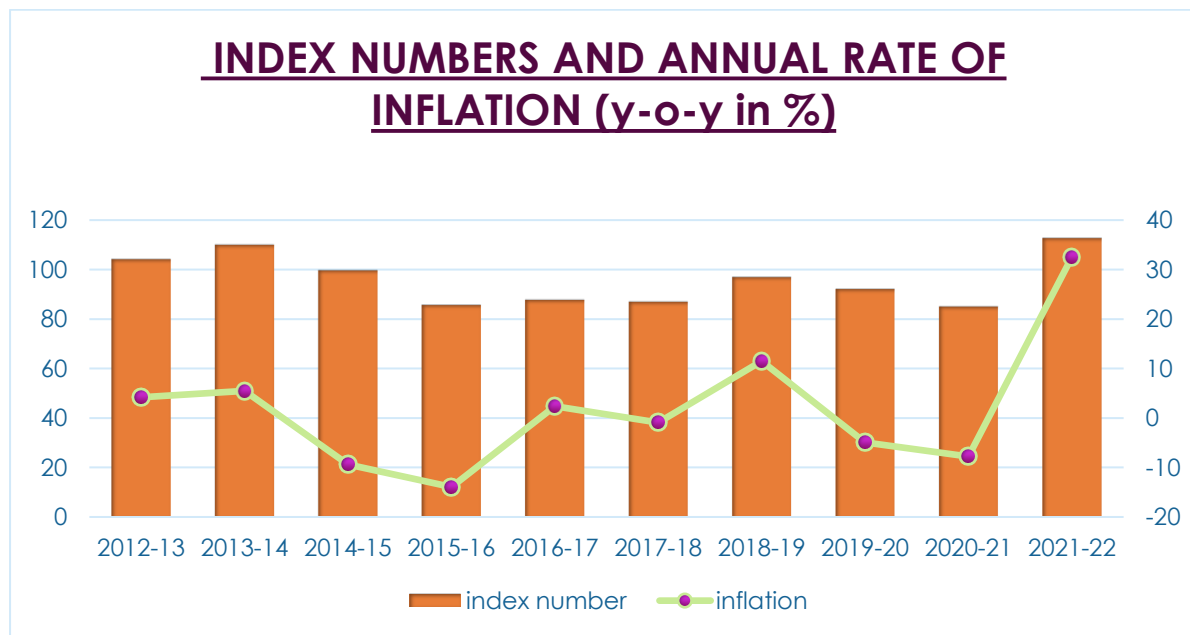


Figure no.24

Interpretation:

Based on clustering result it seems that commodities in cluster 3 have experienced some fluctuations in price index over the past decades but this fluctuation is not very large.

Additionally, most of the commodities in this cluster have a price index that is less than 100 the commodities that belongs to cluster 3 with price index that are not subject to large swings or significant volatility.

By observing the inflation rate for cluster 3 the following interpretation can be made:

The inflation rate was high in the initial years with rate of 4.2% in financial year 2012-2013 and 5.48% in 2013-2014 indicates that Price index of commodities were increasing during this period. Followed by Negative inflation in year 2014-2015 and High-level deflation of 13.93% in financial year 2015-16 implies for decrease in price index of commodities during these subsequent years again for year 2016-17 recovery can be observed as inflation rate is positive again. Mixed inflation and deflation in subsequent years whereas high inflation rate of 32.53% in financial year 2021-22 was observed which is significantly higher than any of previous years.

VISUALIZING AND COMPARING RESULT OF ALL 3 CLUSTERS IN SINGLE PLOT:

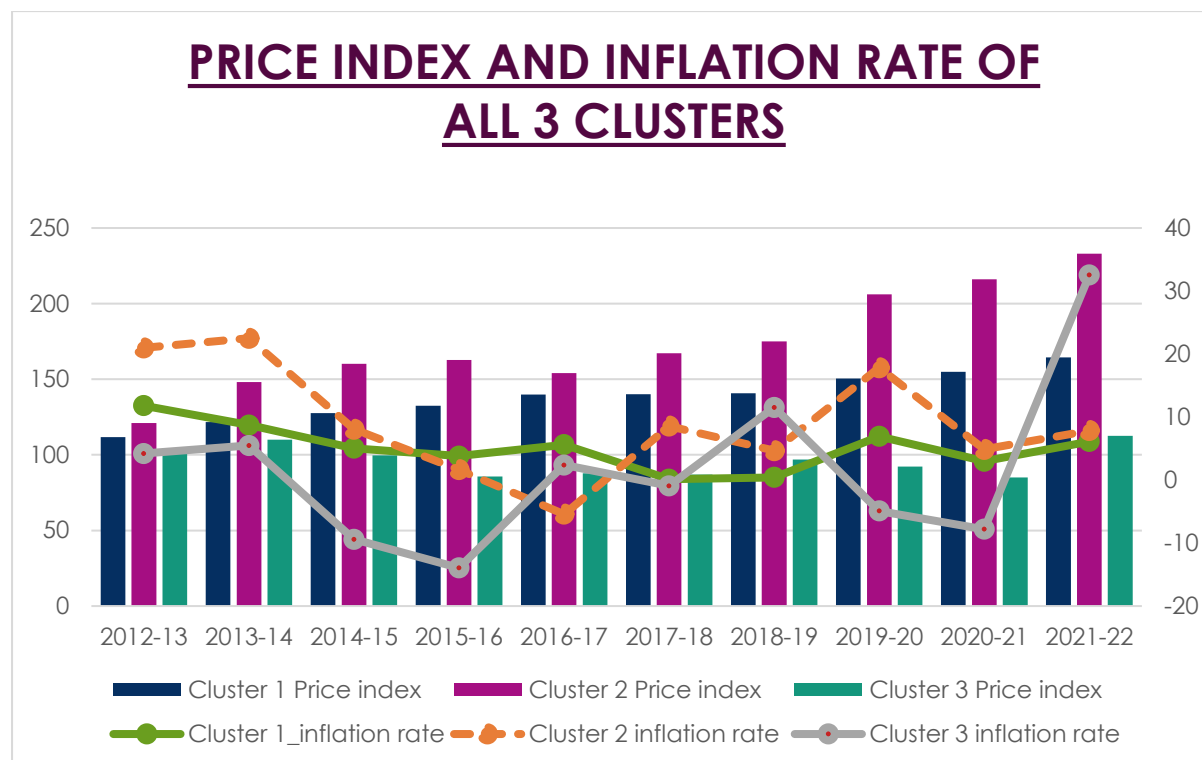


Figure no.25

FUEL & POWER CATEGORY

Following analysis is for the Fuel & Power Category.

Descriptive Statistics:

summary(d)

INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
Min. : 98.9	Min. : 92.8	Min. : 93.6	Min. : 54.30	Min. : 58.10
1st Qu.:101.1	1st Qu.:103.5	1st Qu.:102.6	1st Qu.: 75.12	1st Qu.: 71.33
Median :106.3	Median :111.2	Median :104.3	Median : 83.35	Median : 90.60
Mean :105.8	Mean :111.0	Mean :105.6	Mean : 88.39	Mean : 89.12
3rd Qu.:109.7	3rd Qu.:118.8	3rd Qu.:109.8	3rd Qu.:103.58	3rd Qu.:105.20
Max. :114.9	Max. :126.3	Max. :118.8	Max. :120.80	Max. :126.30

INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
Min. : 68.80	Min. : 85.60	Min. : 81.00	Min. : 62.50	Min. : 87.8
1st Qu.: 79.90	1st Qu.: 92.47	1st Qu.: 85.33	1st Qu.: 77.30	1st Qu.:115.2
Median : 95.95	Median :106.20	Median :104.50	Median : 98.65	Median :124.1
Mean : 98.29	Mean :111.90	Mean :110.41	Mean :101.79	Mean :137.8
3rd Qu.:117.35	3rd Qu.:126.83	3rd Qu.:129.75	3rd Qu.:131.28	3rd Qu.:147.8
Max. :134.10	Max. :152.40	Max. :172.80	Max. :141.80	Max. :221.9

Variation among price index of different commodities in Fuel and power Category for every financial year from 2012-13 to 2021-22:

INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017	INDEX20172018
26.29600	88.48933	51.75629	389.42517	461.18829	491.07529
INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022		
512.63200	755.01050	867.01850	1461.64362		

Box Plot

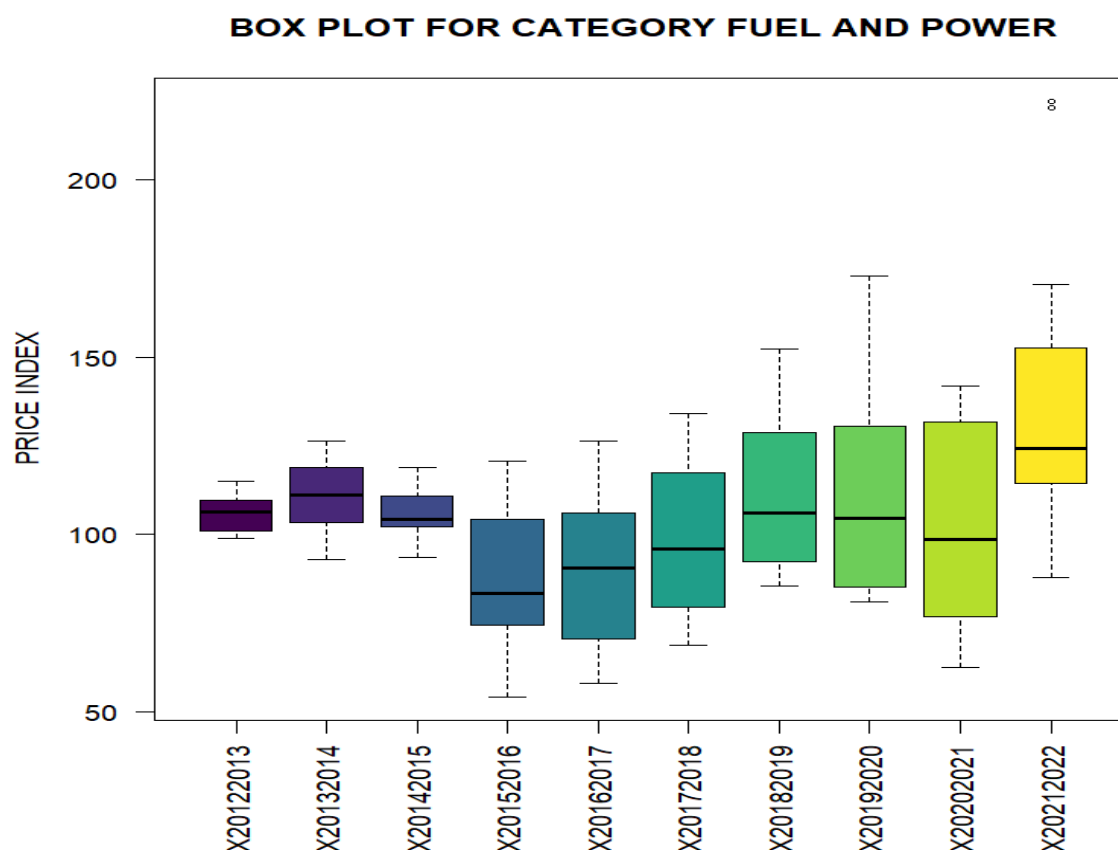


Figure no.26

Interpretation on distribution of financial year 2021-22:

Minimum price index for financial year 2021-22 is 87.8 whereas max is 221.9

The mean value of 137.8 indicates the average price index for all commodities in fuel and power category during this financial year. Q1 at 115.2 indicates that out of 16 commodities 25% that is 4 commodities had price index which is below 115.2 whereas Q2 at 124.1 indicating that 50% of 16 that will be 8 so price index of 8 commodities which is below 124.1 and Q3 is 147.8 indicating price index of 12 commodities out of 16 which is below 147.8 .

Since mean > median implies distribution is skewed toward right , there must be outliers.

Anomaly detection:

Kerosene=221.9

Petroleum coke=220.1

ASSESSING CLUSTERING TENDENCY

Before applying any clustering method on our data , it is important to evaluate whether the data set contains meaningful (i.e., random structures) or not.

Testing of hypothesis:

H0: DATA SET CONTAINS NO MEANINGFUL CLUSTERS

VS

H1: DATA SET CONTAIN MEANINGFUL CLUSTERS

Calculated Hopkins Statistics:

[1] 0.9439458

Calculated p value for Hopkins statistics

[1] 1.2361e-11

Conclusion: P-value for Hopkins statistics implies for rejection of null hypothesis and therefore we conclude that Data set is significantly clusterable.

Next task is to find optimal number of clusters. Therefore, estimating optimal number of clusters.

ELBOW METHOD

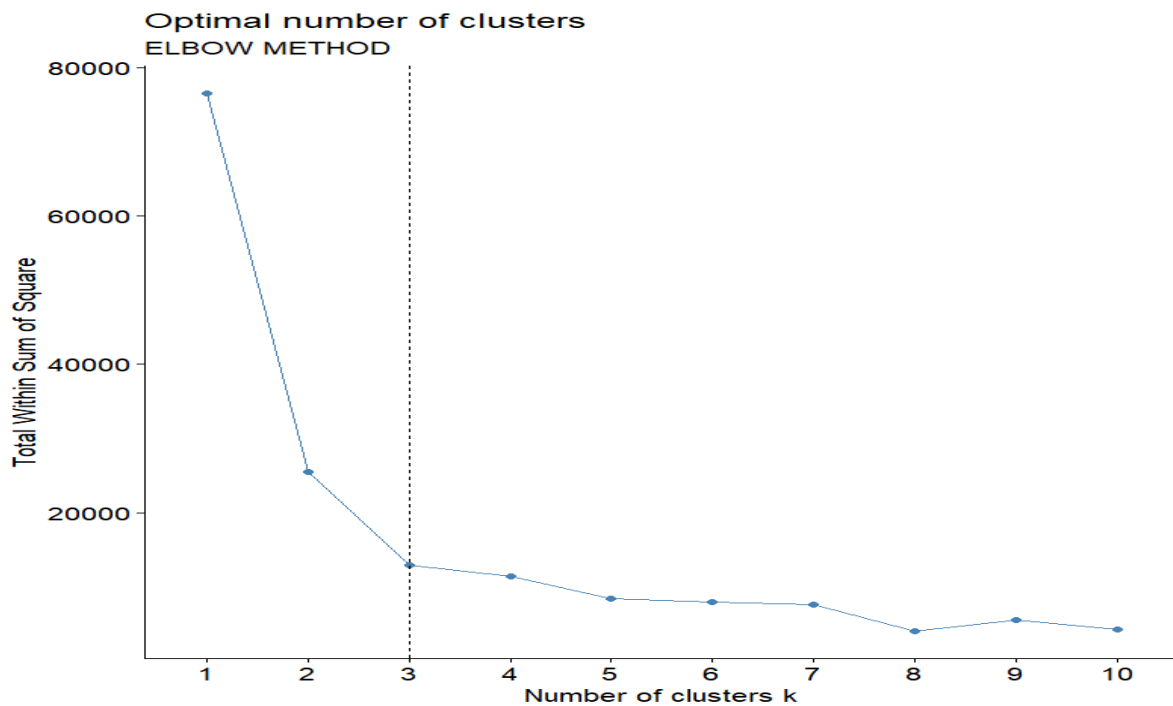


Figure no.27

Elbow method : 3 clusters solution suggested

SILHOUETTE METHOD

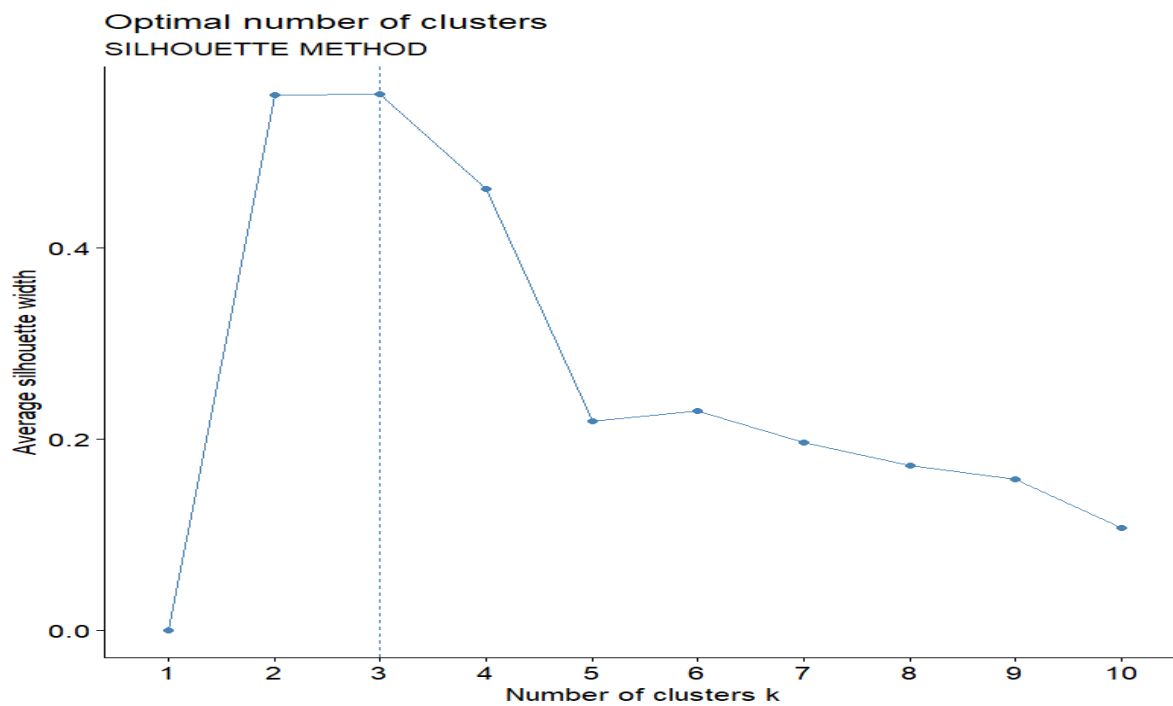


Figure no.28

Silhouette method : 3 clusters solution suggested.

According to these observations it is possible to define $K=3$ as the optimal number of clusters in the data.

Within cluster sum of square is the measure of dispersion of data points within a cluster so this dispersion should be small (clusters should be cohesive or homogeneous) whereas between cluster sum of squares is the dispersion between the clusters and this dispersion should be high (clusters should be separated from each other).

By considering this we can evaluate the optimum number of clusters.

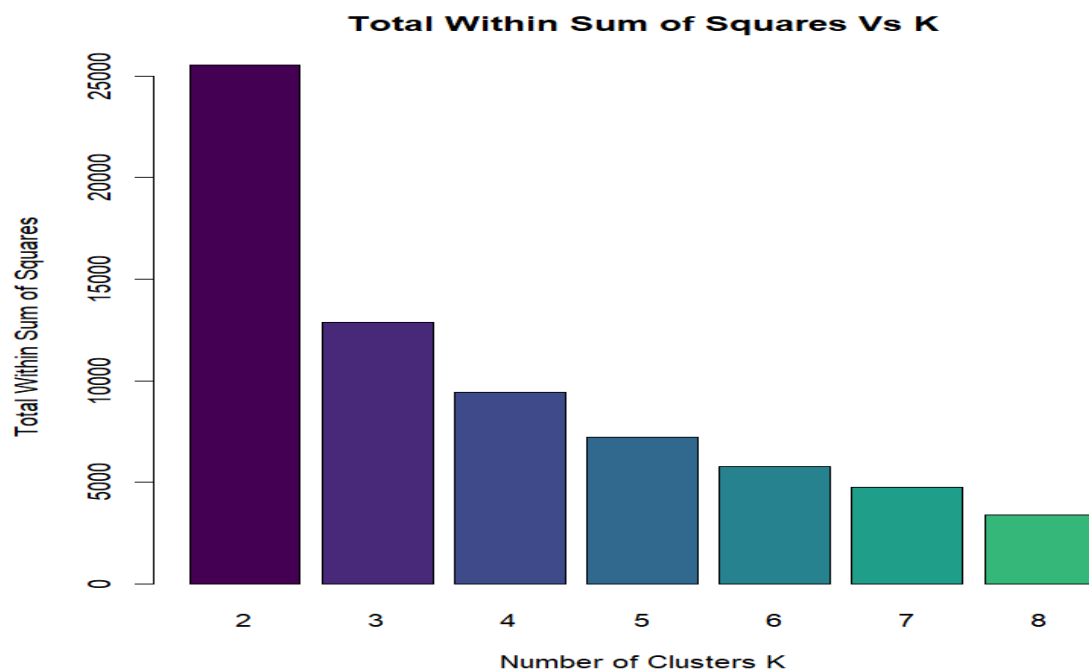


Figure no.29

Observation: After $k=3$ there is small decrease in total ss but after $k=4$ there is not much decrease or change in total within sum of squares.

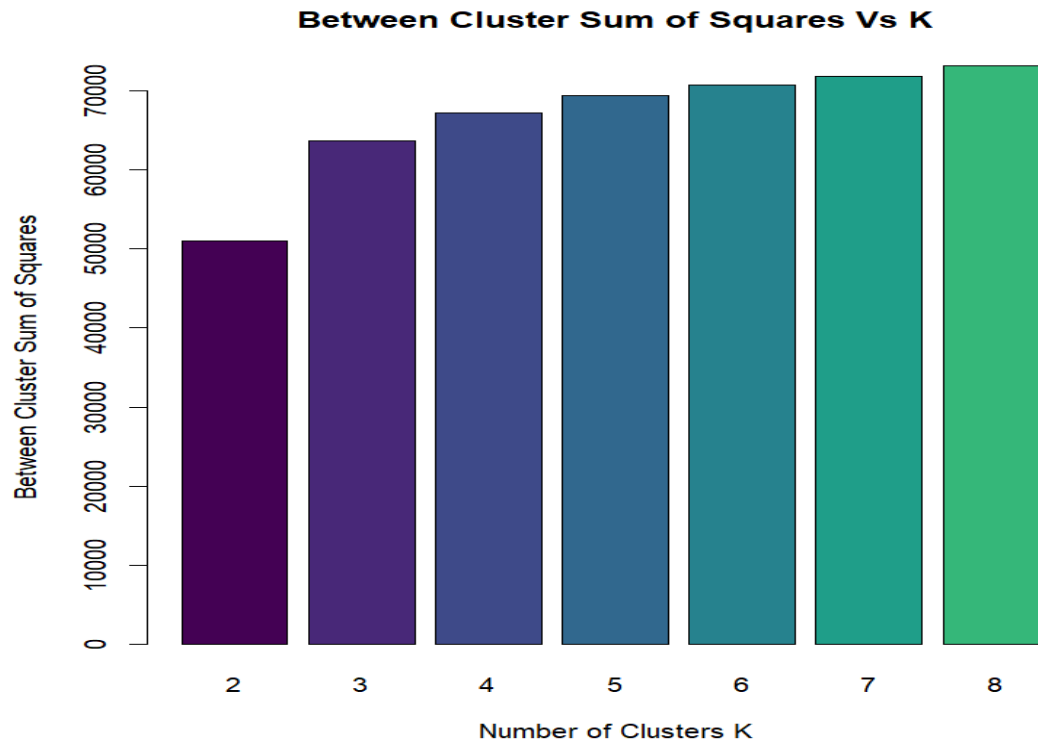


Figure no.30

Observation: After $k=3$ there is not much increase or change in between cluster sum of squares they are almost same.

So, using these 2 plots of within cluster sum of squares and between cluster sum of squares one can finalize $k=3$ for k-means clustering.

Performing k means clustering with $k=3$

K-means clustering with 3 clusters of sizes 2, 8, 6

Cluster means:

	INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
1	103.2500	101.0500	98.9000	83.3500	93.6500
2	108.9125	116.9000	104.7125	74.5750	71.4625
3	102.3667	106.3167	108.9667	108.4833	111.1500

	INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
1	117.5000	151.0500	150.7000	124.6000	221.00
2	78.5500	92.6125	86.8375	74.8750	113.30
3	118.2167	124.5667	128.4167	130.0667	142.65

Clustering vector:

Coking Coal	Non-Coking Coal G1 to G6	Non-Coking Coal G7 to G14
3	2	3
Non-Coking Coal G15 to G17	Lignite	LPG
3	3	2
Petrol	Kerosene	ATF
2	1	2
HSD	Naphtha	Bitumen
2	2	2
Furnace Oil	Lube oils	Petroleum Coke
2	3	1
Electricity		

3

within cluster sum of squares by cluster:

```
[1] 1363.885 5051.954 6471.257
```

(between_ss / total_ss = 83.2 %)

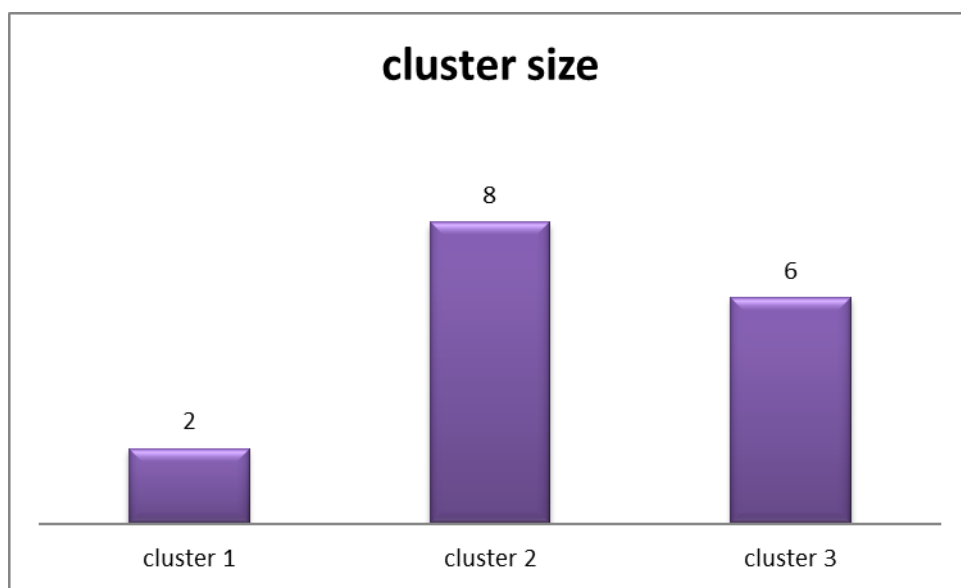


Figure no.31

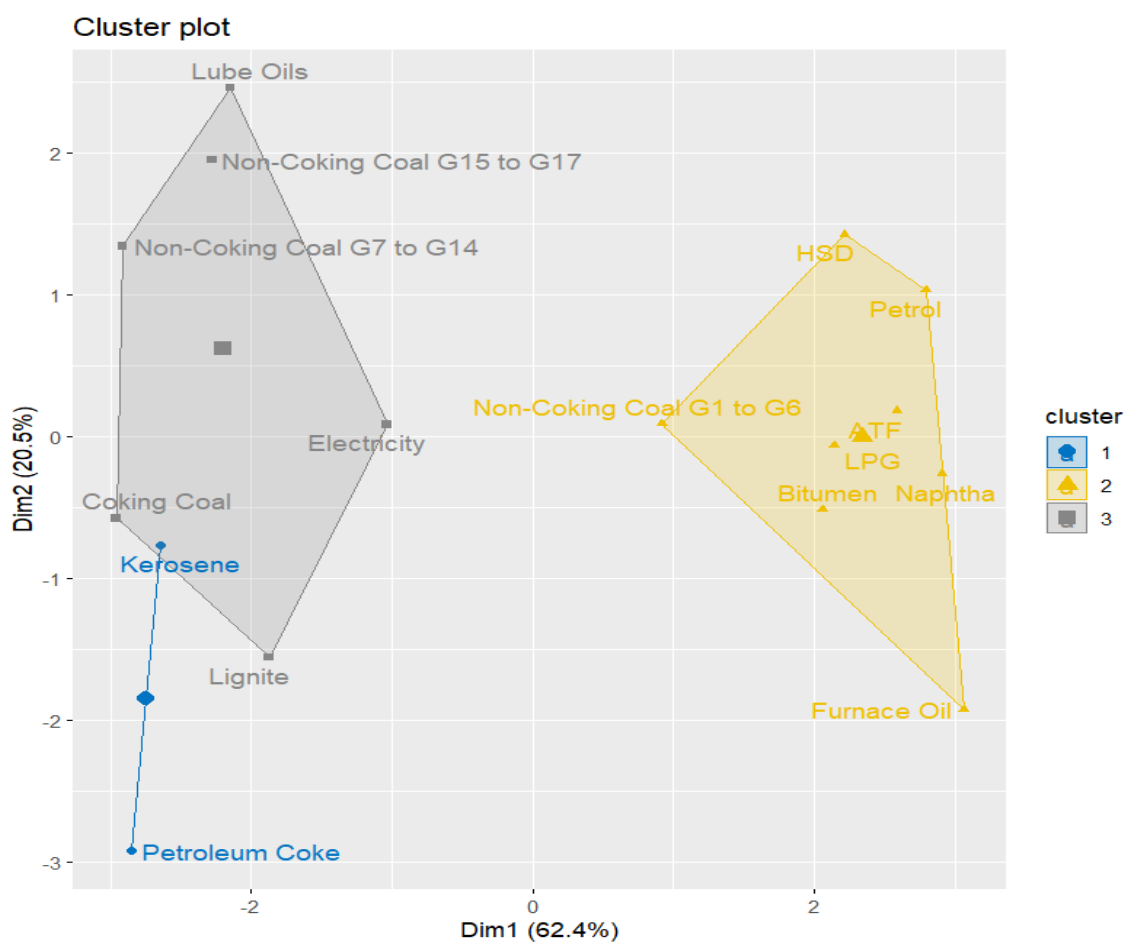


Figure no.32

These two components explain 82.9% of the point variability.

ASSESSING THE QUALITY OF CLUSTERING RESULTS:

```
$widths
               cluster neighbor sil_width
Kerosene             1         3 0.4953452
Petroleum Coke       1         3 0.4492362
Naphtha              2         3 0.7674255
LPG                  2         3 0.7404700
Petrol               2         3 0.7402327
Bitumen              2         3 0.7377787
ATF                  2         3 0.7127723
Furnace Oil          2         3 0.6961418
HSD                  2         3 0.6550314
Non-Coking Coal G1 to G6  2         3 0.4176817
Non-Coking Coal G7 to G14 3         1 0.5952642
Non-Coking Coal G15 to G17 3         2 0.5931426
Coking Coal          3         1 0.5116897
Lube Oils            3         1 0.4728066
Electricity          3         2 0.2097748
Lignite              3         1 0.1763524

$clus.avg.widths
[1] 0.4722907 0.6834418 0.4265051

$avg.width
[1] 0.5606966
```

The silhouette coefficient is a measure of how well each data point in a cluster is separated from the other cluster. The value of silhouette coefficient ranges from -1 to 1 .

Observation with large si (almost1) are very well clustered.

A small si (around 0) means that the observation lies between 2 clusters

Observations with a negative si are probably placed in the wrong cluster.

In our case all the observation has silhouette score greater than 0 implies all the observation are well clustered.

	cluster	size	ave.sil.width
1	1	2	0.47
2	2	8	0.68
3	3	6	0.43

The average Silhouette Width for Cluster 1 with 2 Commodities is 0.47 , for cluster 2 with 8 commodities is 0.68 whereas for Cluster 3 with 6 Commodities

is 0.40. The Total average that is mean of all individual Silhouette Width is 0.43.

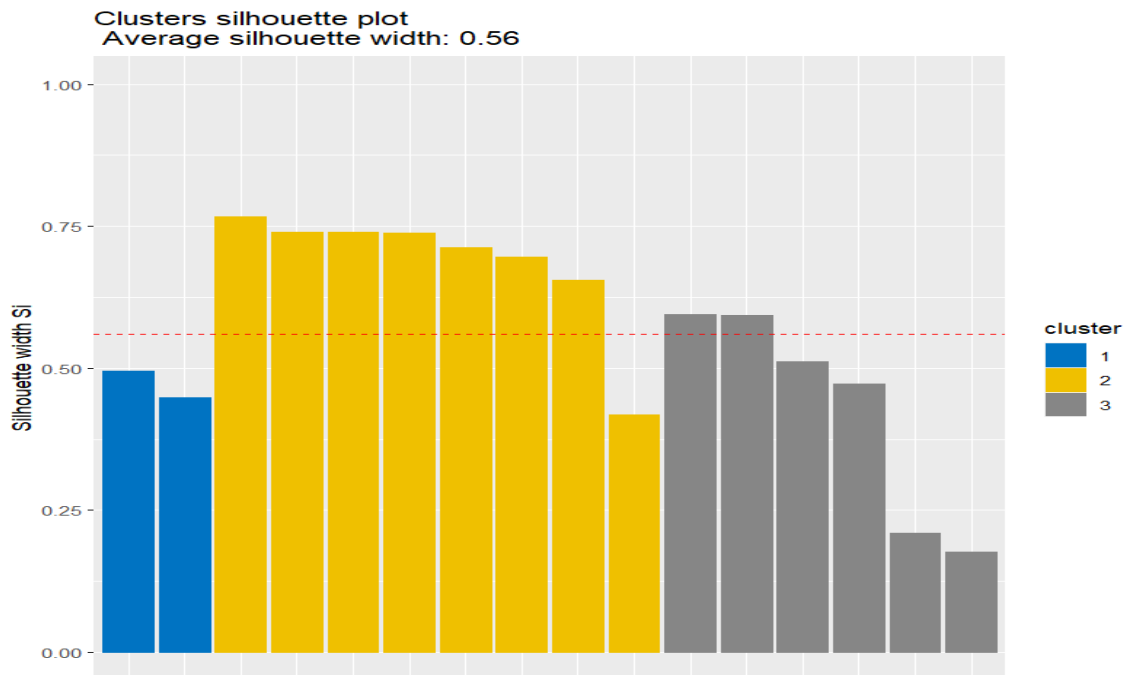


Figure no.33

Observation: In our case the average silhouette coefficient is 0.56 , it indicates that the clustering is reasonably good and effective with well separated clusters.

Visualizing Cluster centres or means:

years	cluster1	cluster2	cluster3	Average
2012-13	103.25	108.91	102.36	105.75
2013-14	101.05	116.9	106.31	110.95
2014-15	98.9	104.71	108.96	105.58
2015-16	83.85	74.57	108.48	88.38
2016-17	93.65	71.46	111.15	89.12
2017-18	117.5	78.55	118.21	98.23
2018-19	151.5	92.61	124.56	111.9
2019-20	150.7	86.83	128.41	110.41
2020-21	124.6	74.87	130.06	101.79
2021-22	221	113.3	142.65	137.76

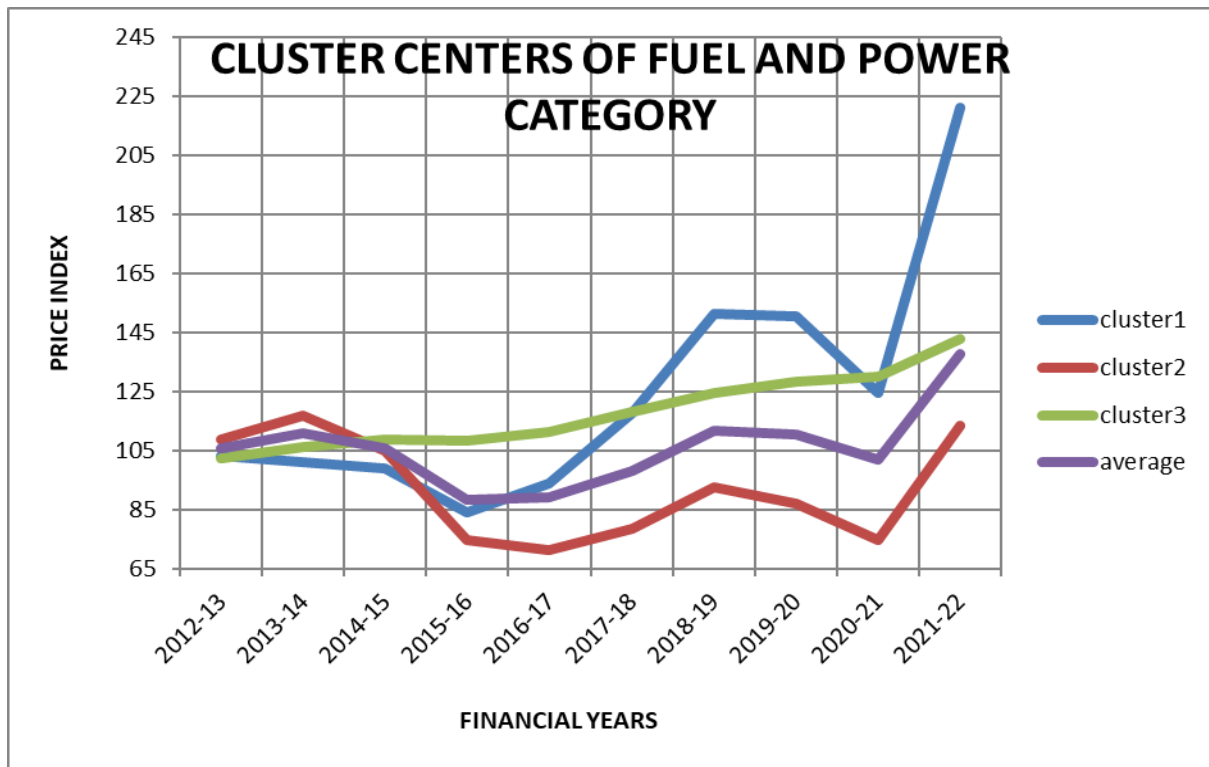


Figure no.34

Conclusion:

The clustering Analysis reveals that the commodities in the fuel and power category can be classified into 3 distinct groups based on their price index behaviour.

Cluster 1 includes only 2 commodities out of 16 commodities coke petroleum and kerosene but it seems that trend of cluster 1 is more volatile than any other clusters despite of only 2 commodities. Till 2015-16 it was lower than the overall average but after 2016-17 it sudden off shows tremendous increase. For financial year 2021-22 the spike in price index is far huge or max as compare to another cluster.

Cluster 2 represent the commodities in fuel and power category which are relatively cheaper than other commodities. More often the price trend decreases over the decades but for financial year 2021-22 the spike in price index is observed.

As compare to cluster 1 and 2 cluster 3 include the commodities whose price index is consistently increasing over the decades.

- Instead of using cluster centers we have used weighted average to represent clusters in our cluster analysis. Since by using Weighted average, we are considering the relative importance or significance of each commodity within the cluster. The weights represent the relative importance of each commodity in the overall cluster.
- The inflation rate for each group of commodities can be interpreted as the percentage change in the average price of commodities in that group over a specified time period.

This will provide insight into which group of commodities experience the most significant price changes.

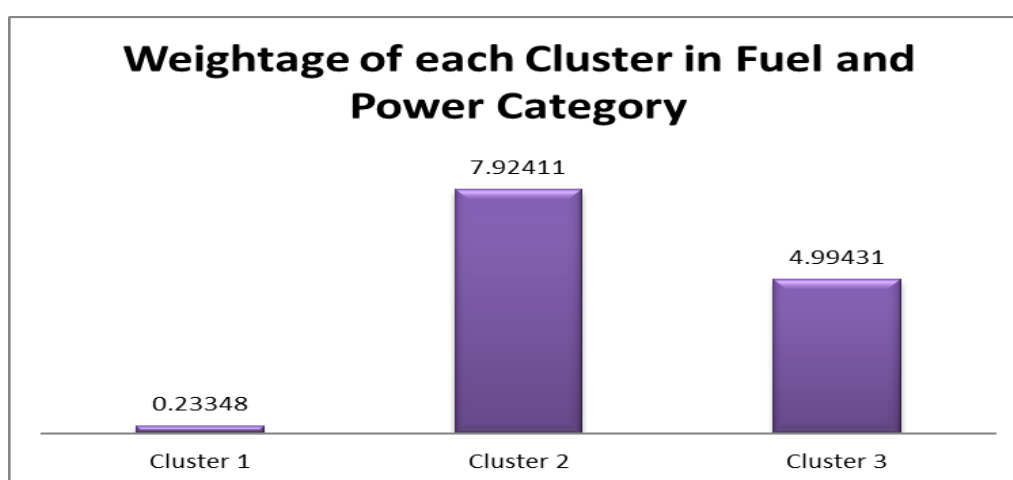


Figure no.35

The three clusters appear to be different in terms of their inflation rates and price index trends.

Cluster 1:

Financial Year	Price index	Inflation rate (% change)
2012-13	105.5622	5.562177
2013-14	106.0047	0.419172
2014-15	101.6626	-4.09611
2015-16	86.38286	-15.0299
2016-17	94.04037	8.864621
2017-18	117.6802	25.13793
2018-19	151.8608	29.04533
2019-20	163.9725	7.97555
2020-21	119.9156	-26.8685
2021-22	221.5405	84.74704

Cluster 1:

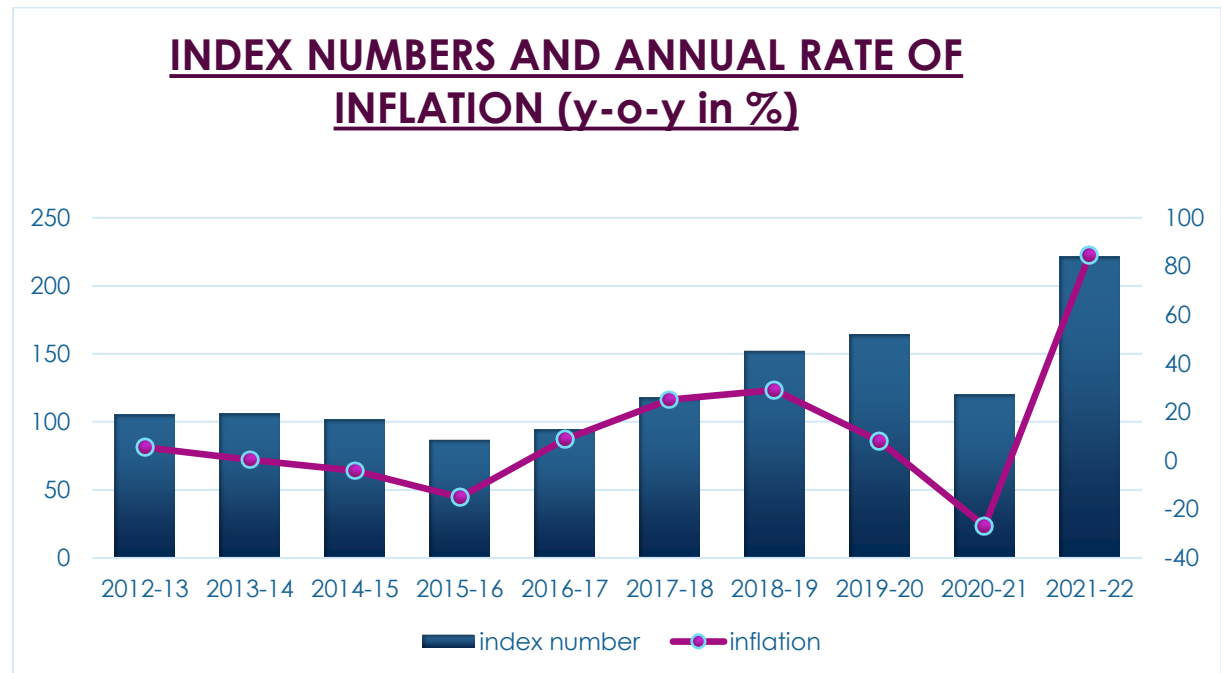


Figure no.36

- There is continuous high amount of fluctuation or ups and down in price index of cluster 1.

Inflation rate:

- In year 2012-13, the inflation rate for cluster 1 was 5.56% indicating moderate increase in price compared to base year of 2011-12 thereafter for year 2013-14 small increase of 0.41% was observed for subsequent financial year 2014-15 and 2015-16 deflation of 4.09% and 15.02% indicating a significant decrease in price compared to previous years. here after for continuous 4 financial years high level of inflation was recorded and for year 2020-21 high level deflation of 26.86% followed by high level inflation of 84.74% in financial year 2021-22.
- Overall conclusion can be met that cluster 1 experienced continuous high amount of fluctuations in inflation over the decades with some years seeing significant increases or decreases in prices index.

Cluster 2

Financial Year	Price Index	Inflation Rate (% change)
2012-13	110.8016	10.80159
2013-14	121.1882	9.374081
2014-15	108.168	-10.7438
2015-16	73.63435	-31.926
2016-17	72.22424	-1.91501
2017-18	80.63183	11.64095
2018-19	93.53913	16.0077
2019-20	88.44158	-5.44964
2020-21	88.44158	-5.44964
2021-22	119.64	35.64

Cluster 2:

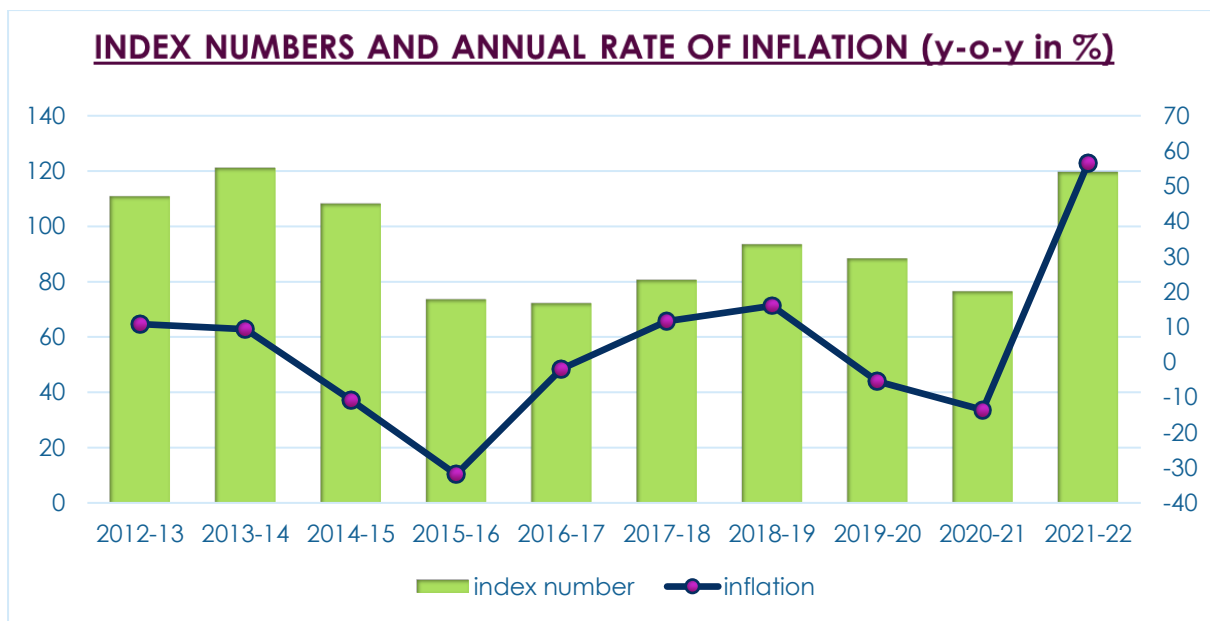


Figure no.37

- **Interpretation:** Initially price index of cluster2 was high with more or less fluctuations but decreased significantly after year 2014-15 and remained below 100 until financial year 2021-22 where price index sudden off spikes to 119.64 which is increase of 56.56% from subsequent previous year.

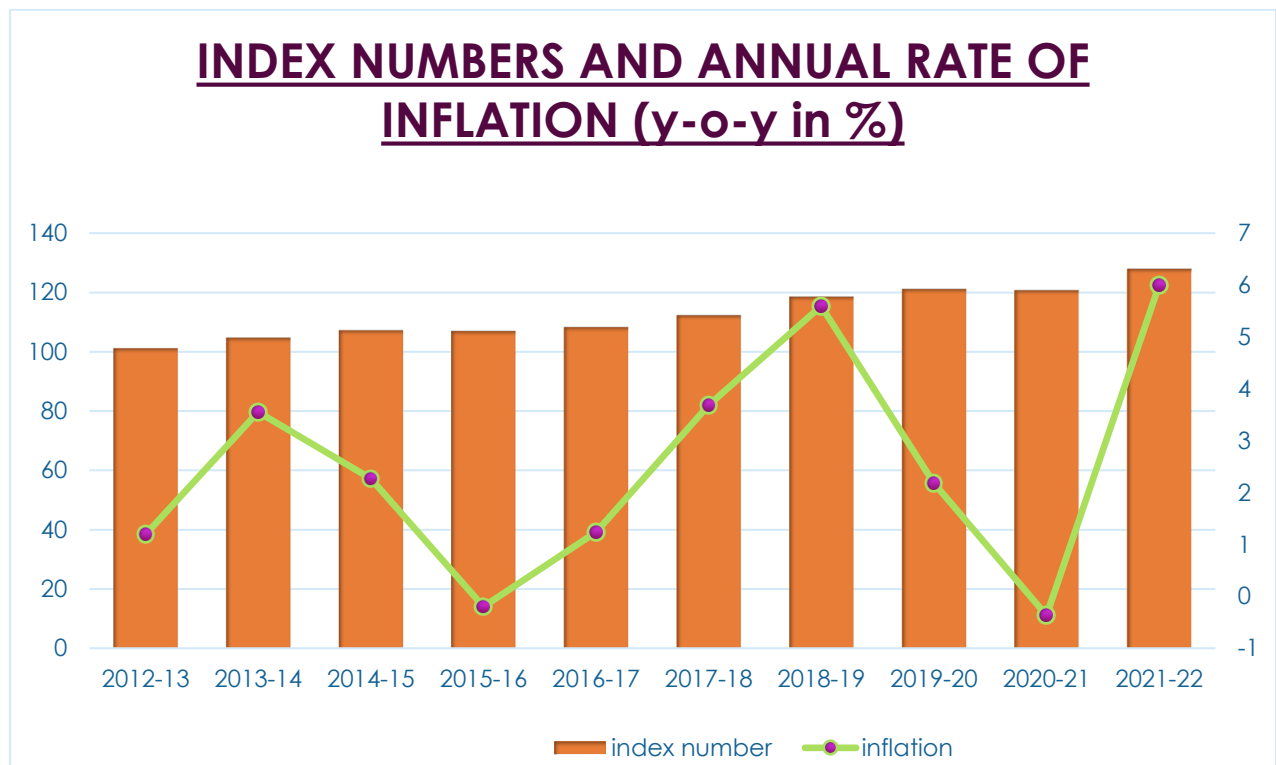
Inflation rate:

- It seems that inflation was relatively high in 2012-13 and 2013-14 with rates of 10.80% and 9.37% respectively. However, inflation then turned

negative in 2014-15 with rate of 10.74% and continued to be negative for preceding 2 years with rate of 31.92% and 1.96% again there is fluctuation for year 2017-18 and 2018-19 with positive inflation rate of 11.64% and 16% followed by negative rate of 5.44% and 13.59% and then for financial year 2021-22 inflation spiked significantly with rate of 56.56%.

Cluster 3

Financial year	Price index	Inflation rate (% change)
2012-13	101.1943	1.194335
2013-14	104.7902	3.553415
2014-15	107.1677	2.268875
2015-16	106.9581	-0.19566
2016-17	108.2816	1.237463
2017-18	112.2761	3.688979
2018-19	118.557	5.594105
2019-20	121.1426	2.180916
2020-21	120.6991	-0.36606
2021-22	127.9326	5.992946

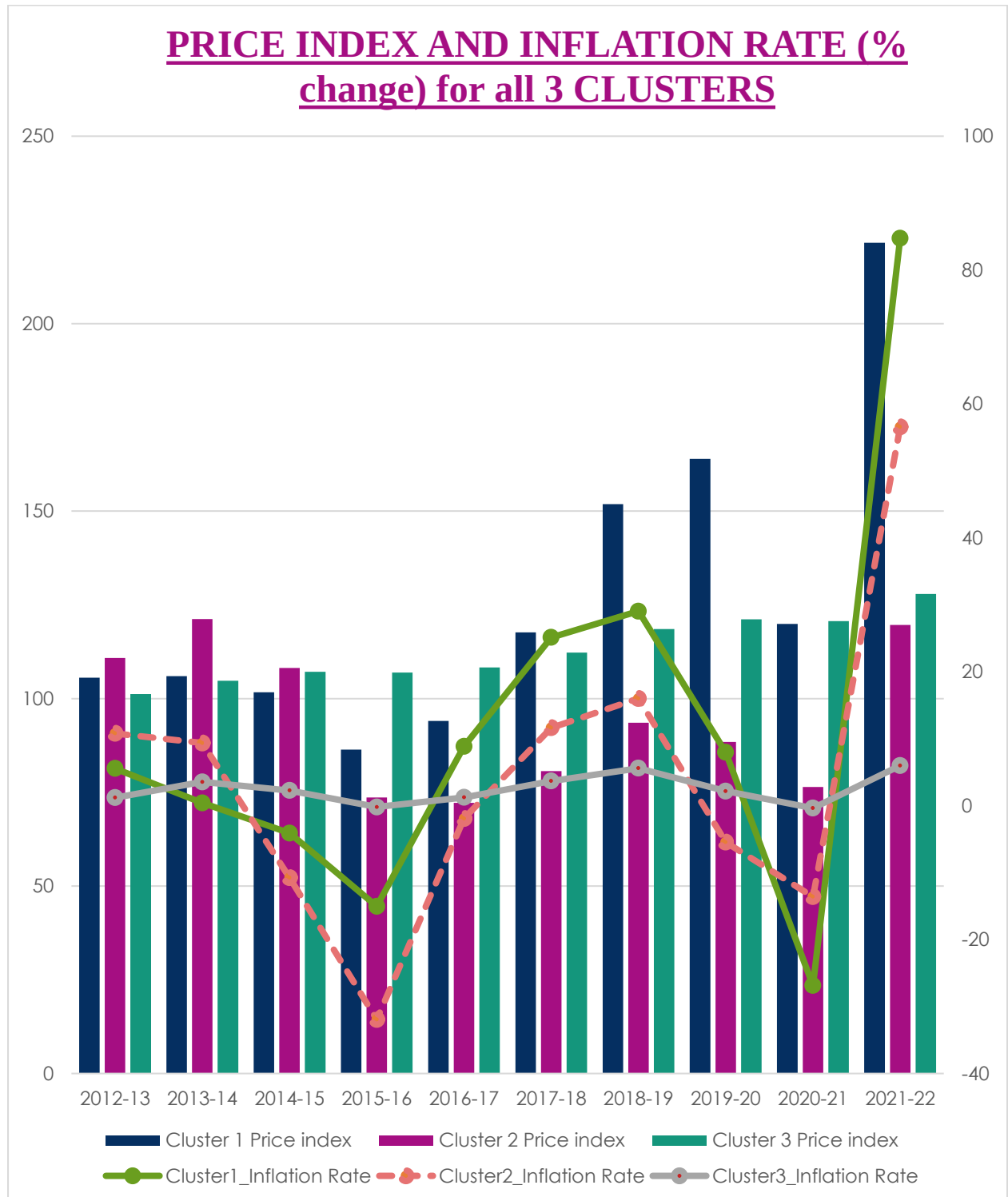
Cluster 3:

- The price index of Cluster 3 is steadily increasing with fewer fluctuations

Interpretation:

- It seems that inflation rate of cluster 3 is more stable compared to cluster 1 and cluster 2. In subsequent financial years 2012-13 to 2014-15, the inflation rate was 1.19%, 3.55%, and 2.26%, which means price increase was comparatively low. Whereas for year 2015-16, deflation of 0.19% was observed, implying that there was a slight decrease in prices hereafter. For 4 years, a slight increase in inflation was observed, followed by deflation of -0.36% in year 2020-21, and then for year 2021-22, inflation of approximately 6% was observed.

VISUALIZING AND COMPARING RESULT OF ALL 3 CLUSTERS IN SINGLE PLOT:



MANUFACTURED PRODUCTS

Following analysis is for the Manufactured Products Category.

Descriptive Statistics:

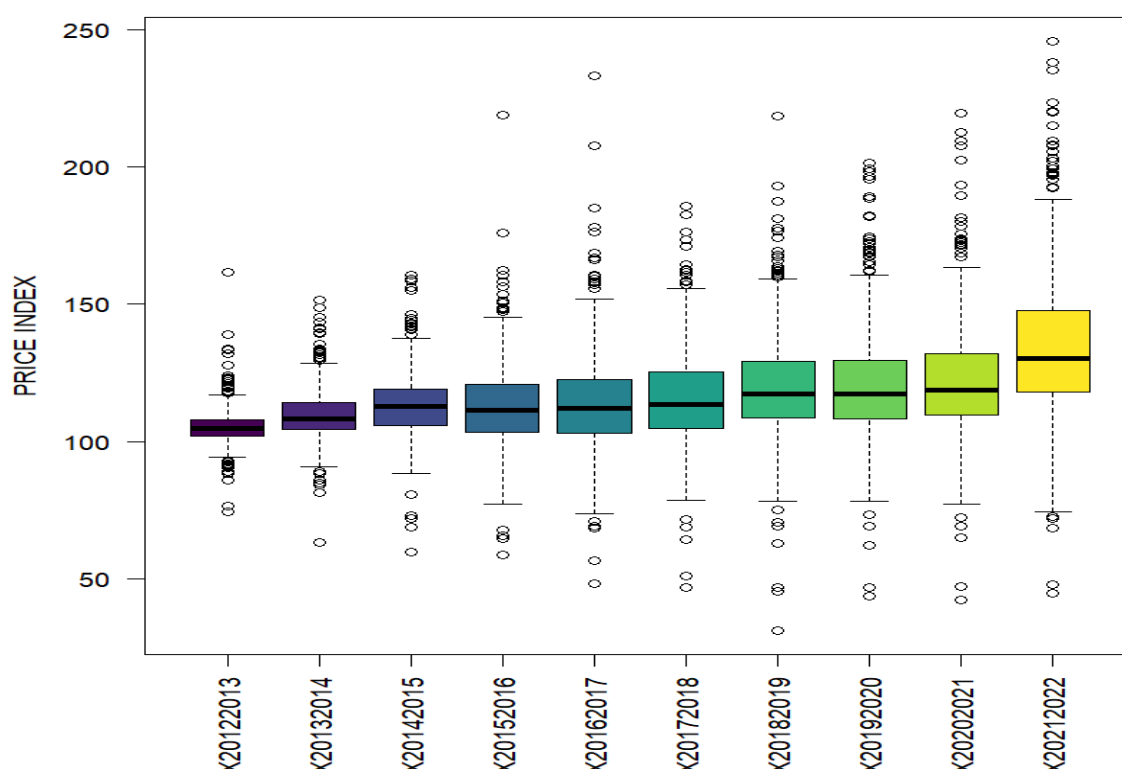
INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
Min. : 74.2	Min. : 63.3	Min. : 59.7	Min. : 58.6	Min. : 48.3
1st Qu.:102.0	1st Qu.:104.3	1st Qu.:105.7	1st Qu.:103.3	1st Qu.:102.9
Median :104.7	Median :108.2	Median :112.7	Median :111.5	Median :112.2
Mean :105.5	Mean :109.7	Mean :113.1	Mean :112.7	Mean :113.7
3rd Qu.:108.0	3rd Qu.:114.0	3rd Qu.:118.9	3rd Qu.:120.8	3rd Qu.:122.6
Max. :161.6	Max. :151.7	Max. :160.7	Max. :218.9	Max. :233.2
INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
Min. : 46.7	Min. : 30.9	Min. : 43.6	Min. : 42.2	Min. : 44.8
1st Qu.:104.8	1st Qu.:108.6	1st Qu.:108.4	1st Qu.:109.6	1st Qu.:118.1
Median :113.5	Median :117.3	Median :117.3	Median :118.6	Median :130.3
Mean :116.0	Mean :119.6	Mean :120.1	Mean :121.8	Mean :133.9
3rd Qu.:125.4	3rd Qu.:129.0	3rd Qu.:129.6	3rd Qu.:132.0	3rd Qu.:147.7
Max. :185.7	Max. :218.6	Max. :201.6	Max. :219.8	Max. :246.0

Variation among price index of different commodities in Manufactured products Category for every financial year from 2012-13 to 2021-22.

INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
47.16558	92.25941	147.46132	246.87547	347.08860
INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022
320.80169	388.26070	440.11409	496.11432	722.28684

Box Plot

BOX PLOT FOR CATEGORY MANUFACTURED PRODUCTS



INTERPRETATION:

Minimum value of price index in financial year 2021-2022 is 44.8 which indicates that there is at least one commodity in manufactured products category whose price index is as low as 44.8 in year 2021-22, whereas max is 246 indicating that there is at least one commodity with price index value as high as 246. Overall average of price index for all commodities in manufactured product category for year 2021-22 is 133.9. The first quartile at 118.1 suggests that 25% of the commodities in manufacturing category have price index below 118.1 whereas median value of 130.3 and Q3 at 147.3 implies that 50% and 75% of the commodities have price index value below 130.3 and 147.3.

For the financial year 2021-2022 the commodities having price index value below

$Q1 - 1.5(IQR) = 73.7$ and above

$Q3 + 1.5(IQR) = 191.7$ is Considered to be outlier.

Outlier detection:

Outliers are Values that are significantly higher or lower than the rest of the data, and they may be indicative of unusual events or trends.

In Our case Outliers are due to Spike or Decline in Price Index of Some Commodities for each Year it may be due to shortage, high demand, low supply, or other market disruption.

As compare to other 2 categories there are many outliers in manufactured product category every year implies that for every year there are certain commodities for which price index is either very low or very high compared to other commodities in that year.

https://docs.google.com/spreadsheets/d/1ZxksiZ5odFdIJczyHn0PipZHdV6vqIDq/edit?usp=drive_link&ouid=109696859131187939010&rtpof=true&sd=true

ASSESSING CLUSTERING TENDENCY

Before applying any clustering method on our data, its important to evaluate whether the data set contains meaningful clusters(i.e., non-random structure) or not.

TESTING OF HYPOTHESIS

H0: DATA SET CONTAINS NO MEANINGFUL CLUSTERS

V/S H1: DATA SET CONTAIN MEANINGFUL CLUSTERS

Calculated value for Hopkins Statistics

[1] 0.9993925

Calculated P-value for Hopkins statistics

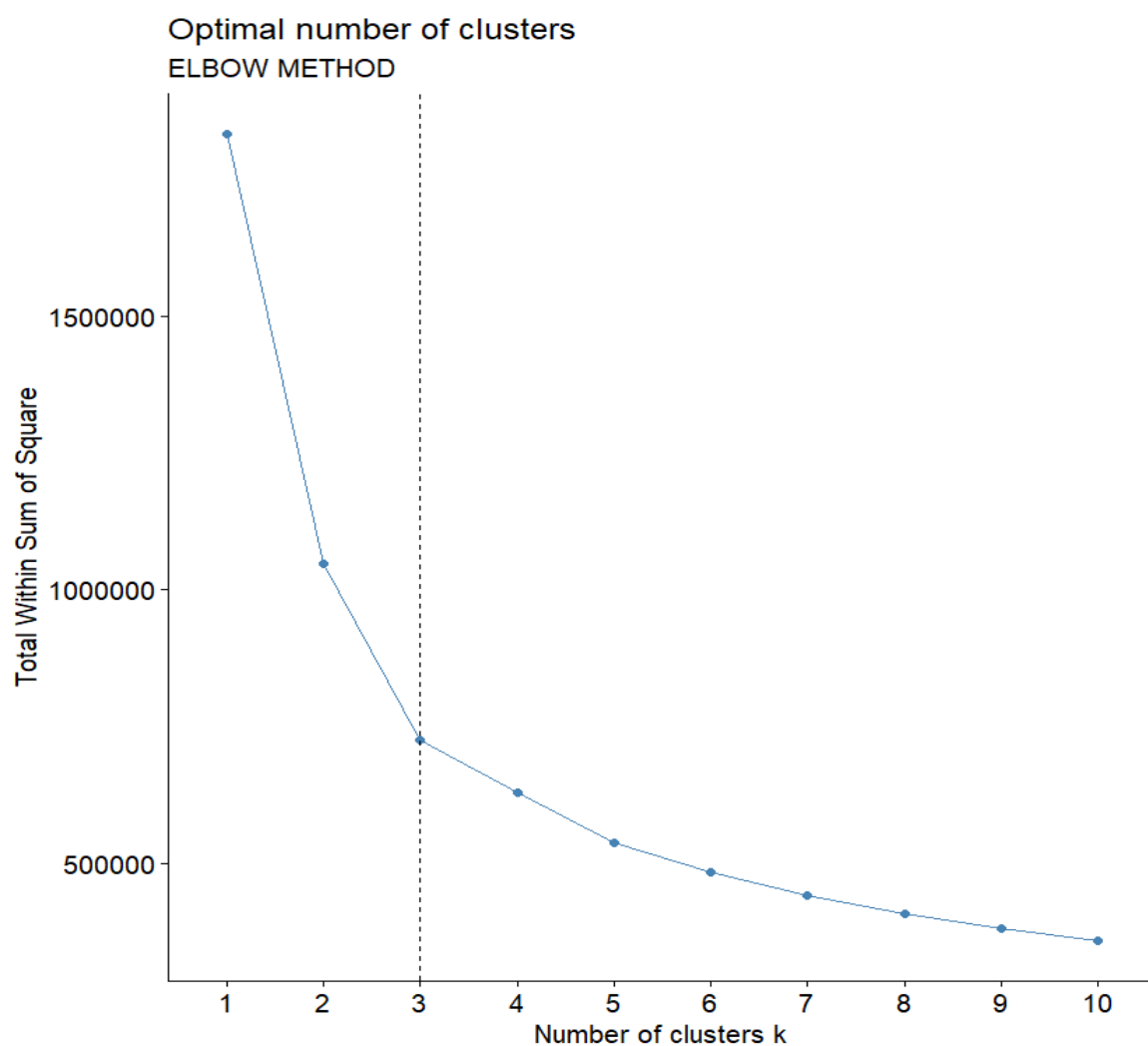
[1] 0

Conclusion: P-value for Hopkins statistics implies for rejection of null hypothesis and therefore we conclude that Data set is significantly clusterable.

NEXT TASK IS TO FIND OPTIMAL NUMBER OF CLUSTER

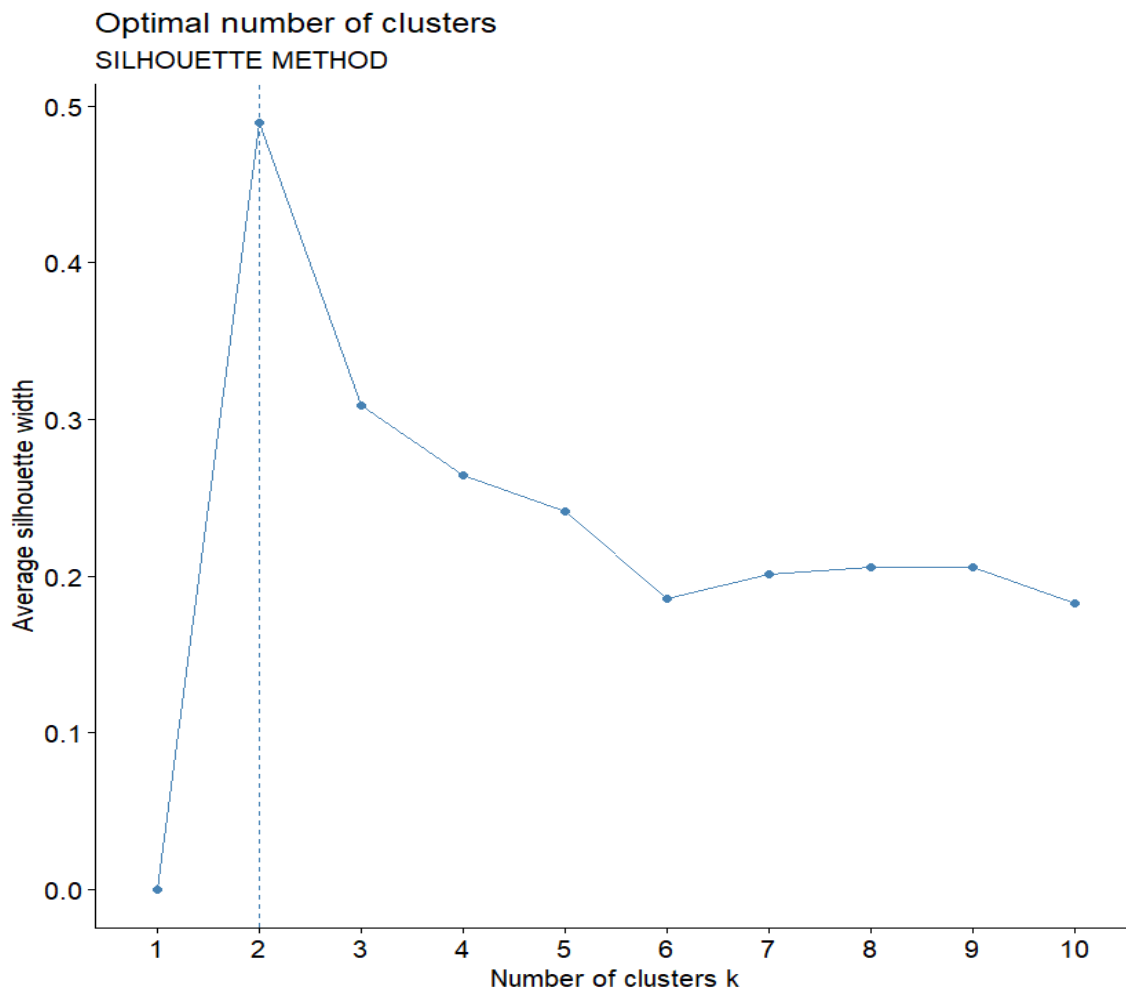
THEREFORE, ESTIMATING OPTIMAL NUMBER OF CLUSTERS

Elbow method:



Elbow method : 3 clusters solution suggested.

Silhouette method:

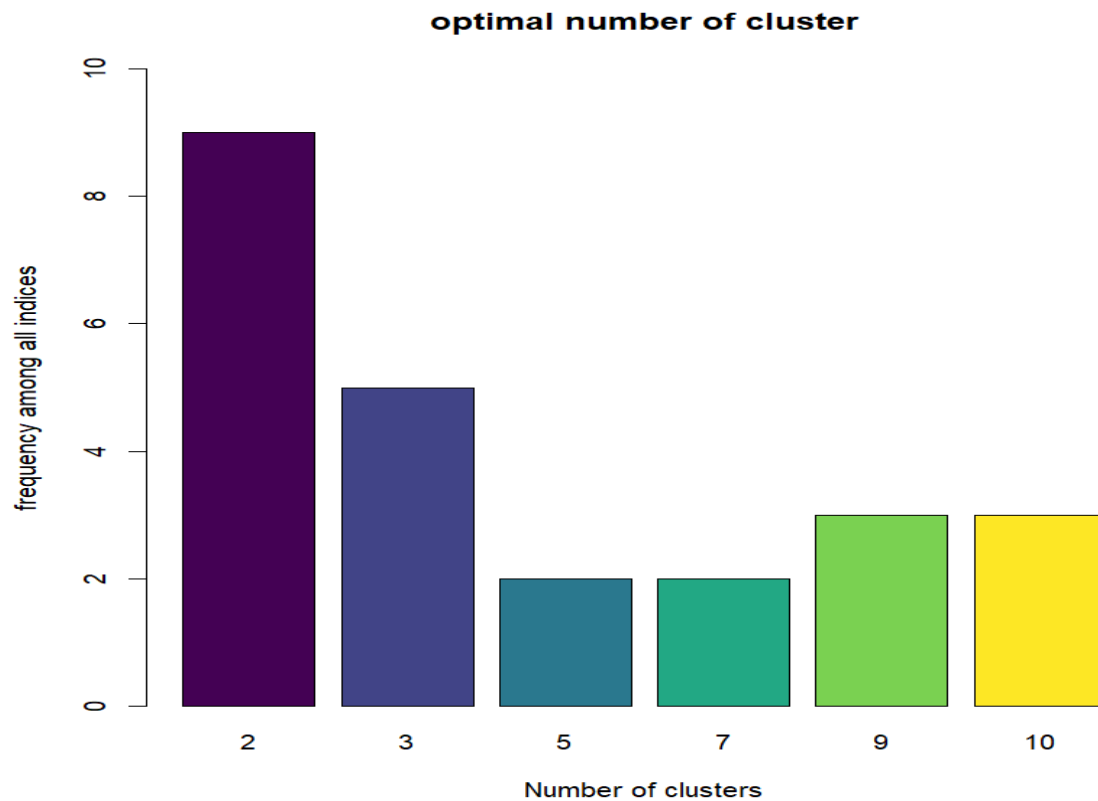


Silhouette method : 2 clusters solution suggested.

30 INDICES FOR CHOOSING THE BEST NUMBER OF CLUSTERS

- * Among all indices:
- * 9 proposed 2 as the best number of clusters
- * 5 proposed 3 as the best number of clusters
- * 2 proposed 5 as the best number of clusters
- * 2 proposed 7 as the best number of clusters
- * 3 proposed 9 as the best number of clusters
- * 3 proposed 10 as the best number of clusters

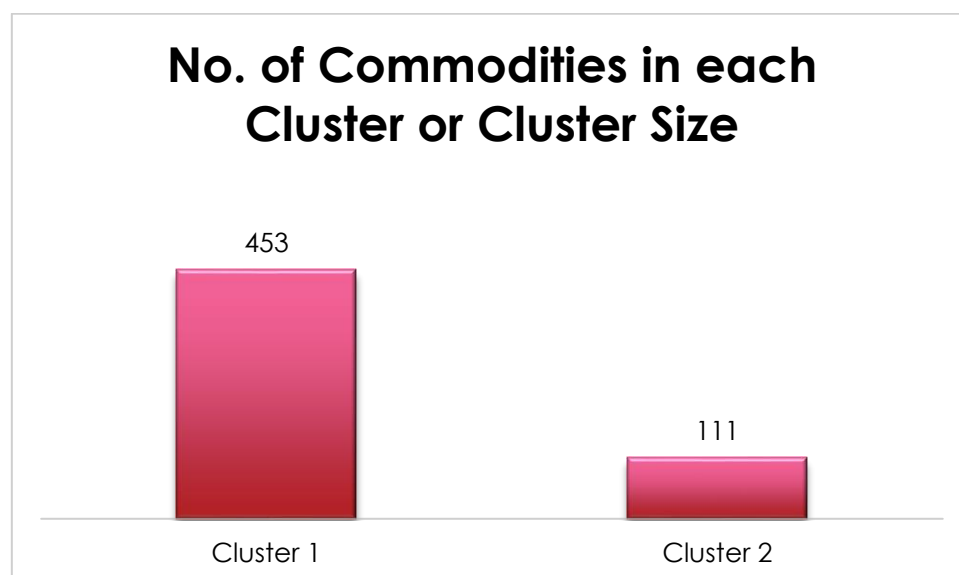
Conclusion: According to the majority rule, the best number of clusters is 2.



OBSERVATION: According to these observations defining $k=2$ as optimal number of clusters for further analysis.

For manufactured products category we have performed Hierarchical agglomerative clustering with Wards linkage and 2 as optimal number of cluster whereas distance measure is Manhattan.

Cluster size



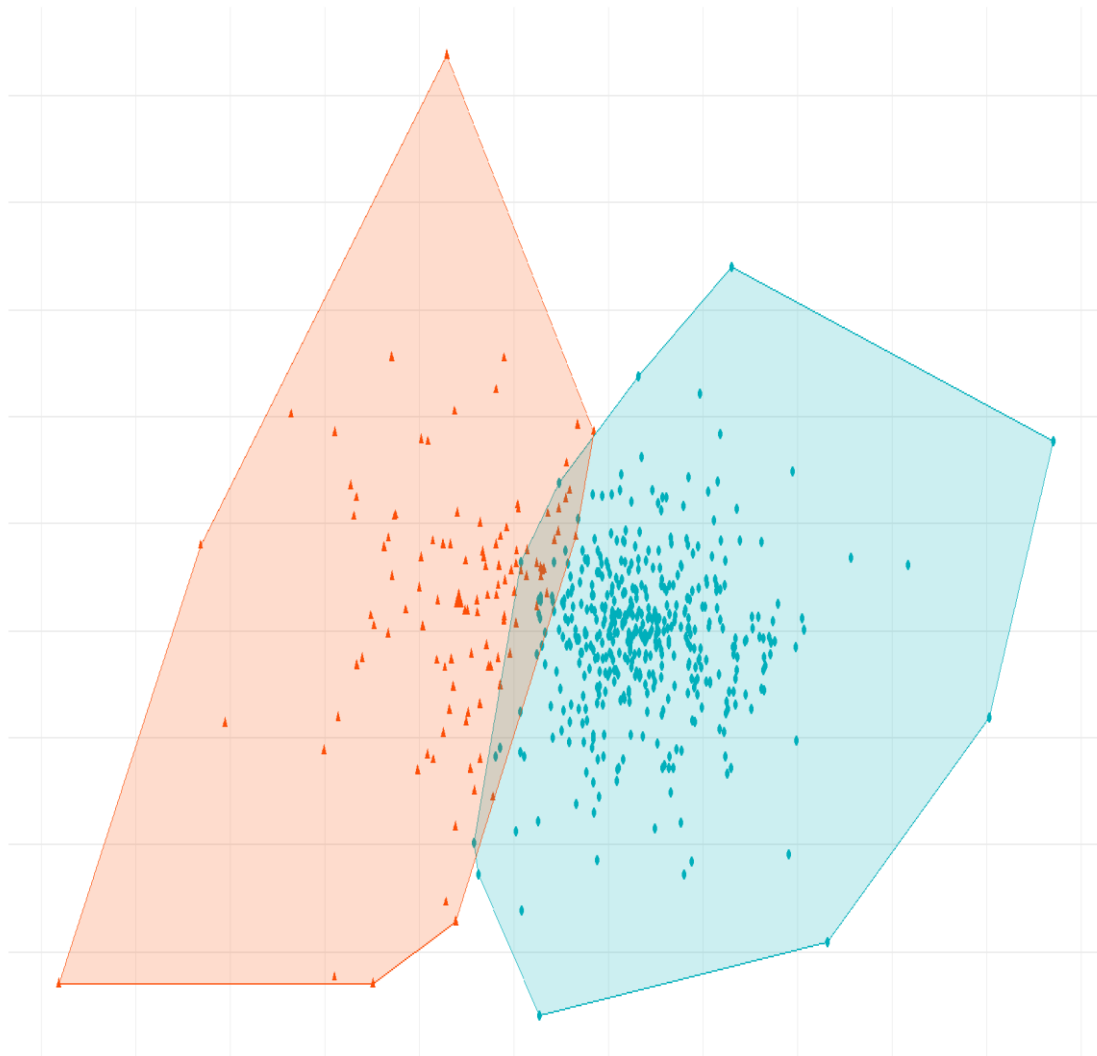
Cluster Centres

	Group	1	INDEX20122013	INDEX20132014	INDEX20142015	INDEX20152016	INDEX20162017
1	1		104.3596	107.4921	109.8342	107.7691	107.6444
2	2		110.2414	118.7784	126.3991	132.5910	138.6360
		INDEX20172018	INDEX20182019	INDEX20192020	INDEX20202021	INDEX20212022	
1		109.6943	112.7263	112.2444	113.8722	125.9892	
2		141.7550	147.5207	151.9441	154.2955	165.9856	

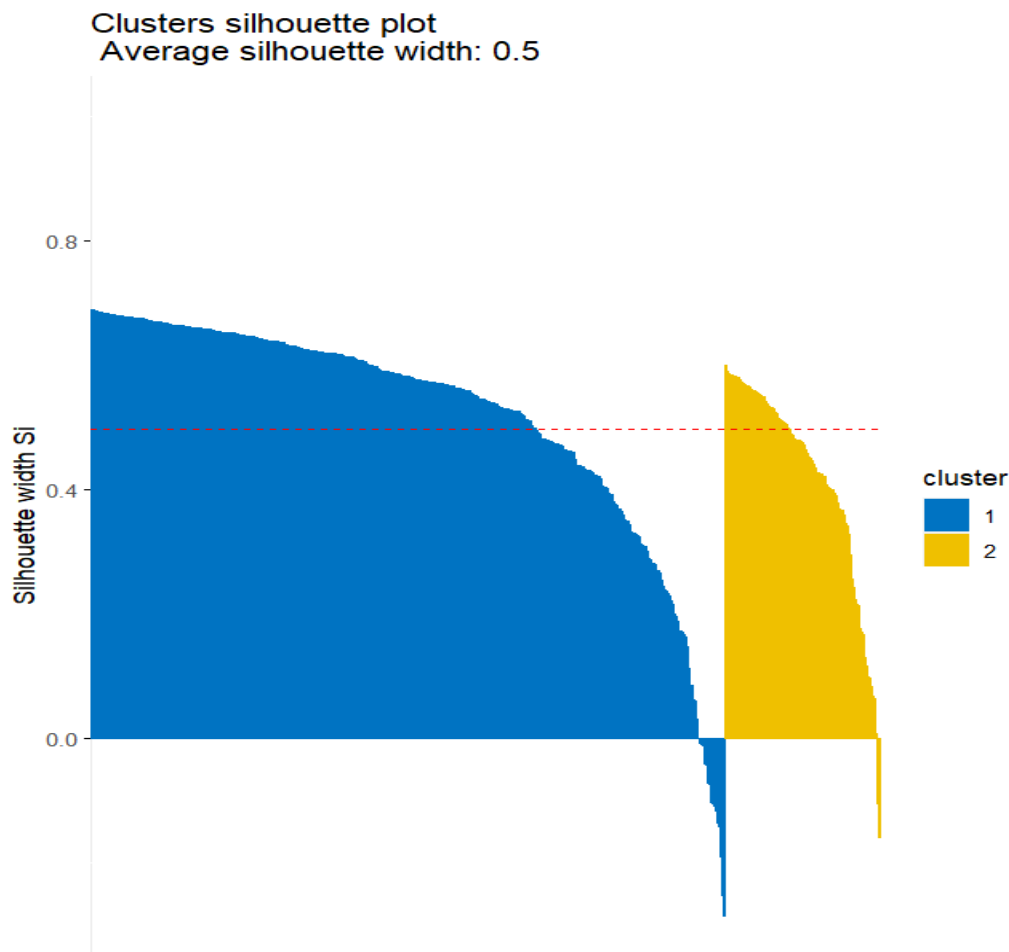
Since there are 564 commodities in manufactured product category so we are unable to visualize the hierarchical clustering of data by dendrogram here the focus is on resulting clusters rather than visualization.

In the below cluster plot, we have just plotted the points to create some visualization of clusters and had not use the labels for the points since there are large number of commodities and all the labels were overlapping.

CLUSTER PLOT:



ASSESSING THE QUALITY OF CLUSTERING RESULTS:



	cluster	size	ave.sil.width
1	1	453	0.52
2	2	111	0.42

The average Silhouette Width for Cluster 1 with 453 Commodities is 0.52 and for cluster 2 with 111 commodities is 0.42. The Total average that is mean of all individual Silhouette Width is 0.5

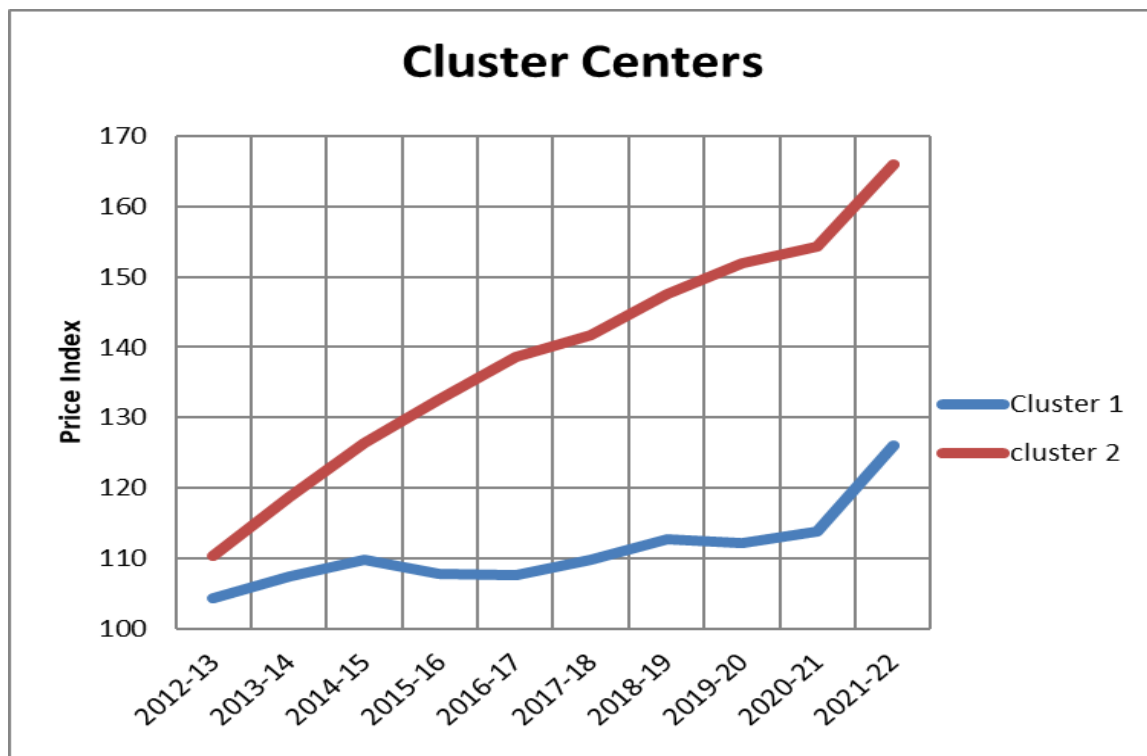
Observation: In our case the average silhouette coefficient is 0.5 , it indicates that the clustering is reasonably good and effective with well separated clusters.

It can be seen that several observations in both the clusters have negative silhouette coefficient. This means that they are not in right cluster. We can find names of these observation and determine the clusters they are closer (neighbour cluster).

	cluster	neighbor	sil_width
Piston ring/Piston and Compressor	1	2	-4.420309e-05
Reprographic machine and scanners	1	2	-7.595473e-03
Electronic Printed Circuit Board (PCB)/micro circuit	1	2	-9.935701e-03
Plastic moulded-others toys	1	2	-1.264377e-02
Medical accessories	1	2	-4.061838e-02
Waterproof footwear	1	2	-4.204170e-02
Powder Milk	1	2	-4.318689e-02
Vaccine for polio	1	2	-7.273679e-02
Sugar Machinery	1	2	-7.485268e-02
Noodles & similar extruded products	1	2	-1.042814e-01
Clock	1	2	-1.056530e-01
Amine	1	2	-1.090785e-01
Wooden box/crate	1	2	-1.125386e-01
Mustard Oil	1	2	-1.170561e-01
Carpets and other floor coverings of textiles	1	2	-1.362483e-01
Aniline (including pna, ona, ocpna)	1	2	-1.426125e-01
Soda ash/washing soda	1	2	-1.907277e-01
Harness, Saddles & Other related Items	1	2	-2.520727e-01
Buffalo meat, fresh/frozen	1	2	-2.862276e-01
Hand tools	2	1	-1.057678e-01
Tooth brush	2	1	-1.591422e-01

For all these commodities the silhouette width is close to zero suggests that these commodities are on or close to the decision boundary between two clusters these could indicate overlapping clusters which is true as observed above from cluster plot.

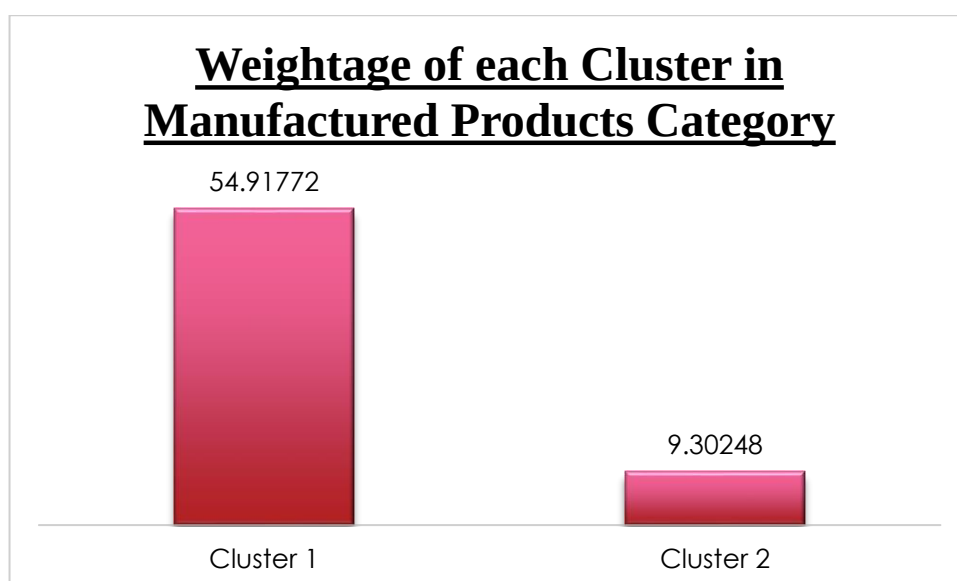
Visualizing cluster centres:



Instead of using cluster centres we have used weighted average to represent clusters in our cluster analysis. Since by using Weighted average, we are considering the relative importance or significance of each commodity within the cluster. The weights represent the relative importance of each commodity in the overall cluster.

The Interpretation are based on the Weighted average of clusters that represent average behaviour of commodities within each cluster.

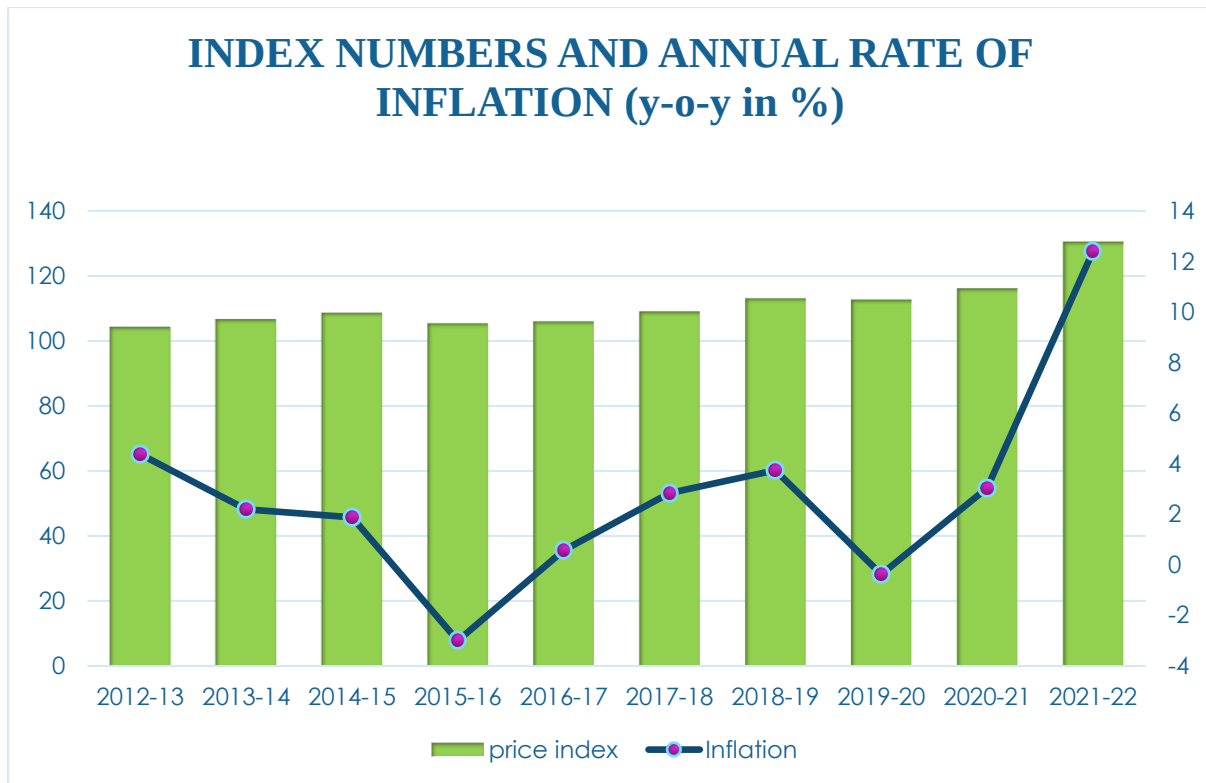
The resulting weighted average represents a representative value for a cluster, considering the contribution of different commodities based on their weights.



The clustering analysis reveals 2 Clusters based on price index trend with 564 commodities in total, Cluster 1 consist of 453 commodities and remaining(111) commodities are in cluster 2. Cluster 1 comprises the majority of commodities.

Cluster 1:

Financial Years	Price Index	Inflation rate (% change)
2012-13	104.37	4.37
2013-14	106.67	2.199
2014-15	108.66	1.87
2015-16	105.41	-3.00
2016-17	106.01	0.58
2017-18	109.02	2.84
2018-19	113.099	3.75
2019-20	112.67	-0.38
2020-21	116.10	3.05
2021-22	130.498	12.398



Interpretation:

The Cluster 1 price index indicates that this cluster contains the product for which the price index over the years shows minor variation as well as there are some ups and downs for a year or two but overall, the price index exhibit a consistent trend without significant increase or decrease except for financial year 2021-22 where a sudden spike can be observed.

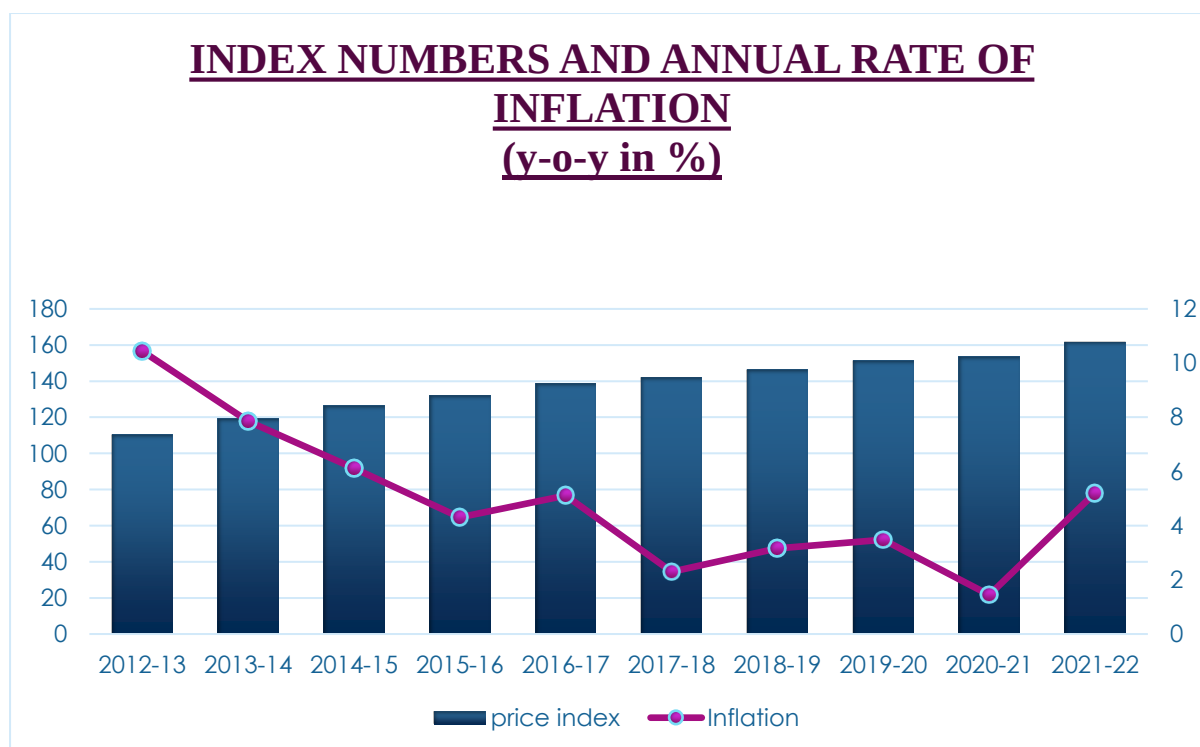
This suggest that the commodities in cluster 1 experience relatively moderate and consistent price fluctuations over decades.

By observing the inflation rate for cluster 1 the following interpretation can be made:

For the initial year 2012-13 the inflation rate was 4.37% whereas for subsequent years 2013-14 and 2014-15 there seems decline in inflation rate to 2.199% and 1.87% followed by deflation period in year 2015-16. afterward cluster 1 experience low inflation rate exception for previous year 2020-21 where it picked up reaching 12.398%.

Cluster 2:

Financial Years	Price index	Inflation rate (% change)
2012-13	110.44	10.44
2013-14	119.12	7.86
2014-15	126.41	6.12
2015-16	131.86	4.31
2016-17	138.60	5.11
2017-18	141.79	2.29
2018-19	146.27	3.16
2019-20	151.37	3.49
2020-21	153.56	1.44
2021-22	161.56	5.21



Interpretation:

The price index of cluster 2 suggest that this cluster comprises commodities with higher price index range and continuous upward trend in price index. The price index of these commodities consistently jump to higher level year after year indicating significant increase in Price index over the decades.

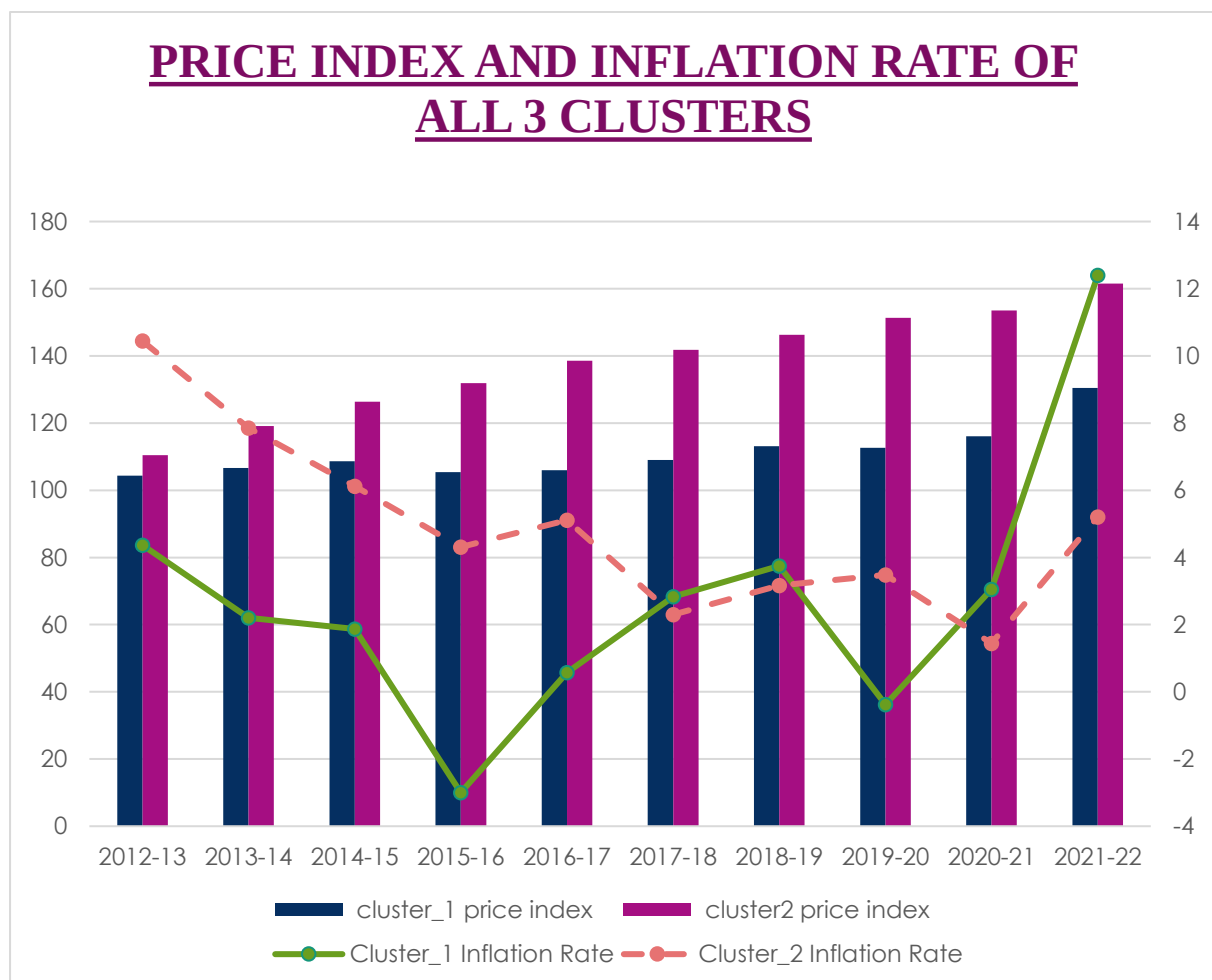
For both the clusters the decline was observed in financial year 2020-21 and after that there is a noticeable spike.

By observing the inflation rate for cluster 2 the following interpretation can be made:

It seems that cluster 2 did not experience any deflation period since last 10 years also it can be observed that the price index of cluster 2 consistently shows increasing trend over the decades.

For initial years the inflation rate was high around 10.44% for 2012-13 and 7.86 % for 2013-14. Overall, the inflation rate for cluster 2 goes on decreasing with some ups and down for a year or two up to the financial year 2020-21 but for the financial year 2021-22 the inflation rate rose to 5.20%.

VISUALIZING AND COMPARING RESULT OF ALL 3 CLUSTERS IN SINGLE PLOT:



Methodology For Stochastic Process Analysis

A Markov chain or Markov process is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event. Informally, this may be thought of as, "What happens next depends only on the state of affairs now." A countably infinite sequence, in which the chain moves state at discrete time steps, gives a discrete-time Markov chain (DTMC). A Markov process is a stochastic process that satisfies the Markov property (sometimes characterized as "memorylessness").

$$P \{X_{n+1}=j | X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0\}$$

$$= P \{X_{n+1}=j | X_n = i\} \text{ for all states } i_0, i_1, \dots, i_{n-1}, i \text{ and } j \text{ in } S \text{ for all } n \geq 0$$

The changes of state of the system are called transitions. The probabilities associated with various state changes are called transition probabilities. The process is characterized by a state space, a transition matrix describing the probabilities of particular transitions, and an initial state (or initial distribution) across the state space. By convention, we assume all possible states and transitions have been included in the definition of the process, so there is always a next state, and the process does not terminate.

$P_{ij} = P \{X_{n+1}=j | X_n = i\}$, then it is the transition probability of moving from state i in n th step, to the state j in $(n+1)$ th step. These probabilities are one-step transition probabilities of Markov Chain. If the one-step transition probabilities are irrelevant of time n then the chain is time homogeneous or stationary Markov Chain. The transition probabilities represented in a matrix form known as transition probability matrix

and written as

$$P = [P_{ij}]_{n \times n}, \text{ where } 0 < P_{ij} < 1$$

All the possible states of the Markov Chain are used as rows and columns, so transition probability matrix is always a square matrix and the row sum is always one.

Properties:

Two states are said to communicate with each other if both are reachable from one another by a sequence of transitions that have positive probability. This is an equivalence relation which yields a set of communicating classes. A class is closed if the probability of leaving the class is zero. A Markov chain is irreducible if there is one communicating class if not the case, then chain is considered as reducible.

A state i is said to be transient if, starting from i , there is a non-zero probability that the chain will never return to i . It is called recurrent (or persistent) otherwise.

A state i has period k if k is the greatest common divisor of the number of transitions by which i can be reached, starting from i . That is:

$$K = \gcd \{n > 0 : P(X_n = i \mid X_0 = i) > 0\}$$

Periodicity, transience, recurrence are class properties — that is, if one state has the property then all states in its communicating class have the property.

A state i is called absorbing if there are no outgoing transitions from the state.

A state i is said to be ergodic if it is aperiodic and positive recurrent. In other words, a state i is ergodic if it is recurrent, has a period of 1, and has finite mean recurrence time. If all states in an irreducible Markov chain are ergodic, then the chain is said to be ergodic.

Stochastic Process Analysis

Objective:

By modeling the whole sale price index of main three categories in WPI data as a Markov chain we can analyze how the Price index changes over time and how likely it is to move from one state to another.

Since the system changes slowly, therefore we have defined accordingly 10 States to capture the long-term behavior of the system.

Where each state represents a different range of values for the price index

We are using past 10 years monthly data to estimate the transition probabilities between states and use this model to predict future probabilities.

Model development for PRIMARY ARTICLE category in WPI:

The price index of primary article category is divided into several interval using frequency distribution. These intervals are distinct and non- overlapping.

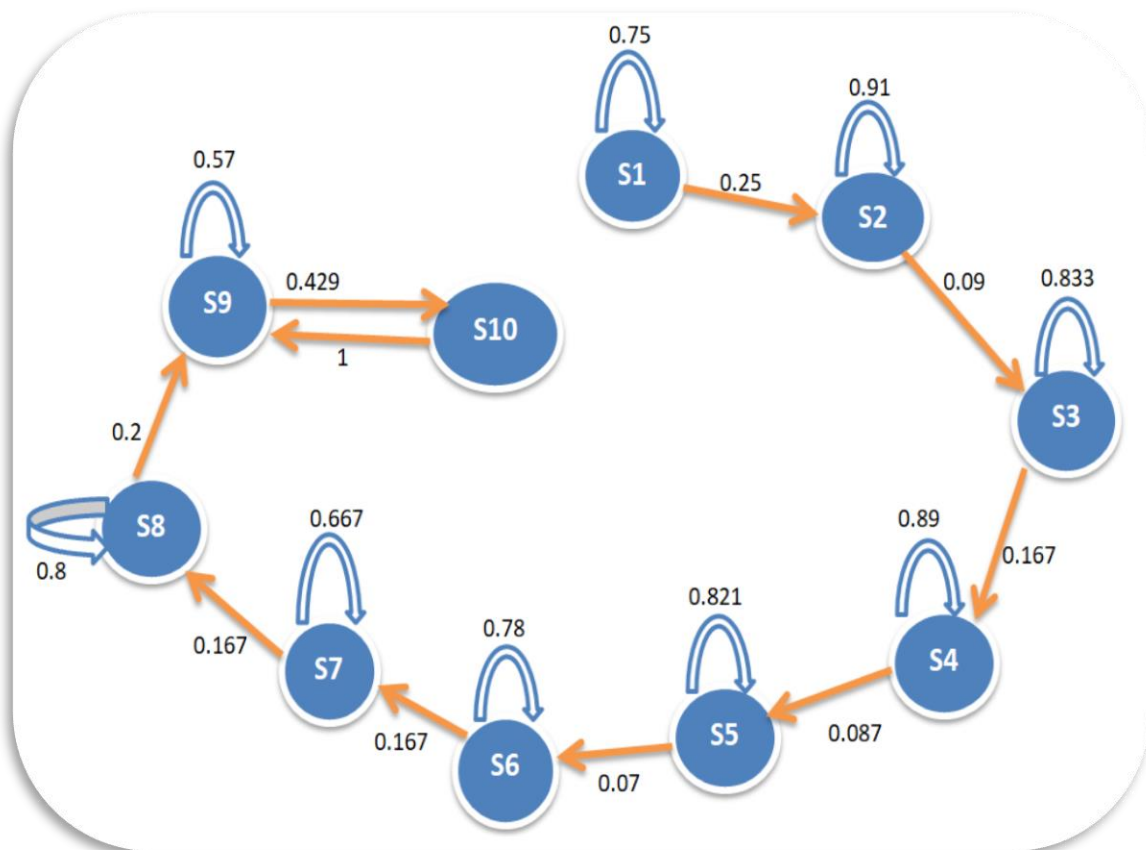
The state space of Markov chain is as follows:

States	Price index range for each state
State 1	91-100
State 2	100-110
State 3	110-120
State 4	120-130
State 5	130-140
State 6	140-150
State 7	150-160
State 8	160-170
State 9	170-180
State 10	180-190

Transition probability matrix (TPM):

	s1	s2	s3	s4	s5	s6	s7	s8	s9	s10
s1	0.75	0.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
s2	0.00	0.91	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00
s3	0.00	0.00	0.83	0.17	0.00	0.00	0.00	0.00	0.00	0.00
s4	0.00	0.00	0.02	0.89	0.09	0.00	0.00	0.00	0.00	0.00
s5	0.00	0.00	0.00	0.11	0.82	0.07	0.00	0.00	0.00	0.00
s6	0.00	0.00	0.00	0.00	0.06	0.78	0.17	0.00	0.00	0.00
s7	0.00	0.00	0.00	0.00	0.00	0.17	0.67	0.17	0.00	0.00
s8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	0.20	0.00
s9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.57	0.43
s10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00

State transition diagram:



Short summary

Primary Article Markov Chain that is composed by:

10 states and 9 classes

One closed class {s9,s10}

Transient classes:

{s1},{s2},{s3},{s4},{s5},{s6},{s7},{s8}

Recurrent class:

{s9,s10}

The above Markov chain is Reducible

Overall description:

From state transition diagram it seems that State i , where $i=1,2,3,4,5,6,7,8$ is Transient state in our context it means that once price index moves to a higher range or next state it is unlikely to move back to a lower range or previous state.

The Probability of each state remaining in its current range is high this suggest that states are stable and it may take a long time to leave this state as the probability of transitioning from one state to another is low indicating that it is relatively unlikely to make a sudden transition.

State 1 to 8 are transient since it is impossible to return to these states after having left them, each of these states form a single class {s1},{s2},{s3},{s4},{s5},{s6},{s7},{s8}

state 9 and 10 communicate since state 9 is accessible from 10 and similarly 10 is accessible from 9 so the state 9 and 10 together forms the class.

Also, can be noticed that set {s9, s10} is close set as cannot go from state 9 to anywhere except state 10 likewise is similar for state 10.

In context of WPI we can say that if WPI value enters closed set it means it cannot transition out it will remain within that set Indefinitely.

Probability of returning from state 10 to state 9 is 1 indication of sudden transition in the next time step.

Henceforth Markov chain is reducible and non-ergodic implies Stationary Distribution does not exist.

But set {9,10} form single closed subset of reducible Markov chain therefore it can be referred to as an irreducible sub-Markov chain as well as aperiodic and recurrent implies sub-Markov chain is ergodic.
we can find stationary distribution for this sub-Markov chain separately.

Tpm of Sub Markov chain

	state9	state10
state9	0.57	0.43
state10	1.00	0.00

Prediction for next 8 months ahead:

The initial state vector is derived by observing the last month price index and by using this vector we estimate the highest probability of next month.

dec=c(0,1)

jan

	state9	state10
[1,]	0.57	0.43

feb

	state9	state10
[1,]	0.7549	0.2451

mar

	state9	state10
[1,]	0.675393	0.324607

apr

	state9	state10
[1,]	0.709581	0.290419

may

	state9	state10
[1,]	0.6948802	0.3051198

june

	state9	state10
[1,]	0.7012015	0.2987985

july

	state9	state10
[1,]	0.6984833	0.3015167

aug

	state9	state10
[1,]	0.6996522	0.3003478

From the above prediction It seems that the state remain same (s9) has the highest probability for next 8 month implies that the price index of primary article category will remain within range of 170-180 with average chance of 70% whereas price index will move or transition within range 180-190 i.e., state 10 has 30% chance on an average.

VALIDATION

Month	P(State 9) (170-180)	P(State 10) (180-190)	Predicted State	Actual Price index
Jan-23	0.57	0.43	State9	174.3
Feb-23	0.7549	0.2451	State9	173
Mar-23	0.67539	0.324607	State9	175
Apr-23	0.709581	0.290419	State9	177.3
May-23	0.6948802	0.3051198	State9	-
Jun-23	0.7012015	0.2987985	State9	-
Jul-23	0.6984833	0.3015167	State9	-
Aug-23	0.6996522	0.3003478	State9	-

The predicted range for price index is almost indistinguishable with actual price index noted down for recent 4 months

State of equilibrium or steady state

state9 state10
[1,] 0.6993007 0.3006993

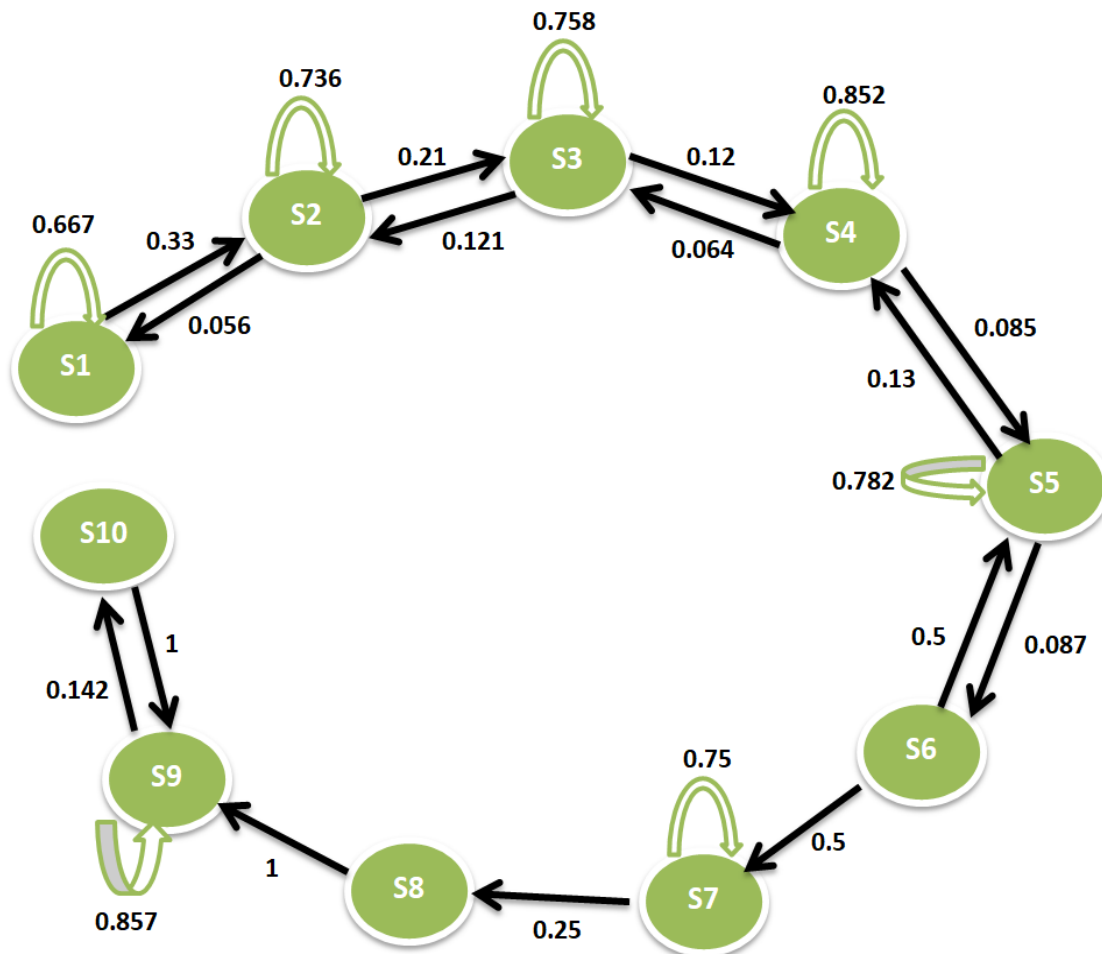
Long term behaviour of the sub-Markov chain implies that irrespective of its initial state price index of primary article will attain the state 9 with probability 0.699 and it will attain state 10 with probability of 0.3006 .Thus, it can be said that there is approximately 70% chance of price index to lie within range of 170-180 and chance of 30% in higher range of 180-190 regardless of its present price index.

The Defined state space for category fuel and power is as follows:

States	Price index range for each state
State 1	70-80
State 2	80-90
State 3	90-100
State 4	100-110
State 5	110-120
State 6	120-130
State 7	130-140
State 8	140-150
State 9	150-160
State 10	160-170

[illegible]

State transition diagram:



Short summary :

FUEL AND POWER Markov chain that is composed by:

10 states and 4 classes

Closed classes:

s9 s10

Recurrent classes:

{s9,s10}

Transient classes:

{s1,s2,s3,s4,s5,s6},{s7},{s8}

The Markov chain is not irreducible

The absorbing states are: NONE

Overall description:

S2 is accessible from s1, s3 from s2 and so on upto s6 is accessible

from s5

$S1 \rightarrow S2 \rightarrow S3 \rightarrow S4 \rightarrow S5 \rightarrow S6$

$S1 \leftarrow S2 \leftarrow S3 \leftarrow S4 \leftarrow S5 \leftarrow S6$

And similar is true in reverse fashion so we can say that all of these states communicate with each other and therefore combined forms a class. State 6 is transient and it is a class property therefore we can say that $\{s1, s2, s3, s4, s5, s6\}$ is a transient class.

If the price index attains the state 6 that is range of 120-130 after one time step it will never be within that range it will either move to higher range that is state 7 of range 130-140 or it will move to lower range of 110-120. For happening of either this or that chance is 50-50 we can even say that state 6 is unstable state.

Each of these states has high probability of being in current range or same state and low probability of transitioning to higher range and comparatively very low for transitioning from current to lower range.

Succeeding state 7 and state 8 forms a single class as well as both states are transient. If the price index attains the state 8 that is range of 140-150 after one time step it will definitely move to higher range of state 9. the state 8 is another unstable state for above Markov chain State 9 and 10 communicates with each other and together form a class and we can say it is closed class as we cannot move out anywhere if price index enters range of either of these 2 states over the time.

As there is more than one class for the above Markov chain hence Markov chain is reducible and stationary distribution does not exist.

The set $\{s9, s10\}$ forms a single closed subset of reducible Markov chain therefore it can be referred to as irreducible sub-Markov chain so we can handle this chain separately as an independent of all other states and can find steady state

This sub-Markov chain is irreducible as well as aperiodic and recurrent this satisfies all the condition to be ergodic where ergodicity implies that the chain has unique stationary distribution, which is probability distribution over the states that remain invariant over time.

Tpm of sub-Markov chain:

	state9	state10
state9	0.86	0.14
state10	1.00	0.00

Prediction for next 8 months ahead:

The initial state vector is derived by observing the last month price index and by using this vector, we estimate the highest probability of next month.

dec=c(0,1)

```

jan
      state9 state10
[1,]    0.86    0.14

feb
      state9 state10
[1,] 0.8796 0.1204

mar
      state9 state10
[1,] 0.876856 0.123144

apr
      state9 state10
[1,] 0.8772402 0.1227598

may
      state9 state10
[1,] 0.8771864 0.1228136

june
      state9 state10
[1,] 0.8771939 0.1228061

july
      state9 state10
[1,] 0.8771929 0.1228071

aug
      state9 state10
[1,] 0.877193 0.122807

```

From the above prediction It seems that the state remain same (s9) has the highest probability for next 8 month implies that the price index of fuel and power category will remain within range of 150-160 with average chance of 87.71% whereas price index will move or transition within range of 180-190 i.e., state 10 has 12.28% chance on an average.

VALIDATION:

Month	P(State 9) (150-160)	P(State 10) (160-170)	Predicted State	Actual Price index
Jan-23	0.86	0.14	State9	155.6
Feb-23	0.8796	0.1204	State9	158.8
Mar-23	0.876856	0.123144	State9	156.8
Apr-23	0.8772402	0.1227598	State9	152.6
May-23	0.8771864	0.1228136	State9	-
Jun-23	0.8771939	0.1228061	State9	-
Jul-23	0.8771929	0.1228071	State9	-
Aug-23	0.877193	0.122807	State9	-

The predicted range for price index is almost indistinguishable with actual price index noted down for recent 4 months.

State of equilibrium or steady state

```

state9 state10
[1,] 0.877193 0.122807

```

Long term behaviour of the sub-Markov chain implies that irrespective of its initial state price index of fuel and power category will attain the state 9 with probability 0.8772 and it will attain state 10 with probability of 0.1228.

MANUFACTURING:

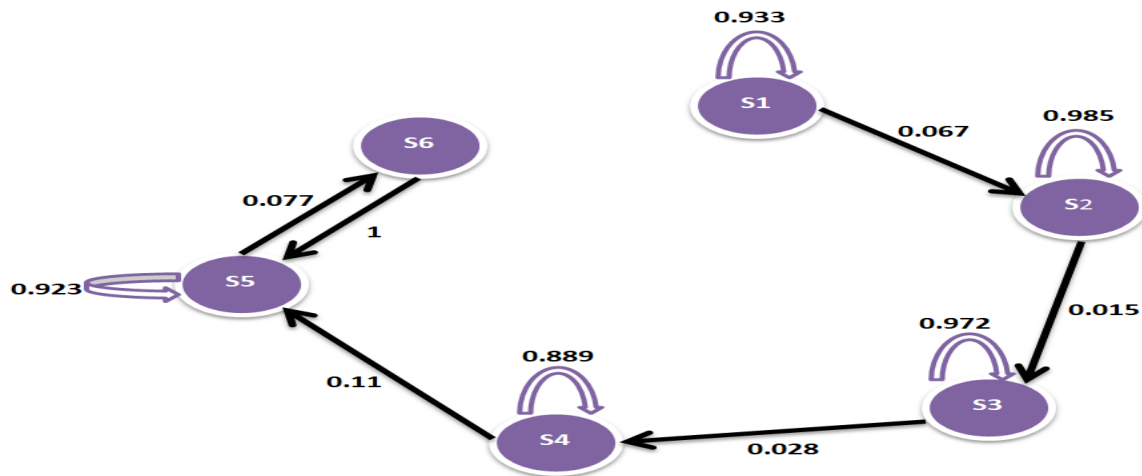
The Defined state space for Manufacturing Category is as follows:

States	Price index range for each state
State 1	95-105
State 2	105-115
State 3	115-125
State 4	125-135
State 5	135-145
State 6	145-155

Transition probability matrix (TPM):

	s1	s2	s3	s4	s5	s6
s1	0.9333333	0.06666667	0.00000000	0.00000000	0.00000000	0.00000000
s2	0.00000000	0.98484848	0.01515152	0.00000000	0.00000000	0.00000000
s3	0.00000000	0.00000000	0.97222222	0.02777778	0.00000000	0.00000000
s4	0.00000000	0.00000000	0.00000000	0.88888889	0.11111111	0.00000000
s5	0.00000000	0.00000000	0.00000000	0.00000000	0.9230769	0.07692308
s6	0.00000000	0.00000000	0.00000000	0.00000000	1.00000000	0.00000000

State Transition Diagram:



Short summary:

Manufacturing Markov Chain that is composed by:

6 states and 5 classes

Closed classes:

s5 s6

Recurrent classes:

{s5,s6}

Transient classes:

{s1},{s2},{s3},{s4}

The Markov chain is not irreducible

The absorbing states are: NONE

Overall description:

For the above Markov chain, the transient states are s1, s2, s3 and s4 since it seems impossible to return any of these states after having left them. Each of these state forms a single class {s1}, {s2}, {s3}, {s4} and state 5 and 6 communicate with each other and since state 5 is recurrent and aperiodic and recurrence and periodicity is class property that is similar properties is true for state6 therefore the combined state5 and 6 {s5,s6} can be consider as class these formation of separate classes in Markov chain implies that the Markov chain is reducible so stationary distribution does not exist.

There is a very high probability for each state except state6 to remain within its current range and low probability of transitioning to higher range or next state in one time step.

By considering this we can say that price index of manufacturing category shows a stable trend over the time.

The set $\{s_5, s_6\}$ also forms a closed class representing the price index range from 135-145 and 145-155. This suggests that once the price index enters either of these range it tends to remain within that range over time.

The set $\{s_5, s_6\}$ forms a single closed subset of reducible Markov chain therefore it can be referred to as irreducible sub-Markov chain so we can handle this chain separately as an independent of all other states and can find steady state.

This sub-Markov chain is irreducible as well as aperiodic and recurrent this satisfies all the condition to be ergodic where ergodicity implies that the chain has unique stationary distribution, which is probability distribution over the states that remain invariant over time.

Tpm of sub-Markov chain:

	state5	state6
state5	0.9230769	0.07692308
state6	1.0000000	0.00000000

Prediction for next 8 months ahead:

The initial state vector is derived by observing the last month price index and by using this vector, we estimate the highest probability of next month.
dec=c(0,1)

jan	state5	state6
[1,]	0.9230769	0.07692308

feb	state5	state6
[1,]	0.928994	0.07100592

mar	state5	state6
[1,]	0.9285389	0.07146108

apr	state5	state6
[1,]	0.9285739	0.07142607

may

```

state5 state6
[1,] 0.9285711 0.07142876

```

```

june

```

```

state5 state6
[1,] 0.9285713 0.07142855

```

```

july

```

```

state5 state6
[1,] 0.9285713 0.07142857

```

```

aug

```

```

state5 state6
[1,] 0.9285713 0.07142856

```

It seems that the state remains same i.e., the price index lies within range of 135-145 has very high probability and very low probability of transitioning or moving to higher range of 145-155 for next 8 months as well as it can be noted that this probability distribution does not differ too much for succeeding months

VALIDATION:

Month	P(State 5) (135-145)	P(State 6) (145-155)	Predicted State	Actual Price index
Jan-23	0.9230769	0.07692308	State 5	141.4
Feb-23	0.928994	0.07100592	State 5	141.6
Mar-23	0.9285739	0.07142607	State 5	141.2
Apr-23	0.9285739	0.07142607	State 5	141.2
May-23	0.9285711	0.07142876	State 5	-
Jun-23	0.9285713	0.07142855	State 5	-
Jul-23	0.9285713	0.07142857	State 5	-
Aug-23	0.9285713	0.07142856	State 5	-

The predicted range for price index is almost indistinguishable with actual price index noted down for recent 4 months.

State of equilibrium or steady state

	state5	state6
[1,]	0.9285714	0.07142858

Long term behaviour of the sub-Markov chain implies that irrespective of its initial state price index of manufacturing category will attain the state 5 with probability 0.9285 and it will attain state 6 with probability of 0.0714 .

Methodology for Time Series Analysis

Holt-Winters method

The Holt-Winters model, also known as the **Triple Exponential Smoothing** model, is a popular time series forecasting technique that incorporates trend, seasonality, and level components into the forecast. It is particularly useful when dealing with data that exhibits both trend and seasonality. The method is named after its developers, Charles C. Holt and Peter Winters.

The Holt-Winters method involves three components: level, trend, and seasonality.

1. **Level:** The level component represents the average value of the series at a given point in time. It can be thought of as the baseline or central tendency of the data.
2. **Trend:** The trend component captures the direction and rate of change in the series over time. It indicates whether the data is increasing, decreasing, or remaining relatively stable.
3. **Seasonality:** The seasonality component refers to repeating patterns or cycles that occur within the data at fixed intervals. It accounts for periodic fluctuations in the series, such as weekly, monthly, or yearly patterns.

The Holt-Winters method involves updating and forecasting these three components iteratively. There are two main variations of the method: additive and multiplicative.

1. **Additive Model:** In the additive model, the components are combined by adding them together. The forecast is calculated by adding the level, trend, and seasonality components.
2. **Multiplicative Model:** In the multiplicative model, the components are multiplied together. The forecast is obtained by multiplying the level, trend, and seasonality components.

Parameters in the Holt-Winters method:

1. **Alpha (α):** It represents the smoothing parameter for the level component. It controls the weight given to the most recent observation in updating the level. A smaller α value gives less weight to recent observations, resulting in a smoother level component.
2. **Beta (β):** It is the smoothing parameter for the trend component. It determines the weight given to the most recent trend observation. Similar to alpha, a smaller β value results in a smoother trend component.

3. Gamma (γ): Gamma is the smoothing parameter for the seasonality component. It controls the weight given to the most recent seasonal observation. A smaller γ value leads to a smoother seasonality component.

By combining the level, trend, and seasonality components, the Holt-Winters method provides accurate forecasts for time series data, considering both the overall patterns and short-term fluctuations.

SARIMA MODEL:

The seasonal ARIMA (SARIMA) model is an extension of the ARIMA model that incorporates seasonality in addition to the autoregressive (AR), integrated (I), and moving average (MA) components. It is specifically designed to handle time series data that exhibit seasonal patterns or cyclic behaviour.

The SARIMA model is denoted as SARIMA(p, d, q)(P, D, Q)[s], where:

p: The order of the autoregressive component for the non-seasonal part.

d: The order of differencing required to make the series stationary for the non-seasonal part.

q: The order of the moving average component for the non-seasonal part.

P: The order of the autoregressive component for the seasonal part.

D: The order of differencing required to make the seasonal series stationary.

Q: The order of the moving average component for the seasonal part.

s: The length of the seasonal period.

The SARIMA model combines the non-seasonal and seasonal components to capture both short-term and long-term dependencies in the data. The non-seasonal components (p, d, q) capture the non-seasonal trends and patterns, while the seasonal components (P, D, Q, s) account for the repeated seasonal patterns.

The SARIMA model is typically fitted to time series data using statistical software packages that estimate the model parameters through maximum likelihood estimation or other optimization techniques.

The model parameters can be determined by analyzing the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots to guide the selection of appropriate orders for the AR, MA, seasonal AR, and seasonal MA terms.

The ACF plot shows the correlation between the series and its lagged values at different lags. It helps to identify the potential lagged relationships in the data. If the ACF plot shows significant spikes or decaying patterns outside the confidence interval, it suggests the presence of autocorrelation and provides an indication of the possible order of the MA component.

Also, for fitting the SARIMA model we need a stationary time series, for that we have different tests to check the stationarity.

Augmented Dickey-Fuller (ADF) Test:

The ADF test is used to determine if a time series is stationary or non-stationary. It tests the null hypothesis that the time series has a unit root (i.e., it is non-stationary) against the alternative hypothesis of stationarity.

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

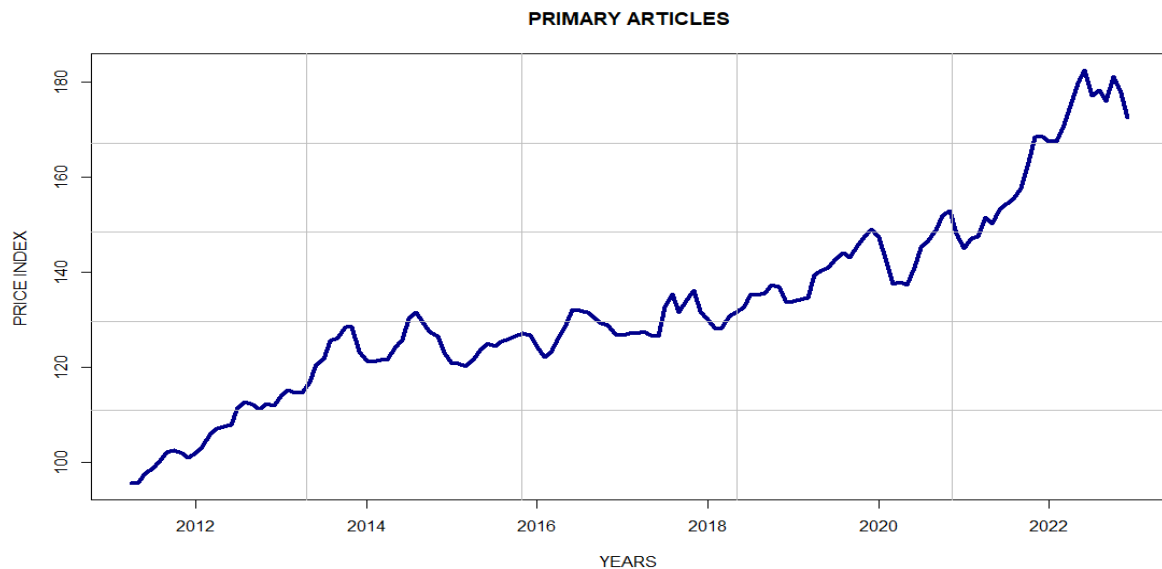
The KPSS test is used to evaluate the null hypothesis that a time series is stationary against the alternative hypothesis of non-stationarity. It tests for the presence of a unit root in the series and assesses the trend stationarity.

The KPSS test is often used in conjunction with the ADF test to assess the stationarity of a time series.

Once the SARIMA model is fitted, it can be used to generate forecasts for future time periods by incorporating both non-seasonal and seasonal patterns observed in the historical data. The SARIMA model is a powerful tool for forecasting time series data with seasonal patterns, enabling the capture and modelling of complex interactions among various data components.

Time Series Analysis

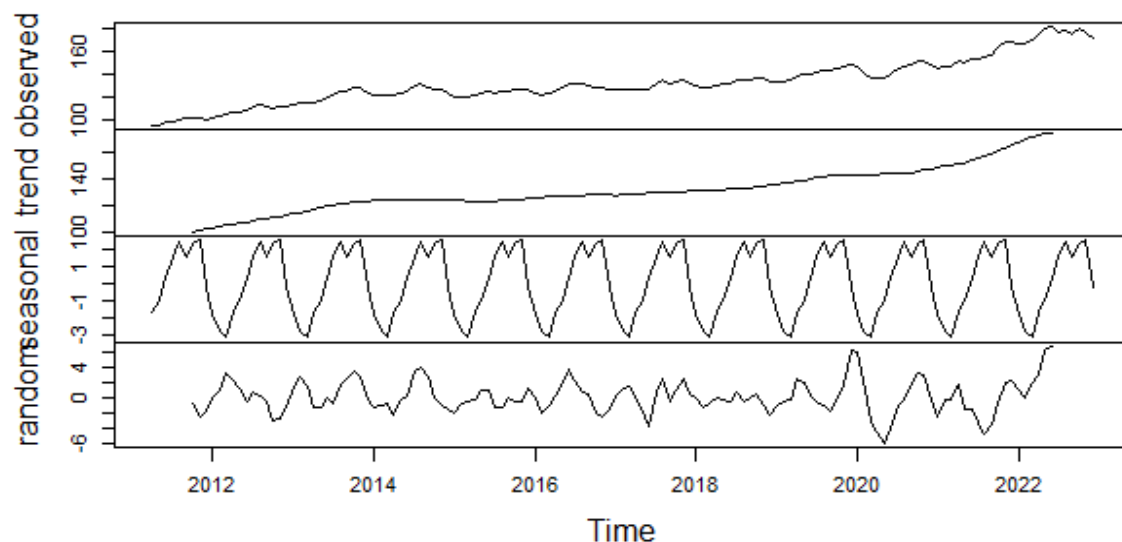
TIME SERIES ANALYSIS FOR PRIMARY ARTICLES



Based on above graph, it can be observed that the series of Primary Articles shows increasing trend with some random fluctuations however, in the middle of financial year 2020-2021 there seems a significant spike.

Decomposition of time series:

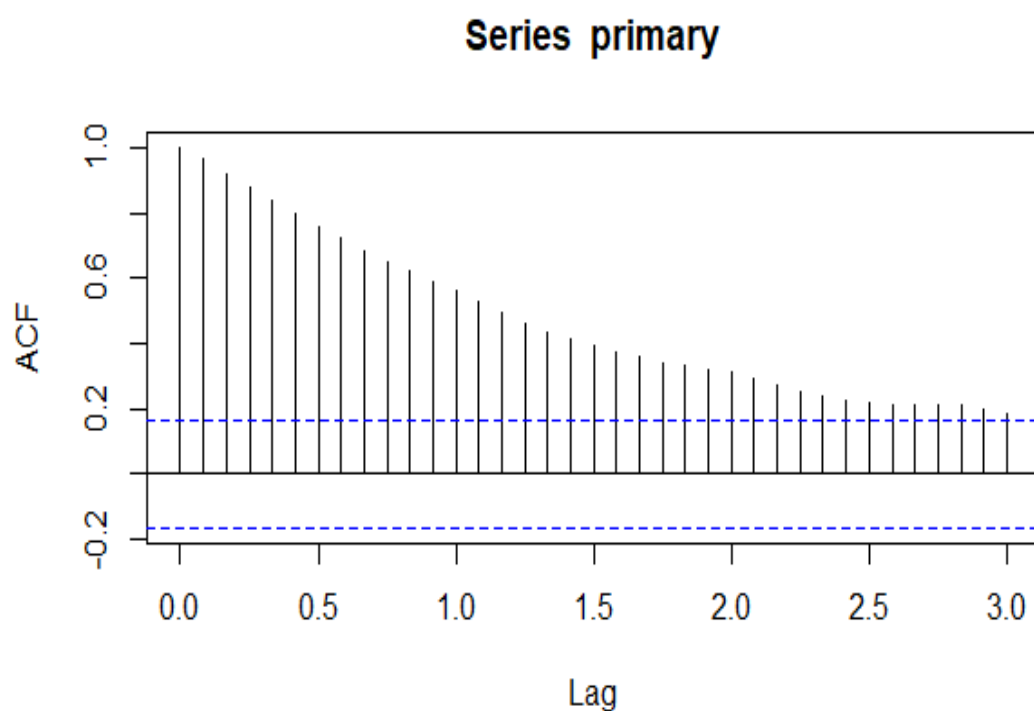
Decomposition of additive time series



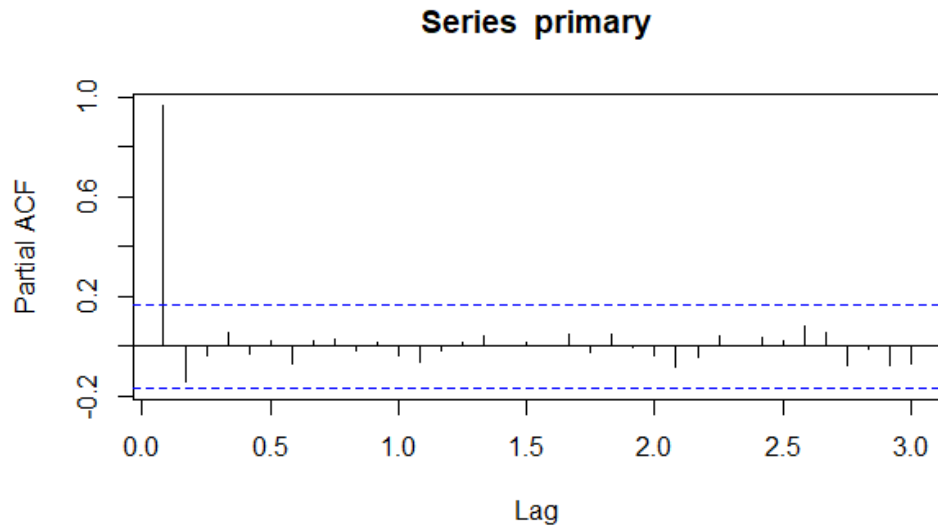
From above decomposition plot we can see that there

- Is a upward trend in series and .
- A distinct seasonal pattern is visible in the middle of the series, characterized by two spikes followed by a decline.
- There are random fluctuations in the series also variation in random component is high after year 2020.

ACF and PACF plots:



From above ACF plot, there is decreasing autocorrelation towards zero as lags are increasing implies series is not stationary.



We can also check the stationarity by using tests as follows :

Augmented Dickey-Fuller Test

Hypothesis:

H0: The series is not stationary

Vs

H1: The series is stationary

Augmented Dickey-Fuller Test

data: primary

Dickey-Fuller = -1.7814, Lag order = 5, **p-value = 0.6674**

alternative hypothesis: stationary

Comment : As the P-value is greater than level of significance $\alpha=0.05$, so we fail to reject the null hypothesis and conclude that series **is Not Stationary**.

KPSS Test :

Hypothesis

H0: The series is stationary

vs

H1: The series is not stationary

KPSS Test for Level Stationarity

data: primary

KPSS Level = 2.5031, Truncation lag parameter = 4, **p-value = 0.01**

Comment : As the p-value is less than level of significance $\alpha=0.05$, we reject the null hypothesis and conclude that series is **not stationary** .

So , to make the series stationary or to detrend the series we need to take differencing of order 1 as we have seen there is an upward trend in the series .

Now , we will check the stationarity after differencing the series .

Augmented Dickey-Fuller Test after differencing :

data: primary_diff

Dickey-Fuller = -5.4611, Lag order = 5, p-value = 0.01

alternative hypothesis: stationary

H0: The series is not stationary

Vs

H1: The series Is stationary

Comment : As the p-value is less than level of significance $\alpha=0.05$, we reject null hypothesis and conclude that series is **stationary** .

Time series models :

We have use holt-winters model as it can handle seasonality and trend in series . Also, model is suitable when the data shows a consistent upward or downward trend over time and also displays repetitive patterns or seasonality at regular intervals.

Forecast method: Holt-Winters' additive method

Model Information:

Holt-Winters' additive method

Call:

hw(y = primary)

Smoothing parameters:

alpha = **0.9999**

beta = **0.023**

gamma = 1e-04

Initial states:

l = 102.7666

b = 0.7534

s = -3.3468 -3.1349 -1.8719 0.0721 2.9889 2.6373

1.4995 2.1387 1.5438 0.6193 -1.066 -2.0799

sigma: 2.1362

AIC AICc BIC

928.8429 933.8186 978.9719

Error measures:

ME RMSE MAE MPE MAPE MASE ACF1

Training set -0.04039968 **2.011382** **1.545349** -0.05562944 1.165462 0.2040847 0.2224543

Interpretation: In above model alpha is larger (0.9999) nearly equals to one means it uses more weights to recent observation making forecast more responsive to recent observation , and beta =0.023 shows less weight to recent trend observation as beta is smoothing parameter to trend component , this means that recent trend changes have less influence on the forecast .

Also , holt winter model has mean absolute error of 1.5453 , which means on average, the forecasted values differ from the actual values by approximately 1.5453 units.

- After exploring various parameter combinations for the Seasonal ARIMA model, the model with the lowest values of AIC , RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error) is selected

Seasonal ARIMA models	RMSE	MAE
(0,1,1)(2,0,0)[12] with drift	2.076	1.52
(0,1,1)(0,1,1)[12]	4.44	1.93
(1,1,0)(2,2,1)[12]	8.46	2.75
(0,1,1)(0,1,2)[12]	4.44	1.92

Hence, the model with lowest value of RMSE and MAE is (0,1,1)(2,0,0)[12] with drift.

Seasonal ARIMA model :

Series: primary

ARIMA(0,1,1)(2,0,0)[12] with drift

Coefficients:

```

      ma1 sar1 sar2 drift
      0.2609 0.1681 0.2695 0.5904
s.e. 0.0787 0.0878 0.0936 0.3581

```

sigma² = 4.466: log likelihood = -302.64
 AIC=615.28 AICc=615.73 BIC=629.99

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-0.0116694	2.075472	1.522449	-0.01705734	1.110918	0.2010605	0.00850476

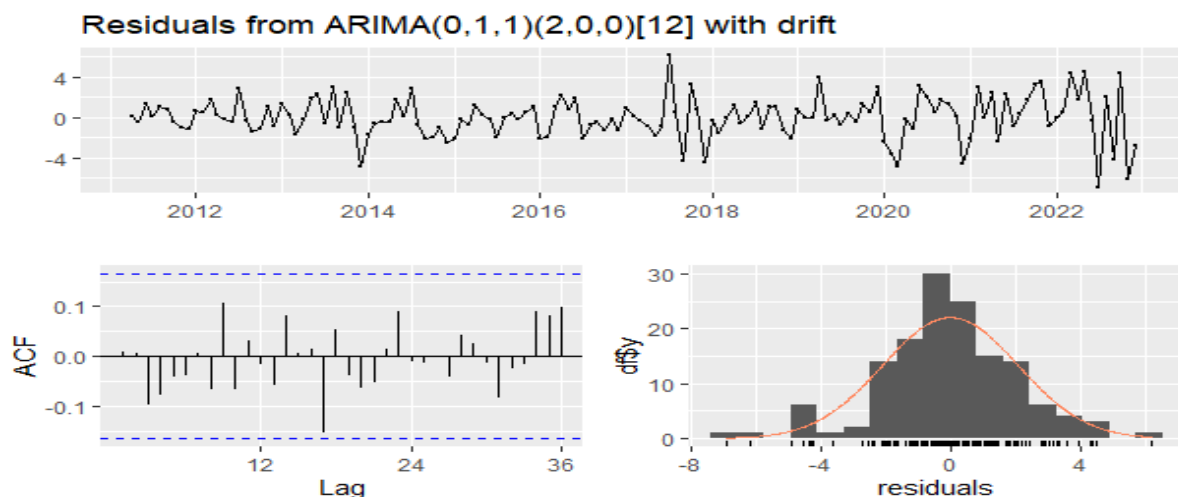
Interpretation: In above model ARIMA(0,1,1) represents the non-seasonal component: no autoregressive term, first-order differencing, and a moving average term of order 1. (2,0,0)[12] represents the seasonal component: two seasonal autoregressive terms, no seasonal differencing, and no seasonal moving average terms with a period of 12 (i.e monthly data).

In our model drift term is present means it will fluctuates around trend specified as 0.5904 means that the model includes a linear trend component that increases or decreases by an average of 0.5904 units per time step.

Also , Seasonal ARIMA model has mean absolute error of 1.5225 , which means on average, the forecasted values differ from the actual values by approximately 1.5225 units.

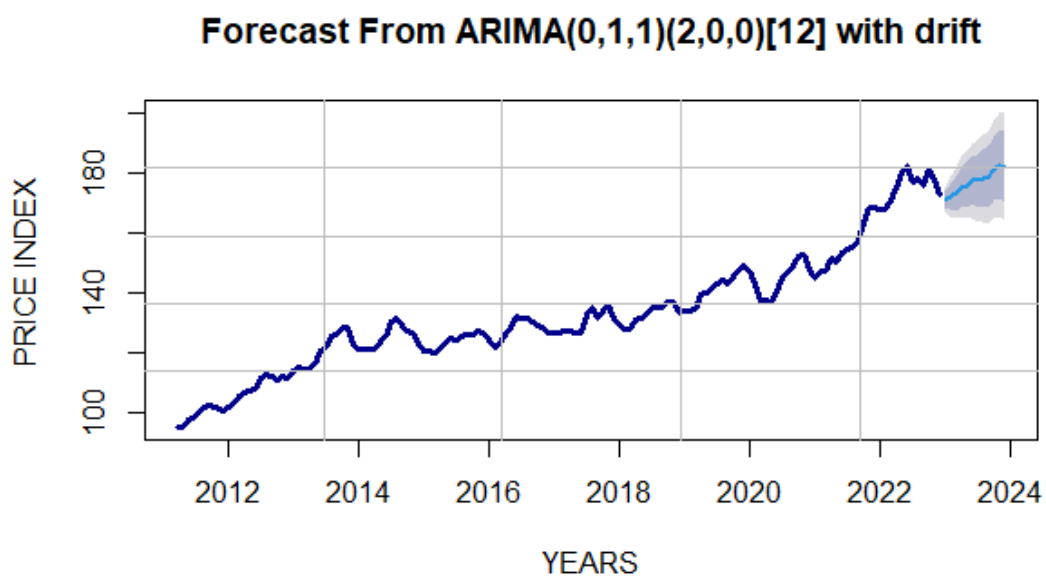
From above Two models RMSE , MAE and AIC, BIC is less for seasonal ARIMA model , so we will choose this model for forecasting .

Diagnostic Measure :



- From above plot we can see that plot of residuals are stationary.
- There is no significant autocorrelation in residuals ,means there is no information left in the residuals.
- Histogram for residuals is bell shaped means residuals are normally distributed with zero mean , if residuals have zero mean then the forecasts are unbiased .

Forecast Using seasonal ARIMA model :



Point Forecasts and Inflation table :

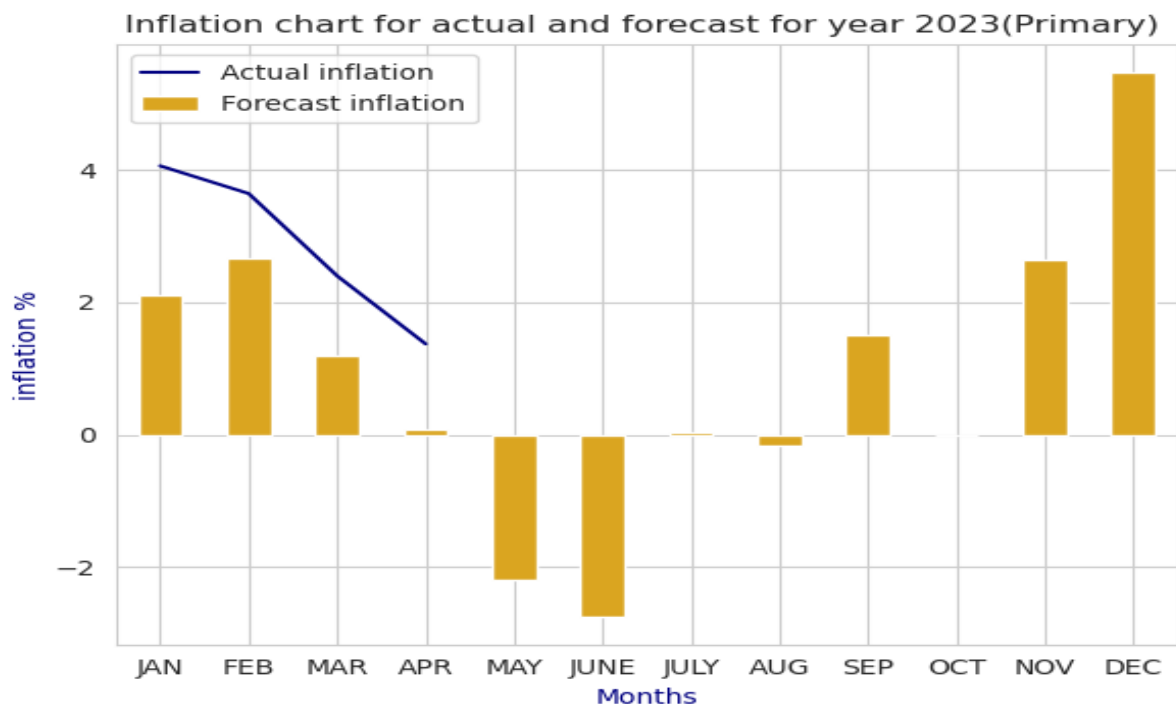
Month	Forecast Price Index	Actual Price Index	Forecast Inflation	Actual Inflation
Jan-23	171.0321	174.3	2.108692	4.059701
Feb-23	171.9570	173.6	2.660891	3.641791
Mar-23	172.9414	175.0	1.194519	2.399064
Apr-23	175.0507	177.3	0.086182	1.372213
May-23	175.8561	----	-2.19347	----
Jun-23	177.3798	----	-2.7523	----
Jul-23	177.1713	----	0.040276	----
Aug-23	178.0015	----	-0.16739	----
Sep-23	178.5500	----	1.506546	----
Oct-23	181.2012	----	0.000663	----
Nov-23	182.4001	----	2.644985	----
Dec-23	181.8413	----	5.476419	----

From above values of Actual price index and Forecasted price index we have **RMSE = 2.380957** and **MAE= 2.304696** , which means on average, the forecasted values differ from the actual values by approximately 2.304696 units.

Validation:

From the stochastic model for Primary Articles Category, we have the result as there is a high chance (70%) that price index will lie within the range of 170-180 and 30% chance that it will lie within range 180-190 , Also from our seasonal ARIMA model we have the forecasts for next 12 month and price index are in the range as predicted from stochastic model .

Inflation Chart for the year 2022-23 Using Forecast Data of Primary Articles

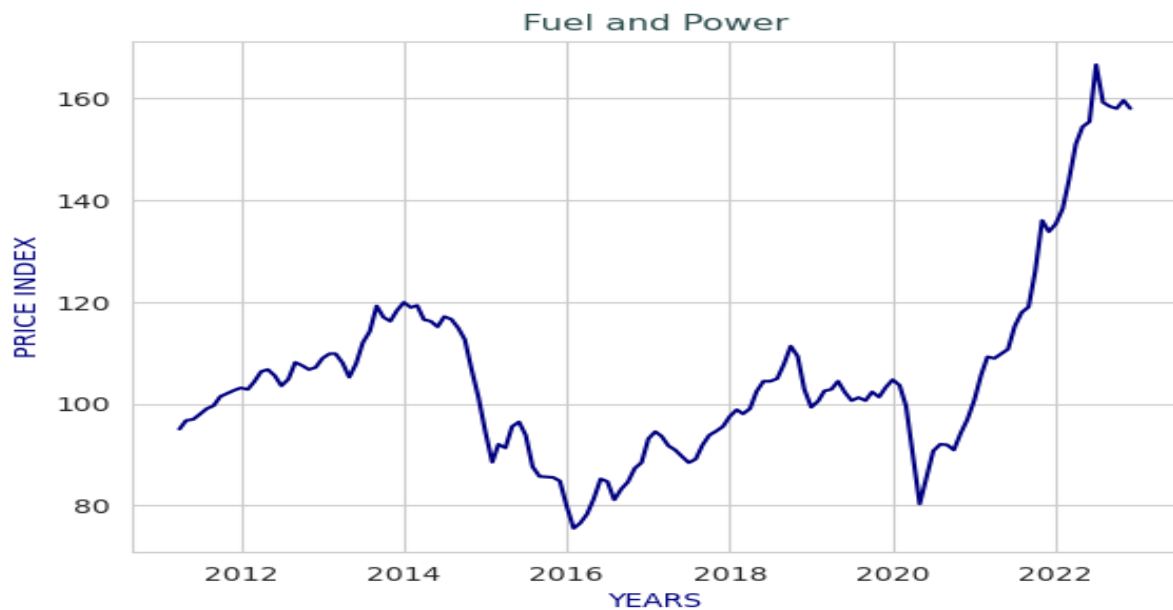


Based on the forecasted and actual inflation data for the year 2022-2023, we can observe the following points:

In January, there was a 4% actual increase in prices compared to the previous year. However, according to our forecasted price index, we anticipated a 2.10% increase in prices for the same period. This indicates that the actual inflation exceeded our initial forecast.

After January, the inflation rates started to decline. Our forecasted price index suggests a continuous decrease in inflation rates from February onwards.

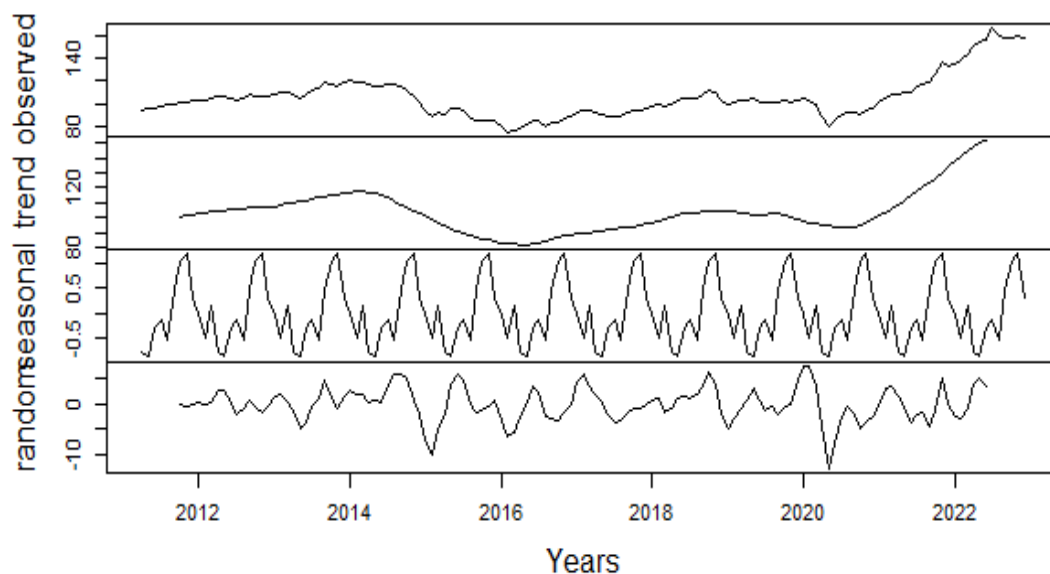
TIME SERIES ANALYSIS FOR FUEL & POWER



The graph shows a significant decrease in the price index for the Fuel and Power category between 2014 and 2016. In early 2016, the price index reached its lowest point, falling below 80. After that, there was a gradual increase in the price index, followed by another decline in 2020. However, from May 2020 onwards, there was a continuous upward trend in the price index, which lasted until 2022.

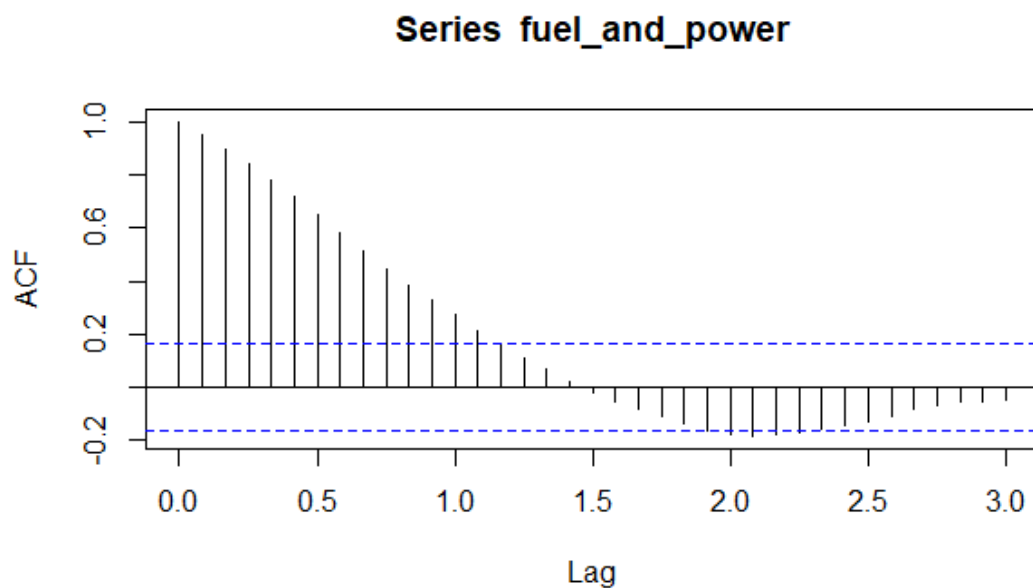
Decomposition of time series :

Decomposition of additive time series

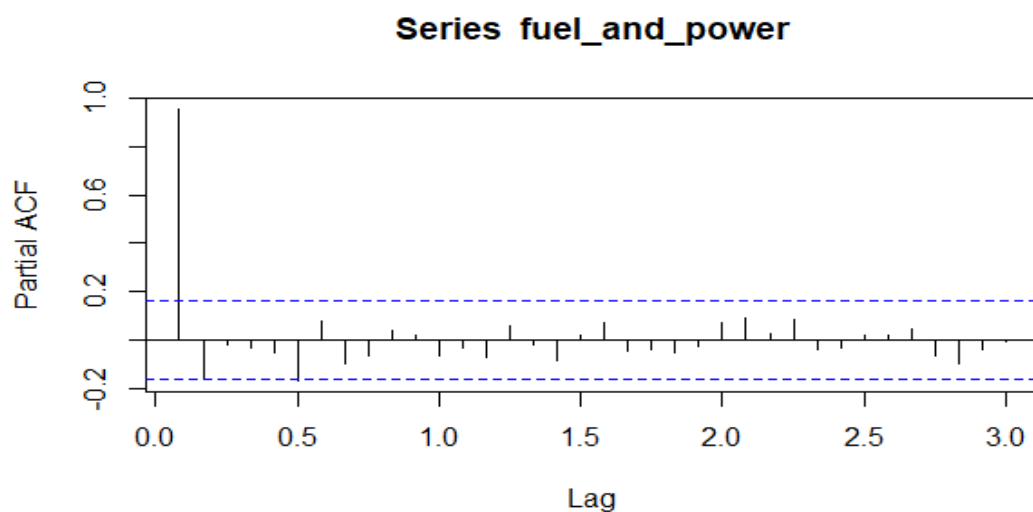


- From above decomposition of time series of fuel and power we can trend is not linear ,there are ups and downs in series followed by an increasing trend from May 2020.
- From the random component we can see variation is high between year 2015 to 2016 and 2020 to 2021 .
- Overall, we can say there is a significant volatility in the data.

ACF and PACF plots:



From Above ACF plot we can see there is decreasing autocorrelation towards zero as lags are increasing, implies that series is not stationary.



We can also check the stationarity by using tests as follows:

Augmented Dickey-Fuller Test

Hypothesis:

H0: The series is not stationary

Vs

H1: The series is stationary

Augmented Dickey-Fuller Test

data: fuel_and_power

Dickey-Fuller = -0.83464, Lag order = 5, **p-value = 0.9563**

alternative hypothesis: stationary

Comment: As the P-value is greater than level of significance $\alpha=0.05$, so we fail to reject the null hypothesis and conclude that series is **Not Stationary**.

KPSS Test :

Hypothesis

H0: The series is stationary

vs

H1: The series is not stationary

KPSS Test for Level Stationarity

data: t

KPSS Level = 0.67419, Truncation lag parameter = 4, **p-value = 0.01589**

Comment : As the p-value is less than level of significance $\alpha=0.05$, we reject the null hypothesis and conclude that series is **not stationary** .

So , to make the series stationary or to detrend the series we need to take differencing of order 1 as we have seen there is a upward and downward trend in the series .

Now , we will check the stationarity after differencing the series .

Augmented Dickey-Fuller Test

Hypothesis:

H0: The series is not stationary

Vs

H1: The series is stationary

Augmented Dickey-Fuller Test

data: diff(fuel_and_power, 1)

Dickey-Fuller = -3.6982, Lag order = 5, **p-value = 0.02695**

alternative hypothesis: stationary

Comment : As the p-value is less than level of significance $\alpha=0.05$, we reject null hypothesis and conclude that series is **stationary** .

Time series models:

We have use holt-winters model as it can handle seasonality and trend in series. Also, model is suitable when the data shows a consistent upward or downward trend over time and displays repetitive patterns or seasonality at regular intervals.

Forecast method: Holt-Winters' additive method

Model Information:

Holt-Winters' additive method

Call:

hw(y = fuel_and_power)

Smoother parameters:

alpha = **0.999**

beta = **0.0942**

gamma = 2e-04

Initial states:

l = 104.6988

b = 0.7731

s = 0.1626 -0.4221 -0.3103 0.4955 1.5274 1.2608
 0.3719 -0.6987 0.2572 -0.833 -0.8224 -0.9889

sigma: 3.3584

AIC AICc BIC
 1056.424 1061.399 1106.552

Error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	0.06302503	3.162097	2.485793	0.07370278	2.419759	0.1787642
ACF1						
		0.2735137				

Interpretation: In above model alpha is larger (0.9999) nearly equals to one means it uses more weights to recent observation making forecast more responsive to recent observation , and beta =0.0942 shows less weight to recent trend observation as beta is smoothing parameter to trend component , This means that recent trend changes have less influence on the forecast .

Also , holt winter model has mean absolute error of **2.485793** , which means on average, the forecasted values differ from the actual values by approximately **2.485793** units.

After exploring various parameter combinations for the Seasonal ARIMA model, the model with the lowest values of AIC , RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error) is selected.

Seasonal ARIMA models	RMSE	MAE
(1,1,0)(0,0,2)[12]	2.89	2.08
(1,1,0)(0,1,1)[12]	4.95	2.65
(1,1,0)(0,2,2)[12]	8.76	3.65

Hence , model with lowest RMSE and MAE value is (1,1,0)(0,0,2)[12]

Seasonal ARIMA model :

Series: fuel_and_power

ARIMA(1,1,0)(0,0,2)[12]

Coefficients:

```

      ar1   sma1   sma2
      0.3441 0.1580 -0.1624
s.e. 0.0805 0.0955 0.0973

```

sigma^2 = 8.633: log likelihood = -348.63

AIC=705.26 AICc=705.56 BIC=717.03

Training set error measures:

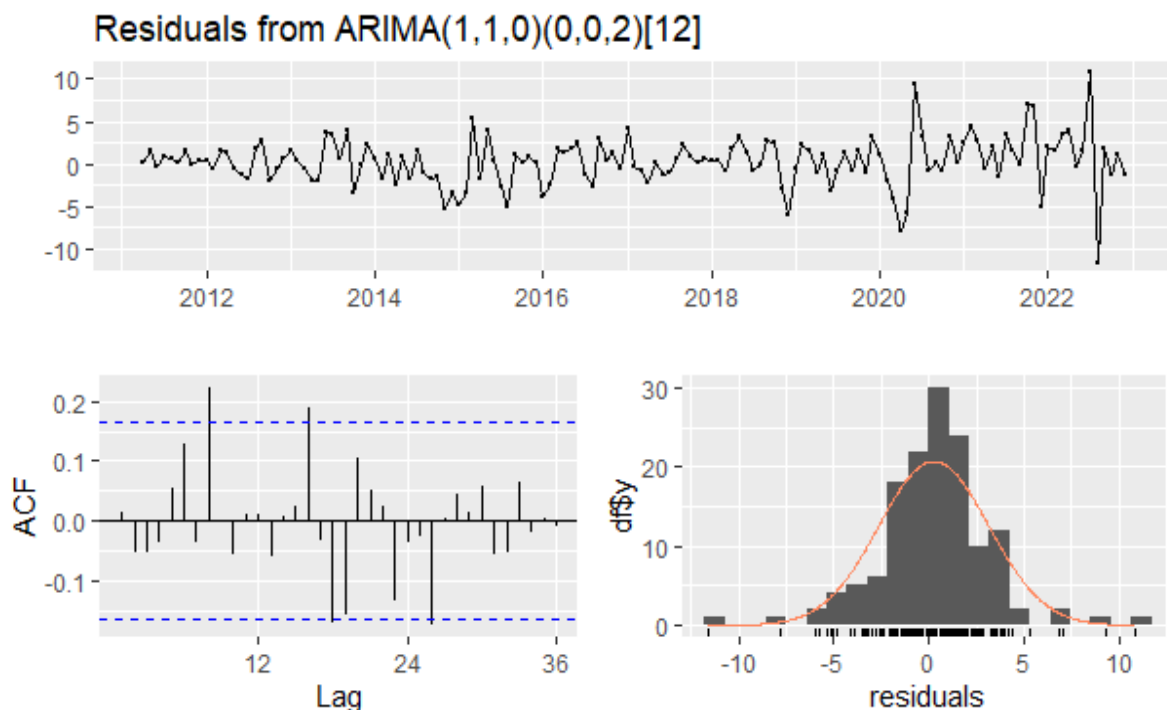
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.262693	2.896282	2.081203	0.2035696	1.981399	0.1496684	0.01315343

Interpretation: In the ARIMA(1,1,0)(0,0,2)[12] model ARIMA(1,1,0) represents non seasonal component and the autoregressive term (AR) of 1 implies that the current value is related to the previous value,. The differencing term (d) of 1 indicates that the series has been differenced once to achieve stationarity. The model does not include any moving average terms (MA) but includes two seasonal MA terms (Q).

Also , Seasonal ARIMA model has mean absolute error of **2.081203** , which means on average, the forecasted values differ from the actual values by approximately **2.081203** units.

From above Two models RMSE , MAE and AIC, BIC is less for seasonal ARIMA model , so we will choose this model for forecasting .

Diagnostic Measure:



The Ljung-Box test for checking the autocorrelation of residuals from seasonal ARIMA model

Hypothesis:

H_0 there is no autocorrelation in the residuals.

V_s

H_1 : There is significant autocorrelation in residuals .

Box-Ljung test

data: residuals

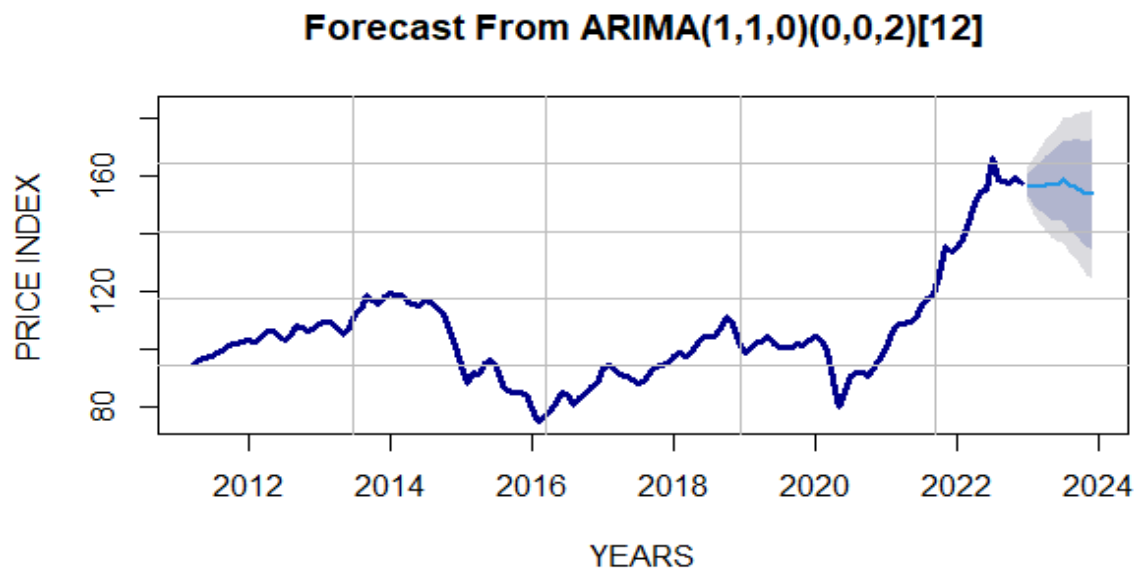
X-squared = 0.024918, df = 1, **p-value = 0.8746**

Comment : As p-value is greater than level of significance $\alpha=0.05$ we fail to reject the null hypothesis and conclude that **there is no autocorrelation in the residuals.**

- From the above test we get there is no autocorrelation in the residuals
- As residuals are stationary ,there is no information left in the residuals .

- Histogram for residuals is bell shaped means residuals are normally distributed with zero mean , if residuals have zero mean, then the forecasts are unbiased.

Forecast Using seasonal ARIMA model:



Point Forecasts and inflation table :

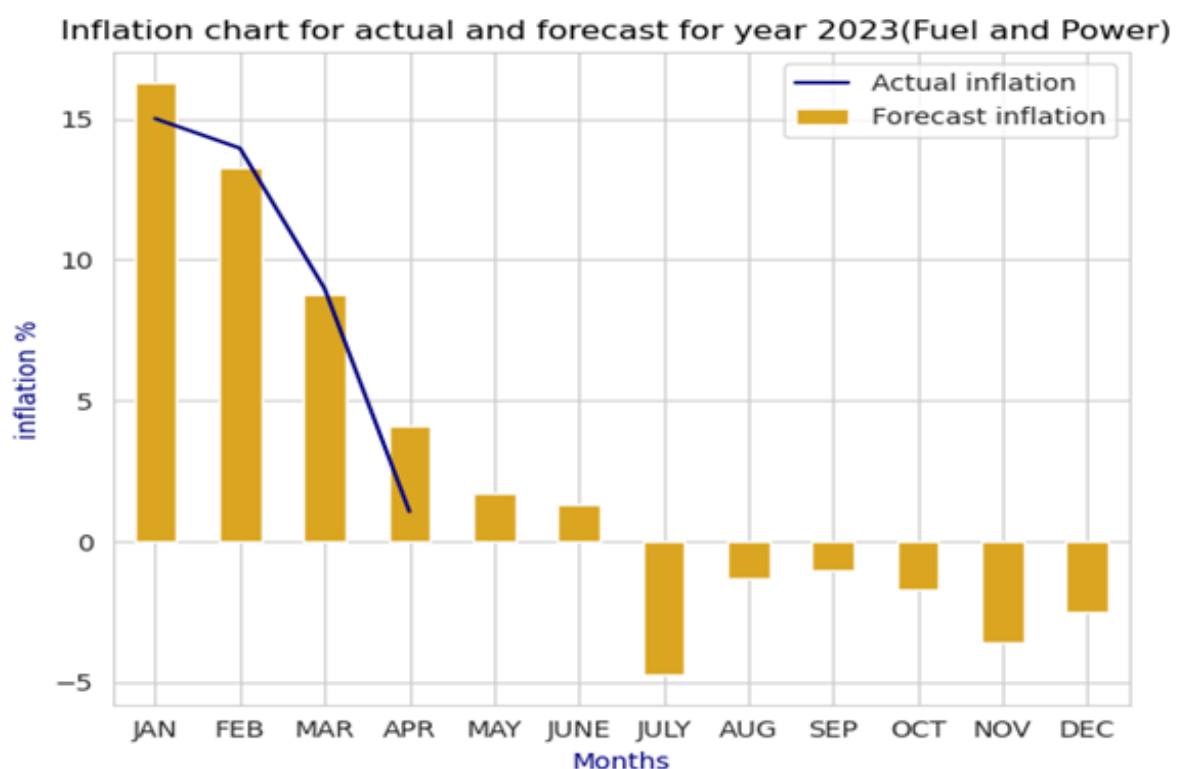
Month	Forecast Price Index	Actual Price Index	Forecast Inflation	Actual Inflation
Jan-23	157.3439	155.6	16.29264	15.0037
Feb-23	156.6422	157.6	13.26264	13.95517
Mar-23	156.5137	156.8	8.765593	8.964559
Apr-23	157.1667	152.6	4.083913	1.059603
May-23	157.0047	----	1.686983	----
Jun-23	157.4234	----	1.302085	----
Jul-23	158.7065	----	-4.73797	----
Aug-23	157.0787	----	-1.3325	----
Sep-23	156.7921	----	-1.01507	----
Oct-23	155.3361	----	-1.68603	----
Nov-23	153.9043	----	-3.56871	----
Dec-23	154.0319	----	-2.51148	----

From above values of Actual price index and Forecasted price index we have **RMSE=2.49476** and **MAE=1.8886**, which means on average, the forecasted values differ from the actual values by approximately **1.8886** units.

Validation:

From stochastic model for Fuel and Power category we have the result that state 9 (**150-160**) has the highest probability and there is an **87.71%** chance that price index will remain in the range 150-160 and 12.28 % chance that price index will be in next state (160-170) , also from our seasonal ARIMA model we have the forecasts for next 12 month and price index are in the range as predicted from stochastic model .

Inflation Chart for the year 2022-23 Using Forecast Data of Fuel and Power category compared with the actual values .



Based on the forecasted and actual inflation data for the year 2022-2023, we can observe the following points

In January we observed high inflation about 16.2% based on forecasted price index and 15% based on actual price index.

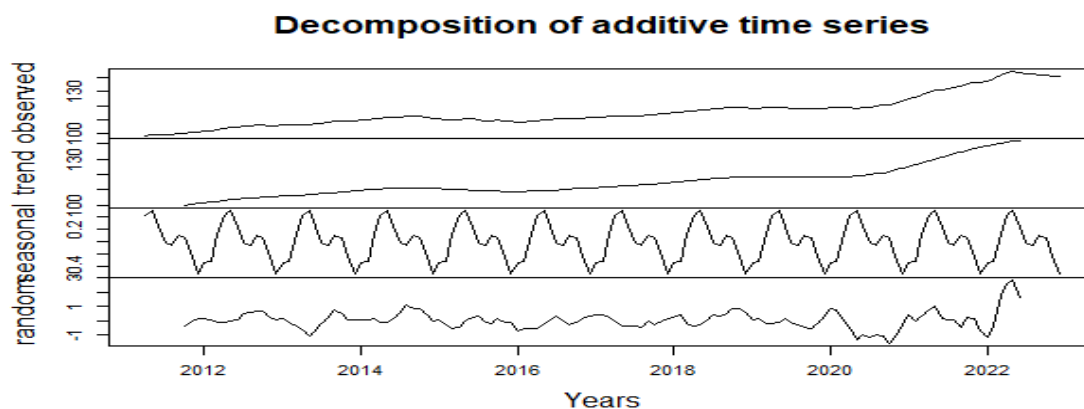
In February inflation is 13.2% based on forecast price index and 13.9% from actual price index which is 2-3% decline in the inflation rates from previous month .After February we can see there is significant decrease in the inflation rates .

TIME SERIES ANALYSIS FOR MANUFACTURED PRODUCTS



The manufacturing category plot reveals a steady trend in the series. From 2014 to 2016, the index prices experienced a decline, followed by a subsequent period of consistent and linear upward movement. However, since May 2020, there has been a significant spike in the index prices, with a continuous increase of 20 to 30% over a span of two years.

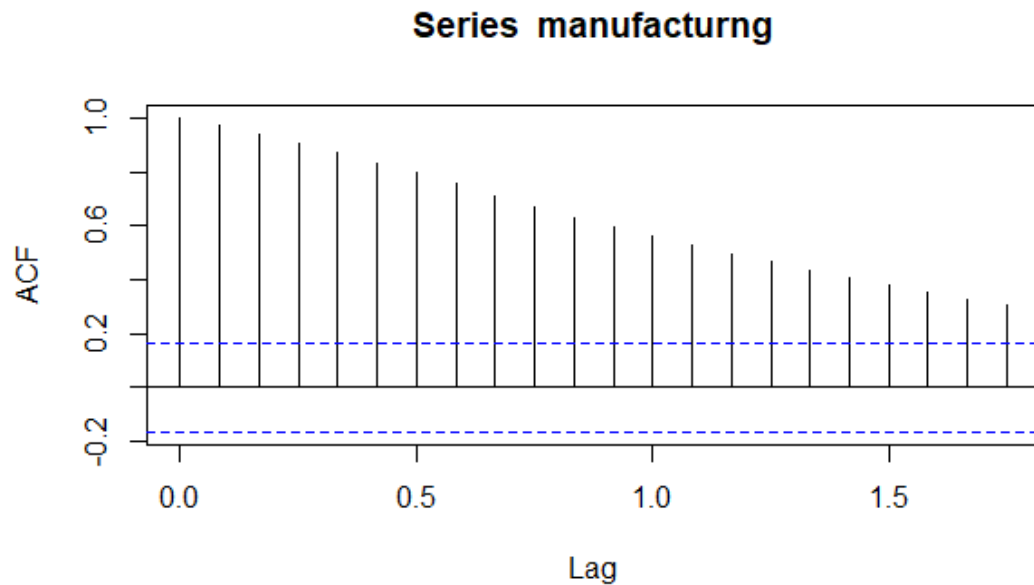
Decomposition of time series :



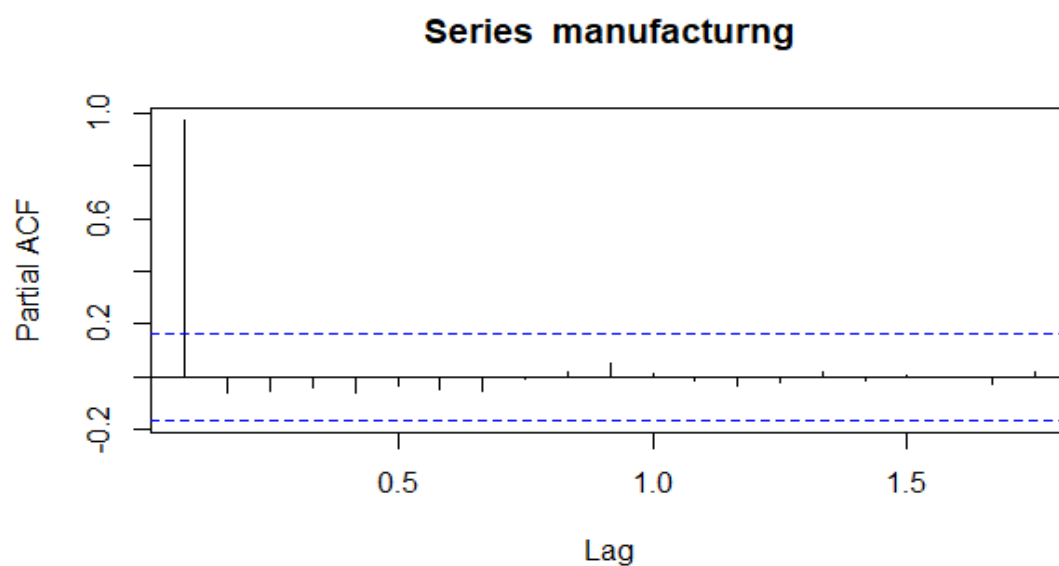
From the above decomposition plot we can see the trend is steady.

The variations are more between the year 2020 and 2022 because prices are increased significantly.

ACF and PACF Plots:



Comment: From Above ACF plot we can see there is decreasing autocorrelation towards zero as lags are increasing, implies that series is not stationary.



We can also check the stationarity by using tests as follows:

Augmented Dickey-Fuller Test

Hypothesis:

H0: The series is not stationary

Vs

H1: The series is stationary

Augmented Dickey-Fuller Test

data: t

Dickey-Fuller = -2.408, Lag order = 5, **p-value = 0.4066**

alternative hypothesis: stationary

Comment : As p-value is greater than level of significance $\alpha=0.05$, we fail to reject the null hypothesis and conclude that series is **not stationary**.

KPSS Test :

Hypothesis

H0: The series is stationary

vs

H1: The series is not stationary

KPSS Test for Level Stationarity

data: manufacturing

KPSS Level = 2.3869, Truncation lag parameter = 4, **p-value = 0.01**

Comment: As p-value is less than level of significance $\alpha=0.05$, we reject the null hypothesis and conclude that series is **not stationary**.

So , to make the series stationary or to detrend the series we need to take differencing of order 1 as we have seen there is a upward trend in the series .

Now , we will check the stationarity after differencing the series .

KPSS Test :

Hypothesis

H0: The series is stationary

vs

H1: The series is not stationary

KPSS Test for Level Stationarity

data: diff(manufacturing, 1)

KPSS Level = 0.30602, Truncation lag parameter = 4, **p-value = 0.1**

Comment: As p-value is greater than level of significance $\alpha=0.05$, we fail to reject the null hypothesis and conclude that series is **stationary without a trend component in the data.**

Time series models:

We have use holt-winters model as it can handle seasonality and trend in series. From the graph there is no significant pattern of seasonality , also model is suitable when the data shows a consistent upward or downward trend over .

Forecast method: Holt-Winters' additive method

Model Information:

Holt-Winters' additive method

Call:

hw(y = t)

Smoothing parameters:

alpha = 0.9991

beta = 0.0603

gamma = 4e-04

Initial states:

$l = 103.1532$

$b = 0.525$

$s = 0.2977 \ -0.0708 \ -0.2194 \ -0.3869 \ -0.2031 \ -0.1817$
 $-0.0082 \ -0.0543 \ -0.2187 \ 0.1808 \ 0.586 \ 0.2786$

sigma: 0.7976

AIC	AICc	BIC
651.0026	655.9782	701.1315

Error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	-0.0079	0.7509	0.4631639	-0.01177	0.3949843	0.1062
	ACF1					
Training set	0.2919222					

Interpretation: In above model alpha is larger (**0.9991**) nearly equals to one means it uses more weights to recent observation making forecast more responsive to recent observation , and beta =**0.0603** shows less weight to recent trend observation as beta is smoothing parameter to trend component , This means that recent trend changes have less influence on the forecast.

Also , holt winter model has mean absolute error of **0.4631** , which means on average, the forecasted values differ from the actual values by approximately **0.4631** units.

- After exploring various parameter combinations for the Seasonal ARIMA model, the model with the lowest values of AIC , RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error) is selected

Seasonal ARIMA models	RMSE	MAE
(1,1,0)(0,0,1)[12] with drift	0.47	0.35
(1,1,0)(2,1,1)[12]	4.10	0.70
(1,1,0)(3,1,0)[12]	4.12	0.72

Hence model with lowest values of RMSE and MAE is **(1,1,0)(0,0,1)[12] with drift.**

Seasonal ARIMA model :

Series: manufacturing

ARIMA(1,1,0)(0,0,1)[12] with drift

Coefficients:

ar1	sma1	drift
0.5997	0.2984	0.3006
s.e. 0.0673	0.1038	0.1268

$\sigma^2 = 0.2317$: log likelihood = -95.55

AIC=199.1 AICc=199.39 BIC=210.86

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	-0.001240	0.4745169	0.3463939	-0.002140	0.2911311	0.07942555
ACF1						
Training set	0.03098451					

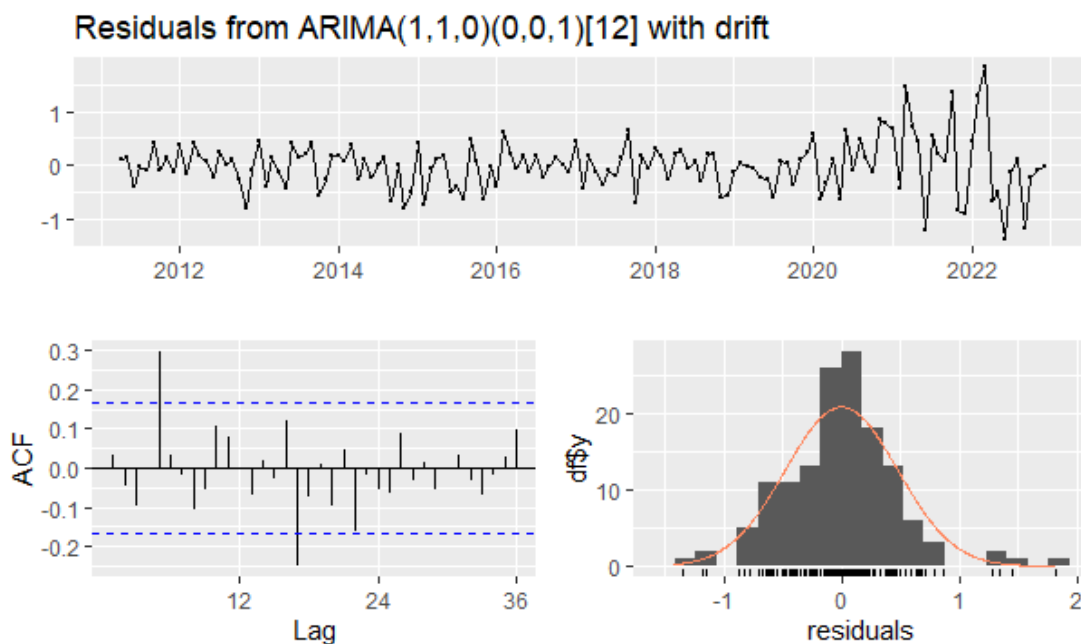
Interpretation :In the SARIMA(1,1,0)(0,0,1)[12] model ARIMA(1,1,0) represents non seasonal component and the autoregressive term (AR) of 1 implies that the current value is related to the previous value. The differencing term (d) of 1 indicates that the series has been differenced once to achieve stationarity. The model does not include any moving average terms (MA) but includes one seasonal MA term (Q).

In our model the drift term specified as **0.3** ,means on an average trend in the series changes by 0.3 units per time step .

Also , Seasonal ARIMA model has mean absolute error of **0.35** , which means on average, the forecasted values differ from the actual values by approximately **0.35** units.

From above Two models RMSE , MAE and AIC, BIC is less for seasonal ARIMA model , so we will choose this model for forecasting .

Diagnostic Measure:



The Ljung-Box test for checking the autocorrelation of residuals from seasonal ARIMA model

Hypothesis:

H0: There is no autocorrelation in the residuals

Vs

H1: There is significant autocorrelation in residuals

Box-Ljung test

data: model\$residuals

X-squared = 0.13827, df = 1, **p-value = 0.71**

Comment : As p-value is greater than level of significance $\alpha=0.05$ we fail to reject the null hypothesis and conclude that **there is no autocorrelation in the residuals.**

- From above plot we can see that plot of residuals is stationary.

- There is no significant autocorrelation in residuals ,means there is no information left in the residuals
- Histogram for residuals is bell shaped means residuals are approx. normally distributed with zero mean , if residuals have zero mean, then the forecasts are unbiased .

Forecast Using seasonal ARIMA model:

Forecast From ARIMA(1,1,0)(0,0,1)[12] with drift



Point Forecasts :

Month	Forecast Price Index	Actual Price Index	Forecast Inflation	Actual Inflation
Jan-23	141.1105	141.4	2.850251	3.061224
Feb-23	141.6222	141.6	1.959857	1.943844
Mar-23	142.5944	141.2	0.206892	-0.77301
Apr-23	143.0978	141.2	-0.62655	-1.94444
May-23	143.3773	----	-0.98252	----
Jun-23	143.2627	----	-0.30434	----
Jul-23	143.2774	----	0.054081	----
Aug-23	143.4414	----	0.168594	----
Sep-23	143.3173	----	0.785702	----
Oct-23	143.2973	----	0.984693	----
Nov-23	143.3769	----	1.326453	----
Dec-23	143.5393	----	1.728769	----

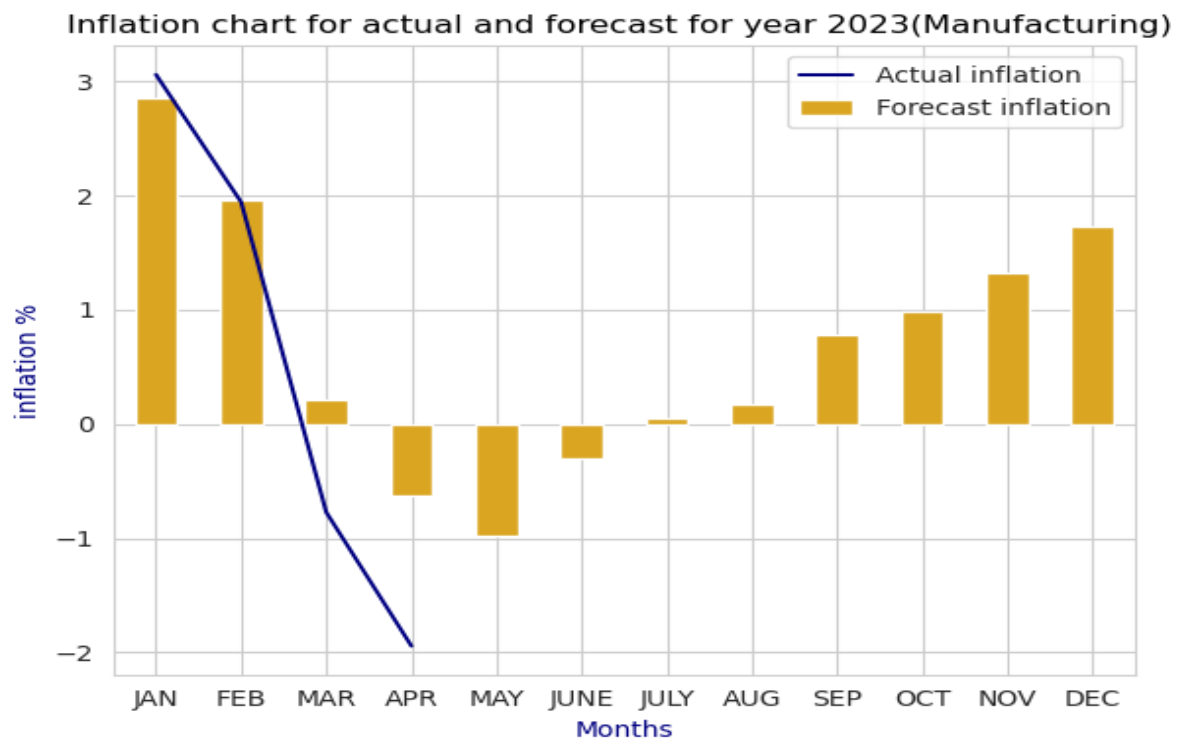
From above values of Actual price index and Forecasted price index we have **RMSE=1.1864** and **MAE=0.9009**, which means on average, the forecasted values differ from the actual values by approximately **0.9009** units,

Validation:

From stochastic model for Manufacturing category, we have the result that price index will lie in state 5 (**135-145**) has the highest probability and there is an **92.30%** chance that price index will remain in the range 135-145 and 7.14 % chance that price index will be in next state (145-155) , also from our seasonal

ARIMA model we have the forecasts for next 12 month and price index are in the range (135-145) as predicted from stochastic model .

Inflation Chart for the year 2022-23 Using Forecast Data of Manufacturing category compared with the actual values .



In January, there was a 3% increase in prices compared to the previous year based on the actual price index. According to the forecasted price index, the anticipated increase was slightly lower at 2.85%, moving to February, there was a slight decrease in inflation rates for both the forecasted and actual price indexes. The forecasted price index projected a decrease of 1.94%, and the actual price index also indicated a similar decline.

From February onwards we observed deflation means decrease in the price index from previous year around 1-2% for March and April months based on actual data.

Overall Findings

CLUSTER ANALYSIS

For Clustering analysis, we have considered financial years that is from 2012-13 to 2021-22 as a variable and objects as commodities. Here the objective was to grouped the commodities together that exhibit similar price index trend over time. Since WPI is divided into 3 categories and as these 3 categories contains different commodities with different weightage therefore, we had applied clustering algorithm separately on these 3 categories and instead of cluster centers we have used weighted average to represent the clusters.

In primary article category we have obtained the best clustering results using PAM (partitioning around medoids) algorithm with an optimal number of 3 clusters one interesting finding is that the outliers in this category are grouped into one cluster.

Cluster 1 accommodate the commodities that **shows steady upward trend** of price index with more or less fluctuations.

Cluster 2: The outliers (commodities that show spike or decline in price index) are being grouped in this cluster that **shows continuous fluctuations** in price index over the span of 10 years.

The commodities in **cluster 3** have price index that is **less than 100** for almost 7 financial years out of 10 so we can say that the cluster 3 includes the commodities which are **relatively cheaper**.

Additionally, the average silhouette coefficient was found to be 0.35

In fuel and power category we have obtained the best clustering results using K-Means algorithm with an optimal number of 3 clusters one interesting finding is that the outliers in this category was only for financial year 2021-22 the commodity petroleum coke and kerosene were found to be outliers since the price index of these commodities shows significant spike for this year.

Cluster 1 contains only these 2 commodities and the price index trend of these cluster is more volatile than any other cluster.

Cluster 2 represent the commodities in fuel and power category which are relatively cheaper but shows a spike for financial year 2021-22.

Cluster 3 represent the commodities whose price trend shows steady upward trend without significant fluctuations.

The average silhouette coefficient was found to be 0.56

In Manufactured products category we have obtained the best clustering results using Hierarchical Agglomerative Clustering with Wards linkage and 2 as optimal number of Cluster and distance measure was Manhattan. There are on an average 29 to 30 outliers for each of the financial year and these outliers forms one separate cluster including 111 commodities whereas all other remaining 453 commodities accommodate in cluster 1.

Cluster 1 represent the commodities that shows steady trend with some ups and down for a year or 2 and a significant spike for year 2021-22. Whereas cluster 2 represent the commodities with higher price index range and continuous increasing trend over the years.

The average silhouette coefficient is found to be 0.5

Limitations:

Clustering results may not always be stable especially when using certain algorithm or random initialization methods. Small changes in the input data or algorithm setting leads to different cluster assignments. Ensuring the stability and reproducibility is quite challenging and require careful consideration.

Dynamic nature of data: Wholesale Price Index data is dynamic and subject to changes over time therefore periodic re-evaluation and adjustment of clustering techniques may be necessary to accommodate evolving nature of data.

STOCHASTIC PROCESS

By modelling the WPI of main 3 categories (Primary Articles, Fuel & Power, Manufactured Products) as Markov Chain, we have described the overall behaviour of system over the span of past 10 years with the help of transition diagram.

We have predicted 8 months ahead probabilities and found that predicted range for Price Index for all 3 categories of WPI was almost similar with actual Price Index. Later, noted down for recent previous 4 months i.e., for January, February, March, and April.

As well as we have studied long term behaviour of each category i.e., steady state probabilities.

For Primary Articles category, state 9 that is Price Index range 170-180 has the highest probability.

For Fuel & Power category, state 9 that is Price Index range 150-160 has the highest probability whereas for 3 rd category Manufactured Products state 5 that is Price Index range 135-145 has the highest probability.

Limitations:

Limitations of using Markov -Chain to model the WPI is that it assumes that the probability of transitioning between states is constant over time.

However, in reality the transition probabilities may change due to external factors.

TIME SERIES

For time series analysis we have used monthly data from 2011 to 2022. Our Aim was to forecast the price index for year 2023. As we have three categories so we fit different models to these categories and choose the model which gives less error measure.

For All three categories we have used holt winters model and seasonal ARIMA model, The AIC and BIC for Seasonal ARIMA model is less from holt winters model. To select the best seasonal ARIMA model we have used different parameter combinations and combination which gives less error measure that is chosen for forecasting.

As we have used all the data to build the models because there is a significant spike in the prices from 2020 onwards and after trying different models it is hard to capture that spike by using the train data, so for that purpose we have used all the data and validate the forecasting result from the stochastic output.

For primary articles category the Seasonal ARIMA model is $(0,1,1) (2,0,0)[12]$ with drift and this model gives less error measure compared to other models , The RMSE for this model is 2.076 and MAE is 1.52 . By using above SARIMA model, we forecast the values for year 2023 and as per current situation we have the actual values till APRIL 2023, so for these 4 values of forecast and actual we get MAE about 2.304696

For fuel and power category the SARIMA model perform well and gives less error measure and AIC value, by using holt winters model we are getting forecast which are exponentially increasing and by using stochastic result the values should be in a range therefore we use SARIMA model, Now the seasonal ARIMA model is $SARIMA(1,1,0)(0,0,2)[12]$ and this gives less error measure compare to other SARIMA models , The RMSE for this model is 2.89 and MAE is 2.08. . By using above SARIMA model, we forecast the values for year 2023 and as per current data we have the actual values till APRIL 2023, so for these 4 values of forecast and actual we get MAE about 1.8886.

For manufacturing category, the SARIMA model perform gives less error measure compare to Holt winters model and the SARIMA model is chosen by using different parameter combination and the model which gives less error measure is SARIMA(1,1,0)(0,0,1)[12] and RMSE for this model is 0.47 and MAE is 0.35 , By using the forecasted values and actual values of year 2023 till April we get MAE = 0.9009.

Limitations :

- SARIMA models are more suitable for short- to medium-term forecasting rather than long-term trends or when the forecasting horizon extends beyond the available historical data. Accuracy may decrease, and uncertainty in predictions may increase for longer-term forecasts.
- External Factors: SARIMA models focus solely on the time series data and assume that the underlying process is driven by its own historical patterns. They do not explicitly incorporate external factors or exogenous variables that may influence the time series.
- Outliers and Anomalies: SARIMA models assume that the time series data is free from outliers or anomalies that could significantly impact the model's performance.

REFERENCES AND SOFTWARES USED FOR THE ANALYSIS

Following are the softwares that has been used for the analysis of WPI data:

- R-Software
- Python
- MS-Excel
- Minitab

Following are the references and the webpages that we have referred to:

- https://en.m.wikipedia.org/wiki/Markov_chain
- <https://eaindustry.nic.in/>
- <https://tradingeconomics.com/india/producer-prices-change>
- Reference Book: Introduction to Time Series and Forecasting, Second Edition, Peter J. Brockwell & Richard A. Davis.